Statistical Thermodynamics

Syllabus

Module 5

1. Fundamentals – concept of distribution, thermodynamic probability and most probable distribution, ensembles
2. Statistical mechanics for systems of independent particles and its importance in chemistry, thermodynamic probability & entropy, idea of microstates and macrostates, statistical weight factor (g), Sterling approximation
3. Maxwell- Boltzmann statistics. The molecular partition function and its relation to the thermodynamic properties, derivation of third law of thermodynamics.
4. Equilibrium constant and equi-partition principle in terms of partition functions, relation between molecular & molar partition functions, factorisation of the molecular partition function into translational, rotational, vibrational and electronic parts, the corresponding contributions to the thermodynamic properties
5. Evaluation of partition functions and thermodynamic properties for ideal mono-atomic and diatomic gases.

Module 6

1. Heat capacities of solids - classical and quantum theories, Einstein's theory of atomic crystals and Debye's modification.
2. Quantum Statistics: Bose - Einstein distribution law, Bose-Einstein condensation, application to liquid helium; Fermi - Dirac distribution law, application to electrons in metals
3. Relationship between Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac statistics.

Conceptual Map of Statistical Thermodynamics

1. Microscopic Description of a System

A macroscopic system contains an enormous number of particles (~10^{23}).

Instead of tracking each particle individually, statistical thermodynamics connects microscopic states to macroscopic properties.

Two fundamental concepts are introduced.

Microstate

A microstate specifies the exact microscopic configuration of the system.

Examples of microscopic variables:

- positions of particles
- momenta of particles
- quantum numbers of particles

For a classical system:

$$\{\boldsymbol{r_1}, \boldsymbol{p_1}, \boldsymbol{r_2}, \boldsymbol{p_2}, \ldots, \boldsymbol{r_N}, \boldsymbol{p_N}\}$$

Macrostate

A macrostate describes the system using thermodynamic variables.

Examples:

$$(\boldsymbol{T}, \boldsymbol{P}, \boldsymbol{V}, \boldsymbol{N})$$

or

$$(\boldsymbol{E}, \boldsymbol{V}, \boldsymbol{N})$$

Many different microstates correspond to the same macrostate.

2. Thermodynamic Probability

If a macrostate can be realized in W different microstates, then

$$\boldsymbol{W} = \text{number of microstates}$$

This quantity is called thermodynamic probability.

3. Statistical Definition of Entropy

Boltzmann established the connection between microscopic states and entropy:

$$\boldsymbol{S} = \boldsymbol{k_B} \ln \boldsymbol{W}$$

This equation forms the foundation of statistical thermodynamics.

Meaning:

- larger number of microstates
- greater disorder
- higher entropy

4. Most Probable Distribution

In a system containing many particles:

- many different particle distributions are possible
- the system overwhelmingly occupies the most probable distribution

This distribution corresponds to thermodynamic equilibrium.

5. Statistical Distribution Laws

Maximization of thermodynamic probability gives the distribution of particles among energy levels.

Three important statistics arise depending on particle properties.

Maxwell–Boltzmann Statistics

Applies to:

- classical particles
- distinguishable particles
- low density / high temperature systems

Distribution:

$$N_i = N \frac{g_i e^{-\frac{\epsilon_i}{kT}}}{\sum g_i e^{-\frac{\epsilon_i}{kT}}}$$

Example:

ideal gases.

Bose–Einstein Statistics

Applies to:

- indistinguishable particles
- integer spin particles (bosons)

Distribution:

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

Example systems:

- photons
- helium-4 atoms.

Leads to Bose–Einstein condensation.

Fermi–Dirac Statistics

Applies to:

- indistinguishable particles
- half-integer spin particles (fermions)
- Pauli exclusion principle

Distribution:

$$\boldsymbol{N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}}$$

Example systems:

- electrons in metals.

6. Partition Function

The central quantity in statistical thermodynamics is the partition function.

For a molecule:

$$\boldsymbol{q = \sum g_i e^{-\frac{\epsilon_i}{kT}}}$$

Physical meaning:

The partition function measures the number of energy states accessible at temperature $\boldsymbol{T}$.

7. Thermodynamic Properties from Partition Function

All thermodynamic quantities can be derived from the partition function.

Examples:

Internal Energy

$$U = kT^2 \frac{\partial \ln q}{\partial T}$$

Helmholtz Free Energy

$$F = -\mathrm{kTln}\, q$$

Entropy

$$S = \mathrm{kln}\, q + kT \frac{\partial \ln q}{\partial T}$$

Heat Capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

8. Factorisation of Molecular Partition Function

For a molecule:

$$q = q_{trans}\, q_{rot}\, q_{vib}\, q_{elec}$$

These correspond to:

Motion	Contribution
Translational	motion of molecule
Rotational	rotation of molecule
Vibrational	vibration of atoms
Electronic	electronic energy levels

Each contributes separately to thermodynamic properties.

9. Applications

Statistical thermodynamics explains several important phenomena:

Ideal Gas Properties

Derived from translational partition function.

Heat Capacity of Solids

Explained by

- Einstein theory
- Debye theory.

Quantum Phenomena

Examples:

- Bose–Einstein condensation
- electrons in metals (Fermi gas)

Part I – Foundations

1. Introduction to Statistical Thermodynamics
    - Role of statistics in thermodynamics
    - Systems of independent particles
    - Importance in chemistry
2. Microstates and Macrostates
3. Thermodynamic Probability
4. Boltzmann Entropy Relation

$$\boldsymbol{S = k\ln W}$$

5. Stirling Approximation

Part II – Ensembles

6. Concept of Ensemble
7. Microcanonical Ensemble
8. Canonical Ensemble
    - Boltzmann distribution
    - Canonical partition function
9. Grand Canonical Ensemble
10. Isothermal–Isobaric Ensemble

Part III – Classical Statistics

11. Maxwell–Boltzmann Statistics

- Thermodynamic probability
- Most probable distribution
- Boltzmann distribution law

Part IV – Partition Function

12. Molecular Partition Function
13. Relation Between Molecular and Molar Partition Functions
14. Factorization of Molecular Partition Function

- Translational
- Rotational
- Vibrational
- Electronic

15. Thermodynamic Properties from Partition Function

$$U = kT^2 \frac{\partial \ln q}{\partial T}$$

$$S = k\ln q + kT\left(\frac{\partial \ln q}{\partial T}\right)$$

16. Equipartition Theorem
17. Evaluation for Ideal Monoatomic and Diatomic Gases

Part V – Quantum Statistics

18. Failure of Classical Statistics
19. Bose–Einstein Statistics
20. Bose–Einstein Condensation
21. Fermi–Dirac Statistics
22. Application to Electrons in Metals

Part VI – Comparison of Statistics

23. Relationship between MB, BE, and FD statistics

Property	MB	BE	FD
Particle type	classical	bosons	fermions
Distinguishability	distinguishable	indistinguishable	indistinguishable
Occupation	unlimited	unlimited	≤1

Unified distribution form:

$$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + \sigma}$$

Part VII – Heat Capacity of Solids

24. Classical Theory (Dulong–Petit)
25. Einstein Theory
26. Debye Modification

Part I – Foundations

1. Introduction to Statistical Thermodynamics

Thermodynamics describes the properties of macroscopic systems using quantities such as temperature, pressure, volume, and entropy. These quantities represent the collective behaviour of a very large number of particles.

However, thermodynamics itself does not explicitly describe the **microscopic structure of matter**. It does not consider how individual atoms or molecules behave.

Statistical thermodynamics provides the **microscopic foundation of thermodynamics**. It connects the macroscopic properties of a system with the behaviour of its constituent particles.

A macroscopic sample of matter contains an extremely large number of particles (of the order of 10^{23}). Because it is impossible to follow the motion of each particle individually, statistical methods are used to describe the **average behaviour of large numbers of particles**.

The central idea of statistical thermodynamics is that macroscopic properties arise from the **statistical distribution of particles among different energy states**.

Using this approach, many thermodynamic quantities can be explained in terms of molecular behaviour. Examples include

- internal energy
- entropy
- heat capacity
- equilibrium constants
- properties of gases and solids.

Statistical thermodynamics therefore acts as a bridge between **microscopic physics and macroscopic thermodynamics**.

Statistical thermodynamics assumes that matter is made up of a large number of microscopic components that individually obey quantum mechanics but collectively obey statistical laws. Statistical thermodynamics establishes a connection between the microscopic description and the macroscopic description of nature.

Statistical Mechanics of Systems of Independent Particles

Many chemical systems can be approximated as collections of particles that behave independently of one another. In such systems, the energy of the system can be expressed as the sum of the energies of the individual particles, which greatly simplifies the theoretical treatment. This approximation is particularly valid for ideal gases and dilute systems where intermolecular interactions are negligible.

In order to describe how particles are distributed among the available energy levels at thermal equilibrium, statistical mechanics introduces a quantitative description of the populations of energy states. This description forms the basis for calculating macroscopic thermodynamic properties from microscopic molecular data. To obtain a quantitative description of the distribution of molecules among the available energy states, statistical mechanics makes use of the **Boltzmann distribution**, which forms the basis for the development of partition functions.

A system is said to consist of independent particles when the particles do not interact significantly with one another. Under this approximation, the state of one particle does not influence the energy or motion of other particles in the system.

In such systems, the total energy of the system can be expressed as the sum of the energies of the individual particles. If a system contains N particles with individual energies

$$\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots, \varepsilon_N$$

then the total energy of the system is

$$E = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \cdots + \varepsilon_N$$

This additive property of energy is a direct consequence of the absence of interactions between particles.

Physical Situations Where the Approximation is Valid

The independent particle approximation applies well to **ideal gases** and many **dilute gaseous systems**, as molecules move freely with negligible intermolecular forces except during short collisions. This allows simplified statistical analysis, focusing on single-particle behaviour to infer system-wide properties.

The **statistical mechanical treatment of independent particles** plays a fundamental role in chemistry. It explains the thermodynamic behaviour of gases and relates macroscopic thermodynamic quantities to molecular properties.

For example, the distribution of molecules among rotational and vibrational energy levels determines the intensity patterns observed in molecular spectra. Statistical mechanics also provides the theoretical basis for chemical equilibrium by relating equilibrium composition to the energy levels and degeneracies of molecules.

In addition, the temperature dependence of thermodynamic properties such as heat capacity can be understood in terms of the excitation of different molecular motions as temperature increases.

Limitations of the Independent Particle Approximation

Although the independent particle model is extremely useful, it has certain limitations. In many real systems, interactions between particles cannot be neglected.

In **liquids, dense gases, and solids**, intermolecular forces become significant and influence the energies and motions of particles. In such cases, the assumption that particles behave independently is no longer valid.

More advanced statistical mechanical methods are required to account for these interactions.

2. Microstates and Macrostates

Microstate

A **microstate** is a specific configuration of a system that describes the precise positions and momenta of all the individual particles or components that make up the system. Each microstate has a certain probability of occurring during the course of the system's thermal fluctuations.

$$U = \langle E \rangle = \sum_{i=1}^{N} p_i \varepsilon_i$$

This is a microscopic statement of the notion of internal energy

Classical system

A microstate is defined by the set:

$$\{\mathbf{r}_1, \mathbf{p}_1, \mathbf{r}_2, \mathbf{p}_2, \dots, \mathbf{r}_N, \mathbf{p}_N\}$$

where:

- $\mathbf{r}_i$= position vector of particle i
- $\mathbf{p}_i$= momentum vector of particle i
- 6N coordinates (3N coordinates of position, and 3N coordinates of momentum) are needed to describe the system in classical mechanics and calculate its properties.

Thus a microstate corresponds to a single point in a multidimensional space ($6N$-dimensional) known as **phase space**, whose coordinates represent the positions and momenta of all particles.

For a quantum mechanical system, instead of specifying exact positions and momenta, a microstate is defined by the **occupation of quantum energy levels**. If ϵ_i represents the energy of the ith quantum state, then a microstate corresponds to a particular arrangement of particles among these energy states.

Each distinct microscopic arrangement dependent on the quantum numbers n,l,m and m_s, which must be specified for each particle of particles corresponds to a different microstate. This means 4N quantum numbers for a system of N particles.

Macrostate

A macrostate represents the overall state of a system, characterized by macroscopic properties such as temperature, pressure, volume, and energy. These variables provide a general

description of the system without specifying the detailed states of the individual particles. Macrostates are characterized by their extensive properties, which depend on the size or amount of the system. For example, the macrostate of a gas can be defined by its pressure, volume, and temperature, regardless of the specific arrangement and motion of its individual molecules.

A **macrostate** is specified by thermodynamic variables called state functions, which depend only on the initial and final conditions but not on the path followed.

Examples:

$$(N, V, E), (N, V, T), (T, P, N)$$

It does not specify microscopic detail.

A macrostate can correspond to many different microstates.

In a pure substance in equilibrium, the state of the system (macrostate) is defined by three variables, pressure, temperature and number of moles. It is not necessary to specify the volume because there is an equation of state that relates the four thermodynamic variables. For example, in the case of an ideal gas PV=nRT.

In this case, the macrostate of the system is specified by three state variables, since any other thermodynamic magnitude can be obtained from them. Thus, U=U(P,T,n)=U(V,T,n)=....

Giving the macrostate of a system consists, therefore, in specifying the smallest possible number of independent variables that determine its thermodynamic state

Relationship Between Microstates and Macrostates

A single macrostate can correspond to **many different microstates**.

This means that a system having a given set of macroscopic properties may be realized through many different microscopic arrangements of particles.

Example

Consider three distinguishable particles A, B, and C distributed between two boxes.

Possible macrostates are

Macrostate	**Box 1**	**Box 2**
I	3	0
II	2	1
III	1	2
IV	0	3

Each macrostate can be realized through different microstates.

For example, the macrostate (2,1) can occur in three different ways:

$$AB, C$$

$$AC, B$$

$$BC, A$$

Similarly

- macrostate (3,0) has only one microstate
- macrostate (1,2) has three microstates
- macrostate (0,3) has one microstate.

This example illustrates that different macrostates can have different numbers of microstates.

Distribution of Particles Among Energy Levels

Consider a system of N **distinguishable particles** distributed among r **energy levels (quantum states)**.

Let

- N_1 particles occupy energy level ε_1
- N_2 particles occupy energy level ε_2
- N_3 particles occupy energy level ε_3
- …
- N_r particles occupy energy level ε_r

The total number of particles is

$$N = N_1 + N_2 + N_3 + \cdots + N_r$$

This specification $(N_1, N_2, \dots, N_r)$ defines a **distribution** (macrostate).

Number of microstates of a distribution

The number of microscopic arrangements corresponding to the distribution $(N_1, N_2, \dots, N_r)$ is given by the **multinomial expression**

$$W = \frac{N!}{N_1!\ N_2!\ N_3! \dots N_r!}$$

where

- W = number of microstates of the distribution
- $N!$ = factorial of total number of particles
- $N_1!, N_2!, \dots, N_r!$ account for indistinguishable permutations within each group.

This expression arises from combinatorics: the number of ways of arranging N distinguishable particles into groups of sizes $N_1, N_2, \dots, N_r$.

Physical Meaning of W

- W represents the **multiplicity** of a macrostate.
- A macrostate with larger W can be realized in more ways.
- Therefore such macrostates are **more probable**.

This concept forms the basis of statistical thermodynamics.

Degeneracy of Energy Levels

In many physical systems, an energy level may correspond to more than one quantum state. When several distinct states possess the same energy, the level is said to be **degenerate**.

If the energy level ε_i corresponds to g_i distinct quantum states, then g_i is called the **degeneracy of that energy level**. Thus, g_i represents the number of available states having the same energy ε_i.

Statistical Weight

For a given distribution $(n_1, n_2, n_3, \dots)$, the particles can be assigned to the different energy levels in many ways. The number of ways of distributing N distinguishable particles among the levels while keeping $n_1, n_2, n_3, \dots$ fixed is

$$\frac{N!}{n_1!\ n_2!\ n_3! \cdots}$$

This factor accounts for the permutations of particles among the different energy levels.

However, if the energy level ε_i has degeneracy g_i, the n_i particles occupying that level can be arranged among its g_i distinct states in

$$g_i^{n_i}$$

different ways.

Taking this into account for all energy levels, the **statistical weight** (or thermodynamic probability) of the distribution becomes

$$W = \frac{N!}{n_1!\ n_2!\ n_3! \cdots}\, g_1^{n_1} g_2^{n_2} g_3^{n_3} \cdots$$

The statistical weight W represents the **number of microscopic arrangements (microstates)** corresponding to a particular distribution of particles among the energy levels.

The factorial term describes the number of ways of assigning particles to the different energy levels, while the factors $g_i^{n_i}$ account for the different ways in which the particles can occupy the **degenerate states within each energy level**. Consequently, energy levels with larger

degeneracy contribute more strongly to the statistical weight because they provide a greater number of accessible states for the particles.

Probability of a distribution

If several possible distributions exist, the probability of a particular distribution $i$is proportional to the number of microstates associated with that distribution.

Let

$$W_i = \text{number of microstates for distribution } i$$

Then the probability of finding the system in distribution $i$is

$$P_i = \frac{W_i}{\sum_j W_j}$$

where

- P_i= probability of distribution i
- $\sum_j W_j$= total number of microstates for all possible distributions.

Most probable distribution

The probability of a particular distribution is proportional to the number of microstates wi for that distribution

$$P_i \propto W_i$$

The distribution with the **largest multiplicity** is therefore the **most probable distribution**.

3. Thermodynamic Probability

The **thermodynamic probability** of a macrostate is defined as the number of microstates (distinct arrangements) corresponding to that macrostate.

It is denoted by

$$W$$

Thus

$$W = \text{number of microstates corresponding to a macrostate}$$

Although the term contains the word "probability", $W$is not a probability in the usual sense. It is simply a **count of the possible microscopic arrangements**.

The probability of observing a particular macrostate is proportional to the number of microstates corresponding to it.

Thus

$$P \propto W$$

This means that macrostates with larger values of W occur more frequently.

In systems containing a very large number of particles, the macrostate with the largest value of W becomes overwhelmingly more probable than all other macrostates.

- A macrostate with larger W can be realized in more microscopic ways.
- Therefore, the system is overwhelmingly more likely to be found in that macrostate.

Physical meaning

A system containing a very large number of particles will almost always be found in the macrostate with the largest number of accessible microstates. This is why equilibrium corresponds to the **maximum thermodynamic probability**.

Entropy and Thermodynamic Probability

For a system of N distinguishable classical particles distributed among energy levels with occupation numbers n_i, the thermodynamic probability of a given distribution is

$$W = \frac{N!}{\prod_i n_i\,!}$$

The quantity W represents the number of microscopic arrangements corresponding to the specified macroscopic distribution.

Entropy and Probability

A macrostate with larger W is **more probable**, because there are more ways for the system to realize it. Experiments show that an isolated system naturally evolves toward **thermodynamic equilibrium**, and equilibrium corresponds to the **most probable macrostate**.

From second law of thermodynamics, For an **isolated system** (no exchange of energy or matter with surroundings),

$$\Delta S \geq 0$$

- $\Delta S > 0$ for a **spontaneous process**
- $\Delta S = 0$ at **equilibrium**

This means that the entropy of an isolated system **never decreases**.

Maximum Entropy Condition

During any spontaneous process the entropy increases, and when the system reaches equilibrium,

$$\Delta S = 0$$

Since entropy keeps increasing during spontaneous changes and stops changing only when equilibrium is reached, the equilibrium state must correspond to the **maximum possible entropy** consistent with the constraints of the system (fixed E, V, N).

Thus,

$$\text{Equilibrium} \leftrightarrow \text{Maximum entropy}$$

On the other hand, statistical considerations show that equilibrium corresponds to the **most probable macrostate**, that is, the macrostate with the largest value of W.

Therefore,

$$\text{Maximum entropy} \leftrightarrow \text{Maximum thermodynamic probability}$$

This implies that entropy must be related to the thermodynamic probability W. Hence we write

$$S = f(W)$$

where f is some function of W.

Mathematically,

$$S = \text{maximum at equilibrium}$$

or

$$\left(\frac{dS}{dX}\right)_{E,V,N} = 0$$

with

$$\frac{d^2S}{dX^2} < 0$$

for stability.

Functional Form of the Entropy

Entropy must satisfy the thermodynamic property of **additivity**. If two systems are independent and do not interact with each other, the entropy of the combined system is equal to the sum of the entropies of the individual systems. For two independent systems,

$$S_{\text{total}} = S_1 + S_2$$

However, the thermodynamic probabilities of independent systems combine differently. If system 1 has W_1 possible microstates and system 2 has W_2 possible microstates, then every microstate of system 1 can occur together with every microstate of system 2. Hence, the total number of possible microstates of the combined system is

$$W_{\text{total}} = W_1 W_2$$

Since entropy is assumed to be a function of the thermodynamic probability,

$$S = f(W)$$

the above relations require that

$$f(W_1 W_2) = f(W_1) + f(W_2)$$

The only continuous function that satisfies this condition is the logarithmic function. Therefore,

$$S = \mathrm{kln}\, W$$

where $k$is the **Boltzmann constant**.

Boltzmann–Planck Entropy Relation

$$\boxed{S = \mathrm{kln}\, W}$$

This relation is known as the **Boltzmann–Planck entropy formula**. It establishes the fundamental connection between the macroscopic thermodynamic quantity entropy and the number of microscopic arrangements accessible to the system.

Thus entropy provides a measure of the **multiplicity of microstates** corresponding to a given macrostate, and equilibrium corresponds to the state with the **maximum thermodynamic probability**.

Physical Interpretation of Entropy

If a system can exist in many different microstates, it has greater disorder and therefore larger entropy.

Thus

$$\text{larger } W \Rightarrow \text{larger entropy}$$

For example, when a gas expands into a larger volume, the number of possible positions available to the molecules increases. As a result, the number of accessible microstates increases and the entropy of the system increases.

At equilibrium, a system tends to occupy the macrostate with the **maximum thermodynamic probability**, because that macrostate corresponds to the largest number of accessible microstates.

5. Stirling Approximation

Many expressions in statistical thermodynamics involve factorial terms such as $N!$. Factorials arise because we count the number of ways particles can be arranged among energy levels.

For macroscopic systems

$$N \sim 10^{23}$$

so calculating $N!$directly is impossible. Stirling's approximation allows these very large numbers to be handled analytically.

For large values of n,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

In statistical thermodynamics, the logarithmic form is particularly useful:

$$\ln n! \approx n\ln n - n$$

This approximation simplifies many calculations involving thermodynamic probability.

6. Most Probable Distribution

Consider a system containing a large number of particles distributed among different energy levels.

There are many possible ways in which the particles can be distributed among these energy levels. Each possible arrangement corresponds to a particular macrostate with its own thermodynamic probability W.

Among all these possible distributions, one particular distribution has the **maximum value of** W. This distribution is called the **most probable distribution**.

For systems containing only a small number of particles, several distributions may occur with comparable probabilities. However, in macroscopic systems containing an extremely large number of particles, the probability of the most probable distribution becomes overwhelmingly large.

As a result, the system spends almost all its time in the most probable distribution.

Therefore, the macroscopic properties of a system at equilibrium are determined almost entirely by the **most probable distribution of particles among energy levels**.

This idea forms the basis for deriving the statistical distribution laws such as

- Maxwell–Boltzmann statistics
- Bose–Einstein statistics
- Fermi–Dirac statistics.

7. Method of Maximization of Thermodynamic Probability

In statistical thermodynamics, a system containing a large number of particles can be distributed among many possible energy levels in numerous ways. Each possible arrangement corresponds to a particular **macrostate** with an associated **thermodynamic probability** W.

The thermodynamic probability represents the number of microstates corresponding to a given macrostate.

Among all possible macrostates, the system naturally evolves toward the macrostate that has the **largest value of** W. This macrostate is called the **most probable distribution** and corresponds to the state of **thermodynamic equilibrium**.

Therefore, determining the equilibrium distribution of particles among energy levels requires finding the distribution that **maximizes the thermodynamic probability** W.

- equilibrium corresponds to the **most probable distribution**
- maximizing W gives the distribution that the system occupies almost all the time.

Use of Logarithm of Thermodynamic Probability

Since the thermodynamic probability W often contains factorial expressions, it is mathematically convenient to work with its logarithm.

Thus instead of maximizing W, we maximize

$$\ln W$$

because the logarithmic function is monotonic; the value of W is maximum when $\ln W$ is maximum.

Using logarithms also simplifies calculations because factorial expressions can be approximated using **Stirling's approximation**

$$\ln n! \approx n\ln n - n$$

Constraints on the Distribution

The distribution of particles among energy levels is not arbitrary. Two important constraints must always be satisfied.

Particle number conservation

The total number of particles in the system is fixed.

$$\sum_i N_i = N$$

where

- N_i = number of particles in the ith energy level
- N = total number of particles.

Energy conservation

The total energy of the system is also fixed.

$$\sum_i N_i \, \epsilon_i = U$$

where

- ϵ_i = energy of the ith level

- U = total energy of the system.

Method of Lagrange Multipliers

To maximize $\ln W$ subject to these constraints, the **method of Lagrange multipliers** is used.

Two multipliers α and β are introduced to incorporate the constraints.

Thus we maximize the expression

$$\ln W + \alpha \sum_i N_i - \beta \sum_i N_i \, \epsilon_i$$

The condition for maximum probability is

$$d\left(\ln W + \alpha \sum_i N_i - \beta \sum_i N_i \, \epsilon_i\right) = 0$$

Differentiating with respect to N_i gives

$$\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta \epsilon_i = 0$$

This equation determines the equilibrium distribution of particles among energy levels.

Physical Meaning of the Lagrange Multipliers

The constants α and β are determined using the constraints.

It can be shown that

$$\beta = \frac{1}{kT}$$

and

$$\alpha = -\frac{\mu}{kT}$$

where

- T = temperature
- k = Boltzmann constant
- μ = chemical potential.

Thus the equilibrium distribution depends on **temperature and chemical potential**.

Importance of the Maximization Method

The method of maximizing thermodynamic probability forms the foundation of statistical mechanics. It provides a systematic procedure for deriving the equilibrium distribution of particles.

Using this method, one can derive the three important statistical distributions:

- **Maxwell–Boltzmann distribution**
- **Bose–Einstein distribution**
- **Fermi–Dirac distribution**

Each distribution is obtained by applying the same maximization principle but using different expressions for the thermodynamic probability W.

Summary

The equilibrium distribution of particles in statistical thermodynamics is obtained by

1. Writing an expression for the **thermodynamic probability** W.
2. Taking the logarithm to obtain $\ln W$.
3. Applying **Stirling's approximation** to simplify factorial terms.
4. Maximizing $\ln W$ subject to the constraints

$$\sum_i N_i = N$$

$$\sum_i N_i \epsilon_i = U$$

5. Solving the resulting equations using **Lagrange multipliers**.

This procedure leads directly to the statistical distribution laws of equilibrium systems.

Summary of Part I

Concept	**Description**
Microstate	exact microscopic configuration of system
Macrostate	thermodynamic description using macroscopic variables
Thermodynamic probability W	number of microstates corresponding to a macrostate
Boltzmann relation	$S = k_B \ln W$
Stirling approximation	simplifies factorial expressions
Most probable distribution	determines equilibrium properties

These ideas form the **conceptual foundation of statistical thermodynamics**.

Part II – Ensembles in Statistical Thermodynamics

1. Concept of an Ensemble

In statistical thermodynamics, it is often convenient to imagine a very large number of identical systems instead of considering only a single system.

An **ensemble** is defined as a large collection of hypothetical copies of a system, each representing a possible microscopic state consistent with specified macroscopic conditions.

All systems in the ensemble have the same macroscopic properties, such as temperature, volume, and number of particles. However, they may differ in their microscopic configurations.

Thus an ensemble represents all possible microstates that a system can have under given macroscopic constraints.

Each member of the ensemble corresponds to one possible microstate of the system.

Purpose of Using Ensembles

Since it is impossible to observe the microscopic state of a real system at every instant, statistical thermodynamics uses ensembles to determine **average values of physical quantities**.

If A_r represents the value of a physical property in microstate r, and P_r represents the probability of that microstate, then the average value of the property is

$$\langle A \rangle = \sum_r P_r A_r$$

This average value corresponds to the **observable macroscopic quantity**.

For example

- average energy corresponds to internal energy
- average particle number corresponds to macroscopic particle number.

2. Classification of Ensembles

Different ensembles are defined depending on which macroscopic variables remain fixed.

The most important ensembles used in statistical thermodynamics are:

Ensemble	Fixed Variables	Fluctuating Variables	Thermodynamic Potential
Microcanonical	E, V, N	none	entropy
Canonical	T, V, N	energy	Helmholtz free energy
Grand canonical	T, V, μ	energy, particle number	grand potential

Ensemble	Fixed Variables	Fluctuating Variables	Thermodynamic Potential
Isothermal–isobaric	T, P, N	energy, volume	Gibbs free energy

Each ensemble corresponds to a particular physical situation.

3. Microcanonical Ensemble

(Constant E, V, N)

Definition

The **microcanonical ensemble** represents a collection of identical systems that are completely isolated from their surroundings.

Physical Situation

An **isolated system**:

- No energy exchange
- No particle exchange
- Fixed volume

Fundamental Postulate

All accessible microstates with total energy E are equally probable.

In this ensemble, the following quantities remain fixed:

- total energy E
- volume V
- number of particles N

Thus the macrostate of the system is specified as

$$(E, V, N)$$

Since the system is isolated, there is no exchange of energy or particles with the surroundings.

Fundamental Postulate of Statistical Mechanics

The fundamental assumption of statistical mechanics states that:

All accessible microstates of an isolated system with fixed energy are equally probable.

If the total number of accessible microstates is

$$\Omega(E, V, N)$$

then the probability of each microstate is

$$P_r = \frac{1}{\Omega}$$

Entropy in the Microcanonical Ensemble

Using Boltzmann's relation

$$S = k_B \ln \Omega$$

where

Ωrepresents the total number of accessible microstates.

Thus entropy measures the number of microscopic states available to the system.

Emergence of Temperature

In the microcanonical ensemble, temperature is defined through the relationship between entropy and energy:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}$$

This equation shows that temperature arises naturally from the statistical description of microscopic states (entropy-energy relation).

4. Canonical Ensemble

(Constant $\boldsymbol{T}, \boldsymbol{V}, \boldsymbol{N}$)

Definition

The **canonical ensemble** describes a system that is in thermal contact with a large heat reservoir.

Physical Situation

System in contact with a large heat reservoir:

- Energy exchange allowed
- Temperature fixed
- Particle number fixed

Energy fluctuates.

In this case

- temperature $T$is fixed
- volume $V$is fixed
- number of particles $N$is fixed

However, the system can exchange energy with the reservoir.

Therefore the energy of the system is allowed to fluctuate.

Boltzmann Distribution

The probability that the system occupies a microstate with energy E_r is given by the **Boltzmann distribution**:

$$P_r = \frac{e^{-\beta E_r}}{Z}$$

where

$$\beta = \frac{1}{k_B T}$$

and Z is the **canonical partition function**.

Canonical Partition Function

The canonical partition function is defined as

$$Z = \sum_r e^{-\beta E_r}$$

where the summation runs over all possible microstates of the system.

The partition function is one of the most important quantities in statistical thermodynamics because **all thermodynamic properties of the system can be derived from it**.

Thermodynamic Properties from the Partition Function

Once the partition function is known, many thermodynamic quantities can be calculated.

Internal Energy

$$U = \langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$$

Helmholtz Free Energy

$$F = -k_B \, \mathrm{T} \ln Z$$

Entropy

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

Heat Capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Energy fluctuations:

$$\langle (\Delta E)^2 \rangle = k_B T^2 C_V$$

Fluctuations become negligible for macroscopic systems.

Thus the partition function acts as a **central generating function for thermodynamic quantities**.

Energy Fluctuations in the Canonical Ensemble

In the canonical ensemble, the energy of the system is not fixed because the system exchanges energy with the reservoir.

The magnitude of energy fluctuations is given by

$$\langle (\Delta E)^2 \rangle = k_B T^2 C_V$$

For macroscopic systems containing a very large number of particles, these fluctuations are extremely small compared to the total energy.

Therefore the canonical ensemble gives results that are essentially identical to those of the microcanonical ensemble for large systems.

6. Grand Canonical Ensemble

(Constant T, V, μ)

Definition

The **grand canonical ensemble** represents a system that can exchange both energy and particles with a reservoir.

Physical Situation

System in contact with:

- Heat reservoir (fixed T)
- Particle reservoir (fixed chemical potential μ)

Both energy and particle number fluctuate.

The fixed variables are

- temperature T
- volume V
- chemical potential μ

The number of particles and the energy of the system are allowed to fluctuate.

Probability Distribution

The probability that the system is in a microstate with energy E_r and particle number N is

$$P_{r,N} = \frac{e^{-\beta(E_r - \mu N)}}{\Xi}$$

where

Ξ is the **grand partition function**.

Grand Partition Function

$$\Xi = \sum_{N=0}^{\infty} \sum_{r} e^{-\beta(E_r - \mu N)}$$

Grand Potential

The thermodynamic potential associated with the grand canonical ensemble is the **grand potential**:

$$\Omega = -k_B \, \mathrm{T} \ln \Xi$$

7. Isothermal–Isobaric Ensemble

(Constant T, P, N)

Definition

The **isothermal–isobaric ensemble** describes systems that are maintained at constant temperature and constant pressure.

Physical Situation

System in contact with:

- Heat reservoir (T fixed)
- Pressure reservoir (P fixed)

Volume fluctuates.

The fixed variables are

- temperature T
- pressure P
- number of particles N

The volume of the system is allowed to fluctuate.

Partition Function

The partition function for this ensemble is

$$\Delta = \sum_{r} e^{-\beta(E_r + PV_r)}$$

Thermodynamic Potential

The corresponding thermodynamic potential is the **Gibbs free energy**:

$$G = -k_B \, \mathrm{T} \ln \Delta$$

8. Importance of Ensembles

Ensembles provide a powerful method for connecting microscopic behaviour with macroscopic thermodynamic quantities.

They allow us to

- calculate average values of physical quantities
- derive thermodynamic relations
- analyze systems under different experimental conditions.

Although the different ensembles correspond to different physical situations, they give essentially the same thermodynamic results for large systems.

Summary of Part II

Ensemble	**Fixed Variables**	**Thermodynamic Potential**
Microcanonical	E, V, N	Entropy
Canonical	T, V, N	Helmholtz free energy
Grand Canonical	T, V, μ	Grand potential
Isothermal–Isobaric	T, P, N	Gibbs free energy

Ensemble theory provides the **mathematical framework that connects microscopic states with thermodynamic properties**.

Part III – Maxwell–Boltzmann Statistics

1. Classical Particle Distribution Problem

Consider a system containing **N distinguishable particles** distributed among a set of energy levels with no restriction on the number of particles in any energy level.

Let

$$\epsilon_i$$

= energy of the i^{th}energy level

$$g_i$$

= degeneracy of that level (number of states having energy ϵ_i)

$$N_i$$

= number of particles occupying that level.

The total number of particles is

$$N = \sum_i N_i$$

and the total energy of the system is

$$U = \sum_i N_i \, \epsilon_i$$

Thus two constraints exist:

Particle conservation

$$\sum_i N_i = N$$

Energy conservation

$$\sum_i N_i \, \epsilon_i = U$$

Our objective is to determine **how particles distribute themselves among the available energy levels at equilibrium**.

2. Thermodynamic Probability for Maxwell–Boltzmann Statistics

For **distinguishable particles**, the number of ways of distributing the particles among energy levels is

$$W = \frac{N!}{\prod_i N_i \, !}$$

If an energy level has degeneracy g_i, each particle in that level can occupy any of the $g_i$states.

Thus the number of arrangements becomes

$$W = \frac{N!}{\prod_i N_i\,!} \prod_i g_i^{N_i}$$

This is the **thermodynamic probability for Maxwell–Boltzmann statistics**.

3. Most Probable Distribution

The most probable distribution corresponds to the maximum probability of occurrence. With the increase in the number of particles, the probability of the most probable distribution increases towards the maximum value of unity.

For example, for N = 10, P = 0.503; N = 20, P = 0.547; N = 50, P = 0.786 and N = 100, P = 0.887). In a real system where very large number of particles (e.g. 10^{23}) is involved, the probability of the most probable distribution will be almost equal to one. The fact that the probability of the most probable distribution is nearly unity, the macroscopic properties of a system at equilibrium is largely governed by the most probable distribution.

At equilibrium, the system occupies the distribution that **maximizes the thermodynamic probability** W.

Since logarithms simplify factorial expressions, we maximize

$$\ln W$$

- The macroscopic properties of a system depend upon the various microscopic states of the system.

4. Taking the Logarithm

$$\ln W = \ln N! - \sum_i \ln \quad N_i! + \sum_i N_i \ln g_i$$

Using **Stirling approximation**

$$\ln n! \approx \mathrm{n}\ln n - n$$

we obtain

$$\ln N! \approx \mathrm{N}\ln N - N$$

and

$$\ln N_i! \approx N_i \ln N_i - N_i$$

Substituting

$$\ln\ W = (\mathrm{N}\ln\ N - N) - \sum_i (\,N_i \ln\ N_i - N_i) + \sum_i N_i \ln\ g_i$$

Expanding

$$\ln W = \mathrm{N}\ln N - N - \sum_i N_i \ln N_i + \sum_i N_i + \sum_i N_i \ln g_i$$

Since

$$\sum_i N_i = N$$

the terms $-N$and $+\sum_i N_i$cancel.

Thus

$$\ln W = \mathrm{N}\ln N - \sum_i N_i \ln N_i + \sum_i N_i \ln g_i$$

5. Maximization of $\boldsymbol{W}$

To find the most probable distribution we maximize $\ln W$.

Thus

$$d(\ln W) = 0$$

subject to the constraints

$$\sum_i d\,N_i = 0$$

$$\sum_i \epsilon_i\, dN_i = 0$$

6. Solving: Use of Lagrange Multipliers

We introduce multipliers $\alpha$and $\beta$to include the constraints.

The condition becomes

$$d\left(\ln W + \alpha \sum_i N_i - \beta \sum_i N_i\, \epsilon_i\right) = 0$$

Differentiating with respect to N_i

$$\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta\epsilon_i = 0$$

Evaluate the Derivative

From

$$\ln W = \mathrm{N}\ln N - \sum_i N_i \ln N_i + \sum_i N_i \ln g_i$$

the derivative with respect to $N_i$is

$$\frac{\partial(\ln W)}{\partial N_i} = -\ln N_i + \ln g_i$$

Thus

$$-\ln N_i + \ln g_i + \alpha - \beta\epsilon_i = 0$$

Rearranging

$$\ln g_i - \ln N_i + \alpha - \beta\epsilon_i = 0$$

$$\ln\left(\frac{g_i}{N_i}\right) + \alpha - \beta\epsilon_i = 0$$

$$\ln\left(\frac{g_i}{N_i}\right) = \beta\epsilon_i - \alpha$$

Exponentiating

$$\frac{g_i}{N_i} = e^{\beta\epsilon_i - \alpha}$$

Thus

$$N_i = g_i e^{\alpha - \beta\epsilon_i}$$

This is the **Maxwell–Boltzmann distribution law**.

7. Determination of the Constant $\boldsymbol{\alpha}$

The total number of particles is

$$N = \sum_i N_i$$

Substituting

$$N = \sum_i g_i\, e^{\alpha - \beta\epsilon_i}$$

$$N = e^{\alpha} \sum_i g_i\, e^{-\beta\epsilon_i}$$

Define

$$q = \sum_i g_i\, e^{-\beta\epsilon_i}$$

where $q$is called the **molecular partition function**.

Thus

$$e^{\alpha} = \frac{N}{q}$$

8. Final Form of the Maxwell–Boltzmann Distribution

Substituting

$$N_i = g_i \frac{N}{q} e^{-\beta \epsilon_i}$$

Thus

$$N_i = \frac{N}{q} g_i e^{-\beta \epsilon_i}$$

or

$$\frac{N_i}{N} = \frac{g_i e^{-\beta \epsilon_i}}{q}$$

This equation gives the **fraction of particles occupying energy level** i in any energy state is equal to the term $e^{-\beta \varepsilon_i}$ divided by the sum of all such terms for all the energy states.

Since the physical quantity q governs the partition of molecules in the energy states, it is called the **molecular partition function**. The importance of molecular partition function is that it contains all the information needed to calculate the thermodynamic properties of a system of independent particles

9. Identification of $\boldsymbol{\beta}$

From kinetic theory, the average kinetic energy per particle for an ideal gas is

$$\bar{E} = \frac{3}{2} kT$$

From statistical mechanics

$$\bar{E} = \frac{3}{2\beta}$$

Equating

$$\frac{3}{2\beta} = \frac{3}{2} kT$$

Thus

$$\beta = \frac{1}{kT}$$

Interpretation

- β is called the **inverse thermal energy parameter**.
- It appears in the **Boltzmann distribution**

$$P_i \propto e^{-\beta \varepsilon_i}$$

- Higher temperature → smaller β → more uniform population of energy levels.

10. Final Expression

Therefore the Maxwell–Boltzmann distribution becomes

$$N_i = \frac{N}{q} g_i e^{-\frac{\epsilon_i}{kT}}$$

Physical Interpretation

The Maxwell–Boltzmann distribution shows that

- lower energy levels are more populated
- higher energy levels have exponentially smaller populations
- temperature controls the spread of particles among energy levels.

As temperature increases, particles occupy higher energy states more frequently.

Validity of Maxwell–Boltzmann Statistics

Maxwell–Boltzmann statistics applies when

- particles are distinguishable
- particle density is low
- temperature is sufficiently high.

Under these conditions quantum effects are negligible.

Summary of Part III

Quantity	Expression
Thermodynamic probability	$W = \frac{N!}{\prod N_i!} \prod g_i^{N_i}$
Boltzmann distribution	$N_i = g_i e^{\alpha - \beta \epsilon_i}$

Quantity	Expression
Partition function	$q = \sum g_i e^{-\beta \epsilon_i}$
Final distribution	$N_i = \frac{N}{q} g_i e^{-\frac{\epsilon_i}{kT}}$

Part IV – Molecular Partition Function and Thermodynamic Properties

1. Molecular Partition Function

From Maxwell–Boltzmann statistics we obtained

$$N_i = g_i e^{\alpha - \beta \epsilon_i}$$

Using the particle conservation condition

$$N = \sum_i N_i$$

we get

$$N = \sum_i g_i \, e^{\alpha - \beta \epsilon_i}$$

Rearranging

$$N = e^{\alpha} \sum_i g_i \, e^{-\beta \epsilon_i}$$

Define

$$q = \sum_i g_i \, e^{-\beta \epsilon_i}$$

This quantity is called the **molecular partition function**.

Thus

$$e^{\alpha} = \frac{N}{q}$$

Substituting into the distribution equation

$$N_i = \frac{N}{q} g_i e^{-\beta \epsilon_i}$$

Physical Meaning

The partition function represents the **sum of Boltzmann factors for all energy states**.

Each term

$$e^{-\frac{\epsilon_i}{kT}}$$

gives the **relative probability of occupying that energy level**.

Since

$$\beta = \frac{1}{kT}$$

Molecular Partition Function

$$q = \sum_i g_i \, e^{-\frac{\varepsilon_i}{kT}}$$

Behaviour at Low Temperature

At temperature $\boldsymbol{T}$close to zero, the value of each of the term $\mathbf{exp}\left(-\frac{\boldsymbol{\varepsilon_i}}{\boldsymbol{kT}}\right)$will be close to zero. If $\boldsymbol{\varepsilon_0} = \mathbf{0}$, then leaving the first term, for which $\mathbf{exp}\left(-\frac{\boldsymbol{\varepsilon_0}}{\boldsymbol{kT}}\right)$will have a value of one, the partition function will have a value close to $\boldsymbol{g_0}$

If $\varepsilon_0 = 0$:

$$\lim_{T \to 0} \quad q = g_0$$

Behaviour at High Temperature

At very high temperature, the value of each of the term $\exp\left(-\frac{\varepsilon_i}{kT}\right)$will be close to one, and the value of partition function will be close to the number of available energy states.

Thus, the molecular partition function gives an idea about the average number of states that are thermally accessible to a particle at the temperature of the system. At $T = 0$ K, only the lowest energy state will be accessible while at very high temperature, a large number of higher energy states are accessible.

2. Canonical Partition Function (Molar Partition Function)

For a system containing Nmolecules, the partition function of the entire system is

$$Q = \sum e^{-\beta E_r}$$

where $E_r$is the energy of the r^{th}microstate.

For **independent particles**

$$E_r = \epsilon_1 + \epsilon_2 + \cdots + \epsilon_N$$

Thus

$$Q = (q)^N$$

However, since molecules are indistinguishable, identical configurations are counted multiple times.

To correct this, we divide by $N!$

$$Q = \frac{q^N}{N!}$$

This is the **canonical partition function for a system of indistinguishable particles**.

For **one mole**:

$$N = N_A$$

So

$$Q = \frac{q^{N_A}}{N_A!}$$

This expression is called the **molar partition function** (or macroscopic partition function).

---

3. Derivation of Thermodynamic Quantities

The partition function contains complete thermodynamic information.

We now derive expressions for important thermodynamic quantities.

3.1 Internal Energy

The expression of internal energy is

$$U = \sum_i N_i \, \varepsilon_i$$

From the Boltzmann expression,

$$N_i = \frac{N}{q} g_i e^{-\beta \varepsilon_i}$$

Where

$q = \sum_i g_i e^{-\beta \varepsilon_i}$, $\beta = \frac{1}{k_B T}$

Hence,

$$U = \sum_i \left(\frac{N}{q} g_i e^{-\beta \varepsilon_i} \right) \varepsilon_i$$

Since

$$\left(\frac{\partial e^{-\beta\varepsilon_i}}{\partial\beta}\right)_V = -\varepsilon_i e^{-\beta\varepsilon_i}$$

we write

$$U = -\frac{N}{q}\sum_i g_i\left(\frac{\partial e^{-\beta\varepsilon_i}}{\partial\beta}\right)_V$$

$$= -\frac{N}{q}\frac{\partial}{\partial\beta}\left(\sum_i g_i\, e^{-\beta\varepsilon_i}\right)_V$$

$$= -\frac{N}{q}\left(\frac{\partial q}{\partial\beta}\right)_V$$

$$= -N\left(\frac{\partial \ln q}{\partial\beta}\right)_V$$

In terms of temperature:

$$U = -N\left(\frac{\partial \ln q}{\partial T}\right)_V\left(\frac{dT}{d\beta}\right)$$

Since $T = \frac{1}{k\beta}$,

$$\frac{dT}{d\beta} = -\frac{1}{k\beta^2} = -kT^2$$

Thus,

$$U = NkT^2\left(\frac{\partial \mathrm{l}\,\mathrm{n}\, q}{\partial T}\right)_V$$

First Law Identification

$$dU = dq' + dw$$

Where q' and w stand for heat and work, respectively.

Since $U = \sum_i N_i\, \varepsilon_i$,

$$dU = \sum_i \varepsilon_i\, dN_i + \sum_i N_i\, d\varepsilon_i$$

First term express change in internal energy due to the change in population of particles in the energy levels. This fact may be attributed to the term dq' of Equation.

The second term of Equation expresses the change in internal energy due to the change in the magnitude of energy levels. This fact may be attributed to the term dw of Equation as the magnitude of energy level is affected by the change in the boundary parameters (say, volume) of the system

$$dq' = \sum_i \varepsilon_i \, dN_i$$

$$dw = \sum_i N_i \, d\varepsilon_i$$

3.2 Pressure

Pressure can be obtained from

$$P = -\left(\frac{\partial F}{\partial V}\right)_T$$

Using

$$F = -\mathrm{kTln}\, Q$$

Thus

$$P = kT\left(\frac{\partial \ln Q}{\partial V}\right)_T$$

Alternate derivation

Mechanical work:

$$dw = -p \, dV$$

Thus,

$$-p \, dV = \sum_i N_i \, d\varepsilon_i$$

Using Boltzmann distribution:

$$-p \, dV = \frac{N}{q}\sum_i g_i \, e^{-\beta\varepsilon_i} d\varepsilon_i$$

At constant temperature, the differential change in the molecular partition function q: $q = \sum_i g_i \, e^{-\beta\varepsilon_i}$

$$dq = \sum_i g_i \, e^{-\beta\varepsilon_i}(-\beta) \, d\varepsilon_i$$

Thus,

$$p \, dV = \frac{N}{q}\frac{1}{\beta} dq = \frac{N}{\beta} \mathrm{dln}\, q$$

Hence,

$$p = \frac{N}{\beta}\left(\frac{\partial \ln q}{\partial V}\right)_T$$

Since $\beta = \frac{1}{kT}$,

$$p = NkT\left(\frac{\partial \ln q}{\partial V}\right)_T$$

3.1 Helmholtz Free Energy

In statistical mechanics

$$F = -\mathrm{kT}\ln Q$$

Substitute

$$Q = \frac{q^N}{N!}$$

Thus

$$F = -\mathrm{kT}\ln\left(\frac{q^N}{N!}\right)$$

Using logarithm rules

$$F = -kT(\mathrm{N}\ln q - \ln N!)$$

Using Stirling approximation

$$\ln N! \approx \mathrm{N}\ln N - N$$

we obtain

$$F = -\mathrm{NkT}\ln q + \mathrm{NkT}\ln N - NkT$$

3.2 Internal Energy

Internal energy is defined as the average energy of the system.

From statistical mechanics

$$U = \sum_r P_r E_r$$

Using canonical ensemble probability

$$P_r = \frac{e^{-\beta E_r}}{Q}$$

Thus

$$U = \sum_r \frac{E_r e^{-\beta E_r}}{Q}$$

This can be written as

$$U = \frac{1}{Q}\sum_r E_r\, e^{-\beta E_r}$$

Now consider

$$\frac{\partial Q}{\partial \beta}$$

$$Q = \sum_r e^{-\beta E_r}$$

Differentiating

$$\frac{\partial Q}{\partial \beta} = \sum_r (-E_r) e^{-\beta E_r}$$

Thus

$$-\frac{\partial Q}{\partial \beta} = \sum_r E_r\, e^{-\beta E_r}$$

Substituting

$$U = \frac{1}{Q}\left(-\frac{\partial Q}{\partial \beta}\right)$$

Thus

$$U = -\frac{\partial \ln Q}{\partial \beta}$$

Since

$$\beta = \frac{1}{kT}$$

another useful form is

$$U = kT^2\left(\frac{\partial \ln Q}{\partial T}\right)_V$$

3.3 Entropy

From thermodynamics

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

Using

$$F = -\mathrm{kT}\ln Q$$

Differentiating

$$S = -\left[-\text{kln}\, Q - kT\frac{\partial \ln Q}{\partial T}\right]$$

Thus

$$S = \text{kln}\, Q + kT\frac{\partial \ln Q}{\partial T}$$

Alternate definition

$$dS = \frac{dq'_{rev}}{T}$$

Using the expression for dq', $dq' = \sum_i \varepsilon_i\, dN_i$

$$dS = \frac{1}{T}\sum_i \varepsilon_i\, dN_i$$

Since $\beta = 1/kT$,

$$dS = \beta k \sum_i \varepsilon_i\, dN_i = k\sum_i (\beta\, \varepsilon_i) dN_i$$

From Boltzmann condition:

$$\frac{\partial \ln\, W}{\partial N_i} + \alpha - \beta\varepsilon_i = 0$$

Thus,

$$\beta\varepsilon_i = \frac{\partial \ln\, W}{\partial N_i} + \alpha$$

Substitute:

$$dS = k\sum_i \left(\frac{\partial \ln\, W}{\partial N_i}\right) dN_i + k\alpha \sum_i d\, N_i$$

Since $\sum_i d\, N_i = 0$,

$$dS = k\, d(\ln\, W)$$

$$S = k\ln\, W$$

(Boltzmann–Planck equation)

Entropy of Distinguishable Particles

$$W = N!\prod_i \frac{g_i^{N_i}}{N_i!}$$

$$\ln\, W = \ln\, N! + \sum_i N_i \ln\, g_i - \sum_i \ln\, N_i!$$

Using Stirling approximation:

$$\ln\ W = -\sum_i N_i \ln\left(\frac{N_i}{Ng_i}\right)$$

From the Boltzmann expression,

$$\frac{N_i}{Ng_i} = \frac{e^{-\beta\varepsilon_i}}{q}$$

$$\ln\left(\frac{N_i}{Ng_i}\right) = -\beta\varepsilon_i - \ln\ q$$

Thus,

$$\ln\ W = \beta U + N\ln\ q$$

$$S = k\ln\ W = k\beta U + kN\ln\ q$$

$$= \frac{U}{T} + kN\ln\ q$$

Substitute $U = NkT^2\left(\frac{\partial \ln\ q}{\partial T}\right)_V$

$$S = NkT\left(\frac{\partial \ln\ q}{\partial T}\right)_V + Nk\ln\ q$$

$$S = Nk\left[T\left(\frac{\partial \ln\ q}{\partial T}\right)_V + \ln\ q\right]$$

Entropy of Indistinguishable Particles

$$W = \prod_i \frac{g_i^{N_i}}{N_i!}$$

$$\ln\ W = \sum_i N_i \ln\ g_i - \sum_i \ln\ N_i!$$

$$\ln W = \sum_i N_i \ln g_i - N_i \ln N_i + N_i$$

$$\ln W = \sum_i N_i \ln\left(\frac{g_i}{N_i}\right) + N$$

From Boltzmann expression, $\frac{N_i}{g_i} = \frac{q}{N} e^{\beta\varepsilon_i}$

Or

$$\ln\left(\frac{N_i}{g_i}\right) = \ln\left(\frac{q}{N}\right) + \beta\varepsilon_i$$

Substituting:

$$\ln W = N\ln\left(\frac{q}{N}\right) + \beta U + N$$

Thus,

$$S = k\ln W$$

$$S = Nk\left[\ln\left(\frac{q}{N}\right) + T\left(\frac{\partial \ln q}{\partial T}\right)_V + 1\right]$$

Expression of Helmholtz Free Energy

$$A = U - TS$$

For distinguishable particles:

$$A = -NkT\ln q$$

For indistinguishable particles:

$$A' = -NkT\left(\ln\frac{q}{N} + 1\right)$$

Expression of Gibbs Free Energy

$$G = H - TS$$

$$G = A + pV = -NkT\ln q + pV$$

For indistinguishable particles:

$$G' = -NkT\left(\ln\frac{q}{N} + 1\right) + pV$$

Since $pV = NkT$,

$$G' = -NkT\ln\left(\frac{q}{N}\right)$$

Expression of Enthalpy

$$H = U + pV$$

$$H = NkT^2\left(\frac{\partial \ln q}{\partial T}\right)_V + pV$$

3.5 Heat Capacity

Heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Substituting

$$U = kT^2\frac{\partial \ln Q}{\partial T}$$

$$C_V = Nk\left[\frac{\partial}{\partial T}\left(T^2\frac{\partial \ln\ q}{\partial T}\right)\right]_V$$

Thus heat capacity can be calculated directly from the partition function.

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

$$C_P = \left[\frac{\partial}{\partial T}\left(NkT^2\frac{\partial \ln\ q}{\partial T} + pV\right)\right]_P$$

Factorization of the Molecular Partition Function

1. Molecular Partition Function

For a single molecule, the molecular partition function is

$$q = \sum_i g_i\, e^{-\beta\varepsilon_i}$$

where

- g_i= degeneracy of the i^{th}quantum state
- ε_i= energy of that state
- $\beta = \frac{1}{kT}$
- k= Boltzmann constant
- T= absolute temperature

[Comment:]

The partition function is a weighted sum over all quantum states. The weight $e^{-\beta\varepsilon_i}$is the **Boltzmann factor**, which determines the probability of occupying that state.

For a system of Nnon-interacting molecules,

$$E = NkT^2\left(\frac{\partial \ln\ q}{\partial T}\right)_V$$

[Comment:]

This relation shows why we seek q: once $q$is known, thermodynamic properties follow.

---

2. Molecular Hamiltonian

The Hamiltonian of a molecule is

$$H = T_e + T_n + V_{ee} + V_{nn} + V_{en}$$

where

- T_e= kinetic energy of electrons
- T_n= kinetic energy of nuclei
- V_{ee}= electron–electron repulsion
- V_{nn}= nucleus–nucleus repulsion
- V_{en}= electron–nucleus attraction

[Comment:]

This Hamiltonian couples all motions of the system, so exact separation of translational, rotational, vibrational, and electronic energies is impossible without approximation.

---

3. Born–Oppenheimer Approximation

The **Born–Oppenheimer approximation** uses the fact that

$$m_{nucleus} \gg m_{electron}$$

Thus electrons move much faster than nuclei.

Therefore the molecular wavefunction can be written approximately as

$$\Psi = \Psi_{ele}\Psi_{nuc}$$

and the energy becomes

$$\varepsilon = \varepsilon_{ele} + \varepsilon_{nuc}$$

[Comment:]

This approximation allows us to treat **electronic motion separately from nuclear motion**.

---

4. Separation of Nuclear Motion

The nuclear motion consists of

- translation of the molecule
- rotation of the molecule
- vibration of the nuclei

Thus the molecular energy can be written as

$$\varepsilon_i = \varepsilon_{tr,n} + \varepsilon_{rot,J} + \varepsilon_{vib,v} + \varepsilon_{ele,u}$$

where

- n= translational quantum numbers
- J= rotational quantum number

- v= vibrational quantum number
- u= electronic quantum number

---

5. Factorization of the Partition Function

Substituting the energy expression into the partition function

$$q = \sum_i g_i \, e^{-\beta(\varepsilon_{tr,n}+\varepsilon_{rot,J}+\varepsilon_{vib,v}+\varepsilon_{ele,u})}$$

Using the property of exponentials

$$e^{-(A+B+C+D)} = e^{-A}e^{-B}e^{-C}e^{-D}$$

$$q = \sum_i g_i \, e^{-\beta\varepsilon_{tr,n}}e^{-\beta\varepsilon_{rot,J}}e^{-\beta\varepsilon_{vib,v}}e^{-\beta\varepsilon_{ele,u}}$$

If the motions are independent

$$g_i = g_n g_J g_v g_u$$

Thus

$$q = \sum_n g_n \, e^{-\beta\varepsilon_{tr,n}} \sum_J g_J \, e^{-\beta\varepsilon_{rot,J}} \sum_v g_v \, e^{-\beta\varepsilon_{vib,v}} \sum_u g_u \, e^{-\beta\varepsilon_{ele,u}}$$

Define

$$q_{tr} = \sum_n g_n \, e^{-\beta\varepsilon_{tr,n}}$$

$$q_{rot} = \sum_J g_J \, e^{-\beta\varepsilon_{rot,J}}$$

$$q_{vib} = \sum_v g_v \, e^{-\beta\varepsilon_{vib,v}}$$

$$q_{ele} = \sum_u g_u \, e^{-\beta\varepsilon_{ele,u}}$$

Therefore

$$q = q_{tr}q_{rot}q_{vib}q_{ele}$$

Taking natural logarithm

$$\ln \, q = \ln \, q_{tr} + \ln \, q_{rot} + \ln \, q_{vib} + \ln \, q_{ele}$$

Translational Partition Function

Energy Levels

For a particle in a 3D box

$$\varepsilon_{tr} = \frac{h^2}{8m}\left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}\right)$$

Partition Function

$$q_{tr} = \sum_{n_x}\sum_{n_y}\sum_{n_z} e^{-\beta\varepsilon_{tr}}$$

Substitute energy

$$q_{tr} = \sum e^{-\frac{\beta h^2}{8m}\left(\frac{n_x^2}{a^2}+\frac{n_y^2}{b^2}+\frac{n_z^2}{c^2}\right)}$$

Using

$$e^{-(A+B+C)} = e^{-A}e^{-B}e^{-C}$$

$$q_{tr} = \sum_{n_x} e^{-\frac{\beta h^2 n_x^2}{8ma^2}} \sum_{n_y} e^{-\frac{\beta h^2 n_y^2}{8mb^2}} \sum_{n_z} e^{-\frac{\beta h^2 n_z^2}{8mc^2}}$$

Convert Summation to Integral

For large quantum numbers

$$\sum_{n=1}^{\infty} f(n) \approx \int_0^{\infty} f(n)dn$$

Thus

$$q_{tr} = \int_0^{\infty} e^{-\frac{\beta h^2 n_x^2}{8ma^2}}\, dn_x \int_0^{\infty} e^{-\frac{\beta h^2 n_y^2}{8mb^2}}\, dn_y \int_0^{\infty} e^{-\frac{\beta h^2 n_z^2}{8mc^2}}\, dn_z$$

Evaluate Gaussian Integral

$$\int_0^{\infty} e^{-\alpha x^2}\, dx = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}$$

For the first integral

$$\alpha = \frac{\beta h^2}{8ma^2}$$

$$\int_0^{\infty} e^{-\frac{\beta h^2 n_x^2}{8ma^2}}\, dn_x = \frac{1}{2}\sqrt{\frac{8m\pi a^2}{\beta h^2}}$$

Similarly

$$\int_0^\infty e^{-\frac{\beta h^2 n_y^2}{8mb^2}}\, dn_y = \frac{1}{2}\sqrt{\frac{8m\pi b^2}{\beta h^2}}$$

$$\int_0^\infty e^{-\frac{\beta h^2 n_z^2}{8mc^2}}\, dn_z = \frac{1}{2}\sqrt{\frac{8m\pi c^2}{\beta h^2}}$$

Multiply Integrals

$$q_{tr} = \left(\frac{1}{2}\sqrt{\frac{8m\pi a^2}{\beta h^2}}\right)\left(\frac{1}{2}\sqrt{\frac{8m\pi b^2}{\beta h^2}}\right)\left(\frac{1}{2}\sqrt{\frac{8m\pi c^2}{\beta h^2}}\right)$$

Multiply constants

$$\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{8}$$

Combine square roots

$$q_{tr} = \frac{1}{8}\sqrt{\frac{(8m\pi)^3 a^2 b^2 c^2}{\beta^3 h^6}}$$

$$q_{tr} = \frac{abc}{h^3}\frac{(8m\pi)^{3/2}}{8}\frac{1}{\beta^{3/2}}$$

Since

$$\beta = \frac{1}{kT}$$

$$\frac{1}{\beta^{3/2}} = (kT)^{3/2}$$

Thus

$$q_{tr} = abc\frac{(2\pi mkT)^{3/2}}{h^3}$$

Since $V = abc$

$$q_{tr} = V\left(\frac{2\pi mkT}{h^2}\right)^{3/2}$$

Rotational Partition Function

Rigid Rotor Model

For a **linear molecule**, rotational motion can be modeled using the **rigid rotor approximation**, where the internuclear distance is fixed and the molecule rotates about axes perpendicular to the bond.

The rotational energy levels obtained from solving the Schrödinger equation for a rigid rotor are

$$\varepsilon_{rot,J} = \frac{\hbar^2}{2I} J(J+1)$$

where

- $J = 0,1,2,3, \ldots$is the rotational quantum number
- I= moment of inertia of the molecule
- $\hbar = \frac{h}{2\pi}$

---

2. Degeneracy of Rotational Levels

Each rotational energy level has degeneracy

$$g_J = 2J + 1$$

This arises because the magnetic quantum number $m_J$takes values

$$m_J = -J, -J+1, \ldots, +J$$

giving $2J + 1$possible orientations for each rotational level.

---

3. Rotational Partition Function

The rotational partition function is defined as

$$q_{rot} = \sum_{J=0}^{\infty} g_J \, e^{-\beta \varepsilon_{rot,J}}$$

Substitute $g_J$and $\varepsilon_{rot,J}$

$$q_{rot} = \sum_{J=0}^{\infty} (2J+1) \, e^{-\beta \frac{\hbar^2}{2I} J(J+1)}$$

Now define a characteristic temperature the **characteristic rotational temperature**

$$\theta_{rot} = \frac{\hbar^2}{2Ik}$$

Since

$$\beta = \frac{1}{kT}$$

we obtain

$$\beta \frac{\hbar^2}{2I} = \frac{\theta_{rot}}{T}$$

hus the partition function becomes

$$q_{rot} = \sum_{J=0}^{\infty} (2J+1)\, e^{-\frac{\theta_{rot}}{T} J(J+1)}$$

The quantity

$$\theta_{rot} = \frac{\hbar^2}{2Ik}$$

has **dimensions of temperature** because

$$\frac{\hbar^2}{2I}$$

is an energy and dividing by $k$converts energy to temperature.

Thus

$$\theta_{rot} = \frac{\text{rotational energy spacing}}{k}$$

Physical Meaning

$$\theta_{rot} \approx \frac{\Delta E_{rot}}{k}$$

where $\Delta E_{rot}$is the spacing between rotational energy levels.

Thus it represents the **temperature equivalent of rotational level spacing**.

Interpretation

Case 1: $T \gg \theta_{rot}$

Thermal energy is much larger than rotational level spacing.

Many rotational states are populated.

$$q_{rot} \approx \frac{T}{\sigma \theta_{rot}}$$

Classical behaviour emerges.

Case 2: $T \ll \theta_{rot}$

Thermal energy is insufficient to excite rotational levels.

Almost all molecules remain in the ground rotational state.

$$q_{rot} \approx 1$$

Rotation is **frozen out**.

Typical Values

Molecule	θ_{rot}
H_2	~85 K
N_2	~2.9 K
CO	~2.8 K
HCl	~15 K

Thus at **room temperature (300 K)**

$$T \gg \theta_{rot}$$

so rotational states are fully populated.

5. High Temperature Approximation

For most molecules at ordinary temperatures

$$T \gg \theta_{rot}$$

Thus the spacing between rotational energy levels is very small compared with kT, and many rotational states are populated.

Therefore the summation over J may be approximated by an integral:

$$q_{rot} \approx \int_0^\infty (2J+1)\, e^{-\frac{\theta_{rot}}{T}J(J+1)} dJ$$

---

6. Transformation of Variables

Let

$$w = J(J+1)$$

Differentiate with respect to J

$$\frac{dw}{dJ} = 2J + 1$$

Thus

$$dw = (2J + 1)dJ$$

or

$$(2J + 1)dJ = dw$$

Substitute this into the integral:

$$q_{rot} = \int_0^\infty e^{-\frac{\theta_{rot}}{T}w}\, dw$$

---

7. Evaluation of the Integral

Using the standard exponential integral

$$\int_0^\infty e^{-ax}\, dx = \frac{1}{a}$$

where

$$a = \frac{\theta_{rot}}{T}$$

Thus

$$q_{rot} = \frac{1}{\theta_{rot}/T}$$

$$q_{rot} = \frac{T}{\theta_{rot}}$$

---

8. Symmetry Correction

If the molecule has indistinguishable orientations upon rotation, the partition function must be divided by the **symmetry number** σ.

Thus

$$q_{rot} = \frac{T}{\sigma\theta_{rot}}$$

Examples:

Molecule	**Symmetry number**
HCl	1
CO	1

Molecule	Symmetry number
H_2	2
O_2	2

---

9. More Accurate High-Temperature Expansion

A more accurate expression can be obtained using a series expansion

$$q_{rot} = \frac{T}{\sigma\theta_{rot}}\left[1 + \frac{1}{3}\frac{\theta_{rot}}{T} + \frac{1}{15}\left(\frac{\theta_{rot}}{T}\right)^2 + \cdots\right]$$

---

10. Low Temperature Case

When

$$T \lesssim \theta_{rot}$$

the integral approximation is no longer valid. In this case the partition function must be evaluated using the original summation:

$$q_{rot} = \sum_{J=0}^{\infty}(2J+1)\, e^{-\frac{\theta_{rot}}{T}J(J+1)}$$

Final Result

For a linear molecule at ordinary temperatures

$$q_{rot} = \frac{T}{\sigma\theta_{rot}}$$

where

$$\theta_{rot} = \frac{\hbar^2}{2Ik}$$

Physical Significance

The rotational partition function represents the **number of thermally accessible rotational states of a molecule**.

It determines

- rotational contribution to internal energy

- rotational entropy
- rotational heat capacity

For a linear molecule at high temperature

$$E_{rot} = NkT$$

which corresponds to **two rotational degrees of freedom**, consistent with the equipartition theorem.

Vibrational Partition Function

1. Vibrational Motion of a Molecule

Atoms in a molecule execute vibrational motion about their equilibrium positions. Near the equilibrium bond length, the potential energy curve can be approximated by a quadratic function -the **quantum harmonic oscillator model**.

Near the equilibrium bond length, the potential energy curve of a diatomic molecule can be expanded as a Taylor series. Keeping only the quadratic term,

$$V(x) \approx \frac{1}{2} k_f x^2$$

where

- x= displacement from equilibrium
- k_f= force constant of the bond.

A quadratic potential corresponds exactly to the **harmonic oscillator**.

---

2. Quantum Mechanical Energy Levels

Solving the Schrödinger equation for a harmonic oscillator gives the energy levels

$$E_v = \left(v + \frac{1}{2}\right) h\nu$$

where

- $v = 0,1,2,3, \ldots$is the vibrational quantum number
- ν= vibrational frequency

---

3. Degeneracy of Vibrational Levels

Each vibrational level has degeneracy

$$g_v = 1$$

Explanation

For a harmonic oscillator there is only **one quantum state corresponding to each** v. Therefore no degeneracy factor appears.

---

4. Partition Function Definition

The vibrational partition function is

$$q_{vib} = \sum_{v=0}^{\infty} g_v \, e^{-\beta E_v}$$

Since $g_v = 1$,

$$q_{vib} = \sum_{v=0}^{\infty} e^{-\beta E_v}$$

Substitute the energy levels

$$q_{vib} = \sum_{v=0}^{\infty} e^{-\beta(v+\frac{1}{2})h\nu}$$

---

5. Separate the Constant Term -Zero-Point Energy Term

Rewrite the exponent

$$(v + \frac{1}{2})h\nu = vh\nu + \frac{1}{2}h\nu$$

Thus

$$q_{vib} = \sum_{v=0}^{\infty} e^{-\beta(vh\nu+\frac{1}{2}h\nu)}$$

Using

$$e^{-(A+B)} = e^{-A}e^{-B}$$

$$q_{vib} = \sum_{v=0}^{\infty} e^{-\beta vh\nu} \, e^{-\beta h\nu/2}$$

The second term does not depend on v, therefore it can be factored out of the summation:

$$q_{vib} = e^{-\beta h\nu/2} \sum_{v=0}^{\infty} e^{-\beta vh\nu}$$

---

6. Introduce Vibrational Temperature

Define the **characteristic vibrational temperature**

$$\theta_{vib} = \frac{h\nu}{k}$$

Since

$$\beta = \frac{1}{kT}$$

we obtain

$$\beta h\nu = \frac{h\nu}{kT} = \frac{\theta_{vib}}{T}$$

Thus

$$q_{vib} = e^{-\theta_{vib}/2T} \sum_{v=0}^{\infty} e^{-\frac{\theta_{vib}}{T}v}$$

---

7. Evaluation of the Infinite Series

The summation

$$S = \sum_{v=0}^{\infty} e^{-\frac{\theta_{vib}}{T}v}$$

is an infinite series.

Let

$$x = e^{-\theta_{vib}/T}$$

Thus

$$S = 1 + x + x^2 + x^3 + x^4 + \cdots$$

This is a **geometric series**.

Derivation of the Geometric Series Result

Let

$$S = 1 + x + x^2 + x^3 + \cdots$$

Multiply both sides by x

$$xS = x + x^2 + x^3 + x^4 + \cdots$$

Subtract the second equation from the first:

$$S - xS = 1$$

Factor S

$$S(1 - x) = 1$$

Thus

$$S = \frac{1}{1 - x}$$

Substitute $x = e^{-\theta_{vib}/T}$

$$S = \frac{1}{1 - e^{-\theta_{vib}/T}}$$

---

8. Vibrational Partition Function

Substituting back into the earlier expression

$$q_{vib} = e^{-\theta_{vib}/2T} \frac{1}{1 - e^{-\theta_{vib}/T}}$$

Thus

$$q_{vib} = \frac{e^{-\theta_{vib}/2T}}{1 - e^{-\theta_{vib}/T}}$$

---

9. Choice of Energy Zero

In statistical thermodynamics it is convenient to measure energies relative to the **ground state energy**.

The ground vibrational energy is

$$E_0 = \frac{1}{2} h\nu$$

If this constant energy is subtracted from all levels

$$E_v' = E_v - \frac{1}{2} h\nu$$

then

$$E_v' = vh\nu$$

Using this shifted energy scale the partition function becomes

$$q_{vib} = \sum_{v=0}^{\infty} e^{-\beta v h \nu}$$

Repeating the geometric series derivation gives

$$q_{vib} = \frac{1}{1 - e^{-\theta_{vib}/T}}$$

---

10. Final Expression

Thus the vibrational partition function of a single vibrational mode is

$$q_{vib} = \frac{1}{1 - e^{-\theta_{vib}/T}}$$

where

$$\theta_{vib} = \frac{h\nu}{k}$$

---

11. Extension to Polyatomic Molecules

A molecule with N atoms has

Molecule type	**Number of vibrational modes**
Linear	$3N - 5$
Non-linear	$3N - 6$

If the vibrational modes are independent, the total vibrational partition function is

$$q_{vib} = \prod_{i=1}^{f} \frac{1}{1 - e^{-\theta_{vib,i}/T}}$$

where f is the number of vibrational modes.

---

12. Physical Interpretation

The vibrational partition function describes the **population distribution among vibrational energy levels**.

Case 1: $T \ll \theta_{vib}$

$$e^{-\theta_{vib}/T} \ll 1$$

Thus

$$q_{vib} \approx 1$$

Almost all molecules remain in the ground vibrational state.

Case 2: $T \gg \theta_{vib}$

Higher vibrational levels become populated.

Vibrational energy contributes significantly to

- internal energy
- heat capacity.

Electronic Partition Function

1. Electronic Energy of Molecules

In a molecule, electrons occupy discrete quantum states determined by the **electronic Schrödinger equation**. Each electronic state has an energy

$$E_{ele,i}$$

where $i$denotes the electronic level.

For convenience the energy of the **ground electronic state** is usually taken as zero:

$$E_0 = 0$$

Thus excited electronic levels have energies

$$E_1, E_2, E_3, \ \ldots$$

measured relative to the ground state.

---

2. Degeneracy of Electronic States

Electronic levels may possess degeneracy.

Let

$$g_i$$

be the degeneracy of the i^{th}electronic level.

This degeneracy arises from

- spin multiplicity
- orbital degeneracy.

Thus a given electronic energy level may correspond to several quantum states.

---

3. Definition of Electronic Partition Function

The general definition of a molecular partition function is

$$q = \sum_i g_i \, e^{-\beta E_i}$$

Applying this to electronic states gives

$$q_{ele} = \sum_i g_i \, e^{-\beta E_{ele,i}}$$

where

$$\beta = \frac{1}{kT}$$

---

4. Expand the Summation

Writing the first few terms explicitly

$$q_{ele} = g_0 e^{-\beta E_0} + g_1 e^{-\beta E_1} + g_2 e^{-\beta E_2} + g_3 e^{-\beta E_3} + \cdots$$

---

5. Choose the Zero of Energy

Since we take the ground electronic energy as zero

$$E_0 = 0$$

then

$$e^{-\beta E_0} = e^0 = 1$$

Thus the first term becomes

$$g_0$$

Therefore

$$q_{ele} = g_0 + g_1 e^{-\beta E_1} + g_2 e^{-\beta E_2} + g_3 e^{-\beta E_3} + \cdots$$

---

6. Express Energies Relative to Ground State

Define excitation energies

$$\epsilon_i = E_i - E_0$$

Since $E_0 = 0$,

$$\epsilon_i = E_i$$

Thus

$$q_{ele} = g_0 + g_1 e^{-\beta\epsilon_1} + g_2 e^{-\beta\epsilon_2} + \cdots$$

---

7. Behaviour at Ordinary Temperatures

Electronic excitation energies are typically very large.

Typical values:

Molecule	First electronic excitation
N_2	~7 eV
CO	~6 eV
O_2	~1 eV

Thermal energy at room temperature:

$$kT \approx 0.026\ eV$$

Thus

$$\epsilon_i \gg kT$$

---

8. Effect on the Boltzmann Factor

Consider the exponential term

$$e^{-\epsilon_i/kT}$$

If

$$\epsilon_i \gg kT$$

then

$$e^{-\epsilon_i/kT} \approx 0$$

Thus all excited electronic contributions become negligible.

---

9. Approximate Electronic Partition Function

Therefore

$$q_{ele} \approx g_0$$

---

10. Special Case for Most Molecules

For many molecules the ground electronic state is non-degenerate.

Thus

$$g_0 = 1$$

and

$$q_{ele} \approx 1$$

---

11. Situations Where $q_{ele} \neq 1$

The approximation fails when

- the ground electronic state is degenerate
- electronic excitation energies are small
- the temperature is extremely high.

Examples include

- atoms
- radicals
- high-temperature plasmas.

---

12. Final Expression

Thus the electronic partition function is

$$q_{ele} = g_0 + g_1 e^{-\epsilon_1/kT} + g_2 e^{-\epsilon_2/kT} + \cdots$$

and at ordinary temperatures

$$q_{ele} \approx g_0$$

---

13. Physical Significance

The electronic partition function describes the **population of electronic energy levels**.

At ordinary temperatures

- almost all molecules remain in the **ground electronic state**
- excited electronic states contribute negligibly to thermodynamic properties.

However, electronic partition functions become important in

- high-temperature gases

- combustion
- astrophysical plasmas
- spectroscopy.

Significance of Each Partition Function

Translational Partition Function

Represents the number of accessible **translational states of the molecule in space**.

It determines

- gas pressure
- translational contribution to internal energy
- entropy of gases

Rotational Partition Function

Represents the number of **rotational orientations available to the molecule**.

It contributes to

- rotational internal energy
- rotational heat capacity
- rotational entropy

Vibrational Partition Function

Describes the population of **vibrational energy levels of chemical bonds**.

It becomes significant at higher temperatures and contributes to

- vibrational internal energy
- vibrational heat capacity

Electronic Partition Function

Represents the population of **electronic energy states**.

At ordinary temperatures

- excited electronic states are rarely populated
- Electronic contribution to thermodynamic properties is usually small.

Partition Function	Physical Meaning
q_{tr}	Number of accessible translational states in the volume
q_{rot}	Number of accessible rotational orientations
q_{vib}	Population of vibrational energy levels
q_{ele}	Population of electronic energy states

Conditions for Valid Factorization

Factorization is valid when

1. **Born–Oppenheimer approximation** holds.
2. Translational, rotational, vibrational, and electronic motions are **weakly coupled**.
3. Molecules behave approximately as **rigid rotors and harmonic oscillators**.

5. Significance of Factorisation

Factorisation is extremely useful because it allows each type of molecular motion to be treated independently.

Each contribution affects thermodynamic properties differently.

For example

- translation contributes strongly to pressure and entropy
- rotation contributes to heat capacity
- vibration contributes significantly at high temperatures
- electronic contribution is usually small at ordinary temperatures.

Factorisation means that each type of molecular motion contributes **independently** to the thermodynamic properties of the system.

Thus

Motion	Partition Function	Effect
Translation	$q_{trans} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V$	dominates entropy and pressure
Rotation	$q_{rot} = \frac{8\pi^2 IkT}{\sigma h^2}$	contributes to heat capacity
Vibration	$q_{vib} = \frac{e^{-\frac{h\nu}{2kT}}}{1 - e^{-\frac{h\nu}{kT}}}$	important at high temperatures
Electronic	$q_{elec} = \sum g_i e^{-\frac{\epsilon_i}{kT}}$ $= g_0 + g_1 e^{-E_1/kT} + \ldots$	usually small at ordinary temperatures

$q_{rot} = \frac{T}{\sigma\theta_{rot}}$, $\theta_{rot} = \frac{h^2}{8\pi^2 Ik}$

$q_{vib} = \frac{1}{1-e^{-\theta_{vib}/T}}$. $\theta_{vib} = \frac{h\nu}{k}$

Summary of Part IV

Quantity	**Expression**
Molecular partition function	$q = \sum g_i e^{-\frac{\epsilon_i}{kT}}$
Canonical partition function	$Q = \frac{q^N}{N!}$
Internal energy	$U = -\frac{\partial \ln Q}{\partial \beta}$
Entropy	$S = k\ln Q + kT\frac{\partial \ln Q}{\partial T}$
Pressure	$P = kT\frac{\partial \ln Q}{\partial V}$
Factorisation	$q = q_{trans} q_{rot} q_{vib} q_{elec}$

Quantity	Expression
Translation	$q_{trans} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V$
Rotation	$q_{rot} = \frac{8\pi^2 IkT}{\sigma h^2}$
Vibration	$q_{vib} = \frac{e^{-\frac{h\nu}{2kT}}}{1 - e^{-\frac{h\nu}{kT}}}$
Electronic	$q_{elec} = \sum g_i e^{-\frac{\epsilon_i}{kT}}$ $= g_0 + g_1 e^{-E_1/kT} + \ldots$

Corresponding Contributions to Thermodynamic Properties

Because the molecular partition function factorizes

$$q = q_{tr} q_{rot} q_{vib} q_{ele}$$

taking natural logarithm gives

$$\ln\ q = \ln\ q_{tr} + \ln\ q_{rot} + \ln\ q_{vib} + \ln\ q_{ele}$$

Thermodynamic properties depend on derivatives of $\ln q$. Therefore **each type of molecular motion contributes independently** to thermodynamic quantities.

---

1. Internal Energy

The internal energy of a system of N molecules is

$$E = NkT^2 \left(\frac{\partial \ln\ q}{\partial T}\right)_V$$

Substituting the factorized form

$$\ln\ q = \ln\ q_{tr} + \ln\ q_{rot} + \ln\ q_{vib} + \ln\ q_{ele}$$

$$\frac{\partial \ln\ q}{\partial T} = \frac{\partial \ln\ q_{tr}}{\partial T} + \frac{\partial \ln\ q_{rot}}{\partial T} + \frac{\partial \ln\ q_{vib}}{\partial T} + \frac{\partial \ln\ q_{ele}}{\partial T}$$

Thus

$$E = E_{tr} + E_{rot} + E_{vib} + E_{ele}$$

---

2. Translational Contribution

Translational partition function

$$q_{tr} = V\left(\frac{2\pi mkT}{h^2}\right)^{3/2}$$

Taking logarithm

$$\ln\ q_{tr} = \ln\ V + \frac{3}{2}\ln\ \left(\frac{2\pi mk}{h^2}\right) + \frac{3}{2}\ln\ T$$

Differentiating

$$\frac{\partial \ln\ q_{tr}}{\partial T} = \frac{3}{2T}$$

Thus

$$E_{tr} = NkT^2\left(\frac{3}{2T}\right)$$

$$E_{tr} = \frac{3}{2}NkT$$

Heat Capacity

$$C_{V,tr} = \left(\frac{\partial E_{tr}}{\partial T}\right)_V$$

$$C_{V,tr} = \frac{3}{2}Nk$$

---

3. Rotational Contribution (Linear Molecule)

Rotational partition function

$$q_{rot} = \frac{T}{\sigma\theta_{rot}}$$

Taking logarithm

$$\ln\ q_{rot} = \ln\ T - \ln\ (\sigma\theta_{rot})$$

Differentiating

$$\frac{\partial \ln\ q_{rot}}{\partial T} = \frac{1}{T}$$

Thus

$$E_{rot} = NkT^2\left(\frac{1}{T}\right)$$

$$E_{rot} = NkT$$

Heat Capacity

$$C_{V,rot} = Nk$$

(This holds when $T \gg \theta_{rot}$.)

---

4. Vibrational Contribution

Vibrational partition function

$$q_{vib} = \frac{1}{1 - e^{-\theta_{vib}/T}}$$

Taking logarithm

$$\ln\ q_{vib} = -\ln\left(1 - e^{-\theta_{vib}/T}\right)$$

Differentiation gives

$$E_{vib} = Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$$

Heat Capacity

$$C_{V,vib} = Nk\left(\frac{\theta_{vib}}{T}\right)^2 \frac{e^{\theta_{vib}/T}}{\left(e^{\theta_{vib}/T} - 1\right)^2}$$

---

5. Electronic Contribution

Electronic partition function

$$q_{ele} = g_0 + g_1 e^{-E_1/kT} + \ldots$$

Since

$$E_1 \gg kT$$

$$q_{ele} \approx g_0$$

Thus

$$\ln\ q_{ele} = \ln\ g_0$$

which is independent of temperature.

Therefore

$$E_{ele} \approx 0$$

$$C_{V,ele} \approx 0$$

---

6. Total Internal Energy

For a linear molecule

$$E = \frac{3}{2}NkT + NkT + Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$$

---

7. Total Heat Capacity

$$C_V = \frac{3}{2}Nk + Nk + Nk\left(\frac{\theta_{vib}}{T}\right)^2 \frac{e^{\theta_{vib}/T}}{(e^{\theta_{vib}/T}-1)^2}$$

8. Summary Table

Motion	Partition Function	Internal Energy Contribution	Heat Capacity Contribution
Translational	$q_{tr} = V\left(\frac{2\pi mkT}{h^2}\right)^{3/2}$	$\frac{3}{2}NkT$	$\frac{3}{2}Nk$
Rotational (linear)	$q_{rot} = \frac{T}{\sigma\theta_{rot}}$, $\theta_{rot} = \frac{h^2}{8\pi^2 Ik}$	NkT	Nk (valid when $T \gg \theta_{rot}$)
Vibrational	$q_{vib} = \frac{1}{1-e^{-\theta_{vib}/T}}$, $\theta_{vib} = \frac{h\nu}{k}$	$Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$	$Nk(\theta_{vib}/T)^2 \frac{e^{\theta_{vib}/T}}{(e^{\theta_{vib}/T}-1)^2}$
Electronic	$q_{ele} = g_0 + g_1 e^{-E_1/kT} + \cdots$	≈0	≈0

Total Contributions for a Linear Molecule

Internal Energy

$$E = \frac{3}{2}NkT + NkT + Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$$

Heat Capacity

$$C_V = \frac{3}{2}Nk + Nk + Nk\left(\frac{\theta_{vib}}{T}\right)^2 \frac{e^{\theta_{vib}/T}}{(e^{\theta_{vib}/T} - 1)^2}$$

Characteristic Temperatures

$$\theta_{rot} = \frac{h^2}{8\pi^2 Ik}$$

$$\theta_{vib} = \frac{h\nu}{k}$$

These determine when a molecular motion becomes thermally active.

Temperature Dependence of Heat Capacity

Temperature Range	Active Motions	Heat Capacity
Very low T	Translational only	$\frac{3}{2}Nk$
Moderate T	Translational + rotational	$\frac{5}{2}Nk$
High T	Translational + rotational + vibrational	$\frac{7}{2}Nk$

Key Physical Insight

- Translational levels are **very closely spaced** → active at all temperatures.
- Rotational levels have **moderate spacing** → activated at moderate temperatures.
- Vibrational levels have **large spacing** → activated only at high temperatures.
- Electronic excitations require **very high energy** → usually negligible.

Temperature Dependence of Heat Capacity of a Gas

Different molecular motions become thermally active at different temperatures. As temperature increases, additional degrees of freedom contribute to the heat capacity. This produces a characteristic **step-like increase in** C_V.

Conceptual Behaviour

C_V

|

```
│                Vibrational contribution
│                /
│               /
│              /
│        _________/
│       /
│      /
│     /   Rotational contribution
│    /
│   /
│_______/
│
│ Translational contribution
│
└──────────────────────────────────── T
   low T     moderate T     high T
```

---

1. Low Temperature Region

At very low temperature

$$T \ll \theta_{rot}, \theta_{vib}$$

Only translational motion is active.

$$E_{tr} = \frac{3}{2} NkT$$

$$C_V = \frac{3}{2} Nk$$

Rotation and vibration are **frozen out** because thermal energy is too small to populate higher energy levels.

---

2. Intermediate Temperature Region

When

$$T \gg \theta_{rot}$$

rotational levels become populated.

The rotational contribution becomes

$$E_{rot} = NkT$$

Thus the total internal energy becomes

$$E = \frac{3}{2}NkT + NkT$$

$$E = \frac{5}{2}NkT$$

Heat capacity

$$C_V = \frac{5}{2}Nk$$

This behaviour is observed for most diatomic gases at room temperature.

---

3. High Temperature Region

When

$$T \sim \theta_{vib}$$

vibrational excitation becomes significant.

The vibrational energy contribution is

$$E_{vib} = Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$$

Thus heat capacity increases further.

At very high temperature the vibrational mode approaches the classical limit:

$$E_{vib} \approx NkT$$

so

$$C_{V,vib} \approx Nk$$

---

4. Classical High-Temperature Limit

For a **linear molecule**

Motion	**Contribution to C_V**
Translational	$\frac{3}{2}Nk$

Motion	Contribution to C_V
Rotational	Nk
Vibrational	Nk
Total	

$$C_V = \frac{7}{2}Nk$$

---

5. Physical Interpretation

The gradual increase in heat capacity occurs because:

- **Translational energy levels are very closely spaced** → active at all temperatures.
- **Rotational levels are moderately spaced** → activated at moderate temperatures.
- **Vibrational levels are widely spaced** → activated only at high temperatures.

Thus molecular heat capacity provides direct evidence for **quantization of molecular energy levels**.

Temperature Dependence of Heat Capacity of a Gas

Different molecular motions become thermally active at different temperatures. As temperature increases, additional degrees of freedom contribute to the heat capacity. This produces a characteristic **step-like increase in** C_V.

Part V – Evaluation of Partition Functions and Thermodynamic Properties for Ideal Gases

1. Introduction

In the previous sections we derived the molecular partition function and showed that it can be factorised as

$$q = q_{trans} q_{rot} q_{vib} q_{elec}$$

Each term corresponds to a different type of molecular motion.

Using this factorisation, the thermodynamic properties of gases can be obtained by evaluating each partition function separately.

Because the logarithm of a product is the sum of logarithms,

$$\ln q = \ln q_{trans} + \ln q_{rot} + \ln q_{vib} + \ln q_{elec}$$

Thus the contributions to thermodynamic quantities also become additive.

For example

$$U = U_{trans} + U_{rot} + U_{vib} + U_{elec}$$

$$C_V = C_{trans} + C_{rot} + C_{vib} + C_{elec}$$

2. Thermodynamic Quantities from Molecular Partition Function

For a system containing N molecules

$$Q = \frac{q^N}{N!}$$

The thermodynamic quantities can be expressed in terms of the molecular partition function.

Internal Energy

$$U = NkT^2 \left(\frac{\partial \ln q}{\partial T}\right)_V$$

Entropy

$$S = \mathrm{Nkln}\, q + NkT \left(\frac{\partial \ln q}{\partial T}\right)$$

Pressure

$$P = NkT \left(\frac{\partial \ln q}{\partial V}\right)$$

Heat Capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Because the partition function factorises, the contributions of different motions can be evaluated independently.

3. Ideal Monoatomic Gas

A monoatomic gas (e.g., He, Ne, Ar) has only translational motion because

- rotational motion is absent for single atoms
- vibrational motion does not exist.

Thus

$$q = q_{trans} \times q_{ele}$$

Since $q_{ele} \approx g_0$ and usually $g_0 = 1$,[i]

$$q \approx q_{trans}$$

Translational Partition Function

For one molecule

$$q_{trans} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V$$

Taking logarithm

$$\ln q_{trans} = \ln V + \frac{3}{2}\ln\left(\frac{2\pi mk}{h^2}\right) + \frac{3}{2}\ln T$$

Internal Energy of Monoatomic Gas

Using

$$U = NkT^2\left(\frac{\partial \ln q}{\partial T}\right)$$

and

$$\frac{\partial \ln q_{trans}}{\partial T} = \frac{3}{2T}$$

Thus

$$U = NkT^2\left(\frac{3}{2T}\right)$$

$$U = \frac{3}{2}NkT$$

Heat Capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$C_V = \frac{3}{2}Nk$$

For one mole

$$C_V = \frac{3}{2}R$$

Enthalpy

$$H = U + PV$$

Using the ideal gas equation

$$PV = NkT$$

Thus

$$H = \frac{3}{2}NkT + NkT$$

$$H = \frac{5}{2}NkT$$

For one mole

$$H = \frac{5}{2}RT$$

Heat Capacity at Constant Pressure

$$C_P = \left(\frac{\partial H}{\partial T}\right)$$

$$C_P = \frac{5}{2}Nk$$

Per mole

$$C_P = \frac{5}{2}R$$

Ratio of Heat Capacities

$$\gamma = \frac{C_P}{C_V}$$

$$\gamma = \frac{5/2R}{3/2R}$$

$$\gamma = \frac{5}{3}$$

4. Ideal Diatomic Gas

Diatomic molecules (e.g., N_2, O_2) possess additional degrees of freedom.

These include

- translational motion
- rotational motion
- vibrational motion.

Thus

$$q = q_{trans}q_{rot}q_{vib}$$

(The electronic contribution is usually negligible at ordinary temperatures.)

Translational Contribution

The translational contribution is identical to that for a monoatomic gas.

Thus

$$U_{trans} = \frac{3}{2}NkT$$

Rotational Contribution

For a diatomic molecule, the rotational partition function (high temperature approximation) is

$$q_{rot} = \frac{8\pi^2 IkT}{\sigma h^2}$$

Taking logarithm

$$\ln q_{rot} = \ln T + \text{constant}$$

Thus

$$\frac{\partial \ln q_{rot}}{\partial T} = \frac{1}{T}$$

Using

$$U = NkT^2 \frac{\partial \ln q}{\partial T}$$

we obtain

$$U_{rot} = NkT$$

Vibrational Contribution

For vibrational motion

$$q_{vib} = \frac{e^{-\frac{h\nu}{2kT}}}{1 - e^{-\frac{h\nu}{kT}}}$$

Taking logarithm

$$\ln q_{vib} = -\frac{h\nu}{2kT} - \ln\left(1 - e^{-\frac{h\nu}{kT}}\right)$$

Differentiation gives the vibrational contribution to internal energy.

At ordinary temperatures the vibrational levels are rarely excited, so

$$U_{vib} \approx 0$$

Internal Energy of Diatomic Gas

Thus

$$U = U_{trans} + U_{rot}$$

$$U = \frac{3}{2}NkT + NkT$$

$$U = \frac{5}{2}NkT$$

Heat Capacity of Diatomic Gas

$$C_V = \left(\frac{\partial U}{\partial T}\right)$$

Thus

$$C_V = \frac{5}{2}Nk$$

For one mole

$$C_V = \frac{5}{2}R$$

5. Effect of Vibrational Excitation

At higher temperatures, vibrational motion becomes active.

Each vibrational mode contributes

$$kT$$

to internal energy.

Thus heat capacity increases at high temperatures.

Partition function

$$q_{vib} = \frac{1}{1 - e^{-\theta_{vib}/T}}$$

Internal energy

$$E_{vib} = Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$$

Heat capacity

$$C_{V,vib} = Nk\left(\frac{\theta_{vib}}{T}\right)^2 \frac{e^{\theta_{vib}/T}}{\left(e^{\theta_{vib}/T} - 1\right)^2}$$

---

6. Summary

Gas Type	Internal Energy	Heat Capacity
Monoatomic gas	$U = \frac{3}{2}NkT$	$C_V = \frac{3}{2}Nk$
Diatomic gas (no vibration)	$U = \frac{5}{2}NkT$	$C_V = \frac{5}{2}Nk$

Gas Type	Internal Energy	Heat Capacity
Diatomic gas (high T)	$U = \frac{3}{2}NkT + NkT + Nk\frac{\theta_{vib}}{e^{\theta_{vib}/T} - 1}$	$C_V = \frac{3}{2}Nk + Nk + Nk\left(\frac{\theta_{vib}}{T}\right)^2 \frac{e^{\theta_{vib}/T}}{(e^{\theta_{vib}/T} - 1)^2}$

Room Temperature

Since

$$T \gg \theta_{rot}$$

but

$$T \ll \theta_{vib}$$

vibrational motion is frozen.

This section shows how statistical mechanics explains classical thermodynamic results such as

- the heat capacity of gases
- the effect of molecular structure on thermodynamic properties.

The partition function therefore provides a direct connection between **molecular motion and macroscopic thermodynamic behaviour**.

Failure of Classical Statistics

The Maxwell–Boltzmann distribution provides an accurate description of many physical systems such as ideal gases at ordinary temperatures and low densities. However, classical statistics is based on several assumptions that are not valid for all systems.

In particular, when dealing with systems of **indistinguishable quantum particles** or systems at **very low temperatures or high particle densities**, the predictions of Maxwell–Boltzmann statistics no longer agree with experimental observations.

Under such conditions, a quantum mechanical treatment becomes necessary, leading to the development of **Bose–Einstein** and **Fermi–Dirac statistics**.

Assumptions of Maxwell–Boltzmann Statistics

The derivation of Maxwell–Boltzmann statistics relies on the following assumptions:

1. **Particles are distinguishable**

Each particle can be uniquely labeled and distinguished from others.

2. **Particles are non-interacting**

The energy of one particle is independent of the presence of other particles.

3. **Any number of particles can occupy the same energy state**

There is no restriction on the occupancy of energy levels.

4. **Quantum effects are negligible**

The wave nature of particles is ignored and particles are treated as classical objects.

These assumptions are valid only under certain conditions.

Situations Where Classical Statistics Fails

Maxwell–Boltzmann statistics fails under the following circumstances.

1 Low Temperature

At very low temperatures, the thermal energy kT becomes comparable to the spacing between quantum energy levels.

Under these conditions:

- energy levels cannot be treated as continuous
- quantum effects become significant

As a result, classical statistics is no longer valid.

2 High Particle Density

When the number of particles per unit volume becomes large, the **wavefunctions of particles begin to overlap**.

The de Broglie wavelength of a particle is

$$\lambda = \frac{h}{\sqrt{2\pi mkT}}$$

When the condition

$$n\lambda^3 \approx 1$$

is satisfied (where n is particle number density), quantum effects become important and classical statistics fails.

3 Indistinguishability of Particles

In classical mechanics particles are assumed to be distinguishable.

However, in quantum mechanics identical particles are **fundamentally indistinguishable**. Exchanging two identical particles does not produce a new physical state.

This indistinguishability significantly changes the counting of microstates and therefore alters the statistical distribution.

Quantum Mechanical Restrictions

Quantum mechanics introduces two important restrictions on particle behaviour.

Bose–Einstein particles (bosons)

Particles with **integer spin** are called bosons.

Examples include

- photons
- helium-4 atoms
- phonons.

A characteristic property of bosons is that **any number of particles can occupy the same quantum state**.

Fermi–Dirac particles (fermions)

Particles with **half-integer spin** are called fermions.

Examples include

- electrons
- protons
- neutrons.

Fermions obey the **Pauli exclusion principle**, which states that

No two identical fermions can occupy the same quantum state.

Thus each quantum state can contain **at most one particle**.

Need for Quantum Statistics

Because of these quantum restrictions, the classical counting of microstates used in Maxwell–Boltzmann statistics becomes incorrect.

Therefore two new statistical distributions are required:

- **Bose–Einstein statistics** for bosons
- **Fermi–Dirac statistics** for fermions

These distributions correctly incorporate the quantum mechanical properties of particles.

Transition to Quantum Statistics

Thus the study of particle distributions proceeds as follows:

1. **Maxwell–Boltzmann statistics**

Applicable to classical particles at high temperature and low density.

2. **Bose–Einstein statistics**

Applicable to indistinguishable bosons.

3. **Fermi–Dirac statistics**

Applicable to indistinguishable fermions obeying the Pauli exclusion principle.

The next sections derive these quantum statistical distribution laws.

Part VI – Bose–Einstein Statistics

1. Introduction

Maxwell–Boltzmann statistics assumes that particles are **distinguishable** and that any number of particles can occupy a given energy state. However, this assumption fails for systems of **indistinguishable quantum particles**.

When particles are indistinguishable and possess **integer spin**, they obey **Bose–Einstein statistics**. Such particles are called **bosons**.

Examples of bosons include

- photons
- helium-4 atoms
- phonons in solids.

A characteristic feature of bosons is that **any number of particles can occupy the same quantum state**. Bosons have symmetric wavefunctions and are **not restricted in occupancy**. Many particles can accumulate in the same quantum state.

This leads to phenomena such as **Bose–Einstein condensation**, where a macroscopic number of particles occupy the lowest energy level.

2. Basic Assumptions of Bose–Einstein Statistics

The following assumptions are made:

1. The particles are **indistinguishable**.
2. Any number of particles may occupy a given quantum state.
3. Energy levels may be degenerate.
4. The system is in thermal equilibrium.
5. Total number of particles and total energy are conserved.

Let

$$N_i$$

= number of particles in the i^{th}energy level

$$g_i$$

= degeneracy of that level

$$\epsilon_i$$

= energy of the level.

The constraints are

$$\sum_i N_i = N$$

$$\sum_i N_i \, \epsilon_i = U$$

3. Thermodynamic Probability for Bose–Einstein Statistics

For bosons, particles are indistinguishable and multiple particles may occupy the same state.

The number of ways of distributing $N_i$particles among $g_i$states is

$$W_i = \frac{(g_i + N_i - 1)!}{(g_i - 1)!\, N_i!}$$

For all energy levels

$$W = \prod_i \frac{(g_i + N_i - 1)!}{(g_i - 1)!\, N_i!}$$

This expression gives the **thermodynamic probability** for Bose–Einstein statistics.

Consider for example, Five indistinguishable particles of total energy 4ε distributed over five energy states, each one with five fold degeneracy

Table 4.3.1 Distributions of Five Bosons Amongst Energy Levels

Distribution	*Number of Particles in Energy Levels*					*Thermodynamic Probability* $W = \Pi_i \frac{(N_i + g_i - 1)!}{(g_i - 1)! N_i!}$
	0	ε	2ε	3ε	4ε	
I	4	0	0	0	1	$\left(\frac{8!}{4!4!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{5!}{4!1!}\right) = 350$
II	3	1	0	1	0	$\left(\frac{7!}{4!3!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{4!}{4!0!}\right) = 875$
III	3	0	2	0	0	$\left(\frac{7!}{4!3!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{6!}{4!2!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right) = 525$
IV	2	2	1	0	0	$\left(\frac{6!}{4!2!}\right)\left(\frac{6!}{4!2!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right) = 1\,125$
V	1	4	0	0	0	$\left(\frac{5!}{4!1!}\right)\left(\frac{8!}{4!4!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right)\left(\frac{4!}{4!0!}\right) = 350$

4. Logarithm of Thermodynamic Probability

Taking logarithm

$$\ln W = \sum_i [\ln(g_i + N_i - 1)! - \ln(g_i - 1)! - \ln N_i!]$$

For systems where

$$g_i \gg 1$$

the "−1" terms may be neglected:

For large numbers we apply **Stirling approximation**

$$\ln n! \approx n\ln n - n$$

Thus

$$\ln(g_i + N_i)! \approx (g_i + N_i)\ln(g_i + N_i) - (g_i + N_i)$$

$$\ln g_i! \approx g_i \ln g_i - g_i$$

$$\ln N_i! \approx N_i \ln N_i - N_i$$

Substituting

$$\ln\ W = \sum_i [(g_i + N_i)\ln\ (g_i + N_i) - g_i \ln\ g_i - N_i \ln\ N_i]$$

5. Maximization of Thermodynamic Probability- Most Probable Distribution

At equilibrium the system occupies the distribution that **maximizes** W.

Thus

$$d(\ln W) = 0$$

subject to the constraints

$$\sum_i d\,N_i = 0$$

$$\sum_i \epsilon_i\, dN_i = 0$$

6. Application of Lagrange Multipliers

Introduce multipliers $\alpha$and β.

$$d\left(\ln W + \alpha \sum_i N_i - \beta \sum_i N_i\, \epsilon_i\right) = 0$$

Differentiating

$$\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta\epsilon_i = 0$$

Differentiation

From the expression for

$$\ln\ W = \sum_i [(N_i + g_i)\ln\,(N_i + g_i) - g_i \ln\ g_i - N_i \ln\ N_i]$$

Differentiation

$$\frac{\partial \ln\ W}{\partial N_i} = \ln\,(N_i + g_i) + 1 - \ln\ N_i - 1$$

$$= \ln\,(N_i + g_i) - \ln\ N_i$$

$$\frac{\partial(\ln W)}{\partial N_i} = \ln(g_i + N_i) - \ln N_i$$

Substitute in $\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta\epsilon_i = 0$

$$\ln(g_i + N_i) - \ln N_i + \alpha - \beta\epsilon_i = 0$$

Thus

$$\ln\left(\frac{g_i + N_i}{N_i}\right) = \beta\epsilon_i - \alpha$$

Rearranging

Exponentiating

$$\frac{g_i + N_i}{N_i} = e^{\beta\epsilon_i - \alpha}$$

Thus

$$1 + \frac{g_i}{N_i} = e^{\beta\epsilon_i - \alpha}$$

Solving for N_i

$$N_i = \frac{g_i}{e^{\beta\epsilon_i - \alpha} - 1}$$

7. Introduction of Chemical Potential

Define

$$\alpha = -\frac{\mu}{kT}$$

and

$$\beta = \frac{1}{kT}$$

Substituting

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

This is the **Bose–Einstein distribution law**.

8. Physical Interpretation

The distribution shows that particle population depends on

- energy of the state
- temperature
- chemical potential.

Unlike Maxwell–Boltzmann statistics, the denominator contains **−1**, which allows many particles to occupy the same state.

9. Reduction to Maxwell–Boltzmann Distribution

For dilute systems

$$\frac{g_i}{N_i} \gg 1$$

. $1 + \frac{g_i}{N_i} = e^{-\alpha} e^{\beta \varepsilon_i}$ becomes:

$$\frac{g_i}{N_i} = e^{-\alpha} e^{\beta \varepsilon_i}$$

$$N_i = g_i e^{\alpha} e^{-\beta \varepsilon_i}$$

Using normalization:

$$N = \sum_i N_i = \sum_i g_i \, e^{\alpha} e^{-\beta \varepsilon_i}$$

$$e^{\alpha} = \frac{N}{\sum_i g_i \, e^{-\beta \varepsilon_i}}$$

Hence,

$$N_i = N \frac{g_i e^{-\beta \varepsilon_i}}{\sum_i g_i \, e^{-\beta \varepsilon_i}} = \frac{N}{q} g_i e^{-\beta \varepsilon_i}$$

which is identical to the **Boltzmann distribution law**.

Approximation of Thermodynamic Probability

When $g_i \gg N_i$,

$$\frac{(g_i + N_i - 1)!}{(g_i - 1)!} = g_i(g_i + 1)(g_i + 2) \dots (g_i + N_i - 1) \approx g_i^{N_i}$$

the thermodynamic probability also reduces to a form identical to that of Boltzmann statistics

Thus the Bose–Einstein expression reduces to the Boltzmann form.

Because the number of translational energy states in a macroscopic system is enormously larger than the number of molecules, the probability of two molecules occupying the same state is negligible. Hence classical Maxwell–Boltzmann statistics becomes valid. However, when temperature decreases or density increases, the number of accessible states becomes comparable to the number of particles. Then quantum effects appear and Bose–Einstein statistics must be usedor, using chemical potential μ(where $\alpha = -\mu/kT$and $\beta = 1/kT$):

$$\boxed{N_i = \frac{g_i}{e^{-\alpha} e^{\beta \varepsilon_i} - 1}}$$

or, using chemical potential μ(where $\alpha = -\mu/kT$and $\beta = 1/kT$):

$$\boxed{N_i = \frac{g_i}{\exp\left(\frac{\varepsilon_i - \mu}{kT}\right) - 1}}$$

Summary of Bose–Einstein Statistics

Feature	**Description**
Particle type	bosons
Spin	integer
Occupation rule	unlimited particles per state
Distribution law	$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$
Special phenomenon	Bose–Einstein condensation

Bose–Einstein Condensation

Bose–Einstein condensation is one of the most remarkable predictions of quantum statistics. It occurs in systems of bosons at very low temperatures when a large number of particles occupy the lowest energy state.

At very low temperatures the chemical potential approaches the lowest energy level.

$$\mu \to \epsilon_0$$

Under these conditions a large number of particles accumulate in the ground state.

This phenomenon is known as **Bose–Einstein condensation**.

In this state

- a macroscopic fraction of particles occupy the same quantum state
- the system exhibits unusual quantum behaviour.

Physical Examples

Bose–Einstein statistics is important in many physical systems such as

- photons in blackbody radiation
- helium-4 at low temperatures
- phonons in solids
- ultracold atomic gases.

Bose–Einstein Distribution

From Bose–Einstein statistics, the number of particles in energy level $i$is

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

where

- g_i= degeneracy of the energy level
- ϵ_i= energy of the level
- μ= chemical potential.

Condition on Chemical Potential

For the occupation number to remain finite, the denominator must be positive.

Thus

$$e^{\frac{\epsilon_i - \mu}{kT}} > 1$$

which requires

$$\mu \leq \epsilon_0$$

where $\epsilon_0$is the lowest energy level.

At very low temperatures

$$\mu \to \epsilon_0$$

Separation of Ground State

The total number of particles is

$$N = \sum_i N_i$$

It is convenient to separate the ground state population.

$$N = N_0 + N_{excited}$$

where

- N_0= number of particles in the ground state
- $N_{excited}$= number of particles in excited states.

Maximum Number of Excited Particles

For a large system the energy levels are very closely spaced, so the summation over excited states can be replaced by an integral.

$$N_{excited} = \int_0^\infty \frac{g(\epsilon)\, d\epsilon}{e^{\frac{\epsilon-\mu}{kT}} - 1}$$

where $g(\epsilon)$is the **density of states**.

For free particles in three dimensions

$$g(\epsilon)d\epsilon = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \epsilon^{\frac{1}{2}} d\epsilon$$

Substituting and evaluating the integral gives

$$N_{excited} = V\left(\frac{mkT}{2\pi\hbar^2}\right)^{\frac{3}{2}} \zeta\left(\frac{3}{2}\right)$$

where

$$\zeta\left(\frac{3}{2}\right) \approx 2.612$$

Critical Temperature

Condensation begins when the number of excited particles equals the total number of particles.

$$N = N_{excited}$$

Thus

$$N = V\left(\frac{mkT_c}{2\pi\hbar^2}\right)^{\frac{3}{2}} \zeta\left(\frac{3}{2}\right)$$

Solving for T_c

$$T_c = \frac{2\pi\hbar^2}{mk}\left(\frac{N}{V\zeta\left(\frac{3}{2}\right)}\right)^{\frac{2}{3}}$$

This temperature is called the **Bose–Einstein condensation temperature**.

Fraction of Condensed Particles

For temperatures below T_c

$$N_{excited} = N\left(\frac{T}{T_c}\right)^{\frac{3}{2}}$$

Thus

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}$$

At absolute zero

$$N_0 = N$$

so all particles occupy the ground state.

Physical Interpretation

Bose–Einstein condensation is not a conventional phase transition like gas–liquid condensation.

Instead it represents

- macroscopic occupation of a single quantum state
- quantum behaviour on a macroscopic scale.

Application to Liquid Helium

Helium-4 atoms are bosons. At temperatures below about

$$T \approx 2.17\ K$$

liquid helium undergoes a transition to a **superfluid state**.

This phenomenon is closely related to Bose–Einstein condensation.

Superfluid helium exhibits unusual properties such as

- zero viscosity flow
- fountain effect
- quantized vortices.

Part VII – Fermi–Dirac Statistics

1. Introduction

Bose–Einstein statistics applies to particles with **integer spin (bosons)**.

However, many particles in nature possess **half-integer spin** and obey a different statistical law known as **Fermi–Dirac statistics**.

Particles that obey Fermi–Dirac statistics are called **fermions**.

Examples include

- electrons
- protons
- neutrons
- atoms with half-integer total spin.

A fundamental property of fermions is that they obey the **Pauli exclusion principle**. Because fermions cannot occupy the same quantum state, particles must fill higher energy levels even at very low temperatures. This leads to the concept of **Fermi energy** and determines the behaviour of electrons in metals.

Pauli Exclusion Principle

The Pauli exclusion principle states that

no two identical fermions can occupy the same quantum state simultaneously.

Therefore the number of particles occupying a quantum state can be either

$$0 \text{ or } 1$$

This restriction leads to a distribution law that differs from both Maxwell–Boltzmann and Bose–Einstein statistics.

2. Assumptions of Fermi–Dirac Statistics

The derivation of Fermi–Dirac statistics is based on the following assumptions:

1. Particles are **indistinguishable**.
2. The **Pauli exclusion principle** applies.
3. Each quantum state can contain **at most one particle**.
4. Energy levels may be degenerate.
5. Total particle number and energy are conserved.

Let

$$N_i$$

= number of particles in energy level i

$$g_i$$

= degeneracy of that level

$$\epsilon_i$$

= energy of the level.

The constraints are

$$\sum_i N_i = N$$

$$\sum_i N_i \, \epsilon_i = U$$

3. Thermodynamic Probability for Fermi–Dirac Statistics

For a level with degeneracy g_i, we must choose $N_i$states out of $g_i$available states.

The number of possible arrangements is therefore

$$W_i = \frac{g_i!}{(g_i - N_i)!\, N_i!}$$

For all energy levels

$$W = \prod_i \frac{g_i!}{(g_i - N_i)!\, N_i!}$$

This is the **thermodynamic probability for Fermi–Dirac statistics**.

For example, five identical indistinguishable particles of total energy of 4ε be distributed over five energy levels with energies $0, \varepsilon, 2\varepsilon, 3\varepsilon$and 4ε, each one is fivefold degenerate.

Table 4.4.1 Distributions of Five Fermions Amongst Energy Levels

Distribution	*Number of Particles in Energy Levels*					*Thermodynamic Probability* $W = \Pi_i \frac{g_i!}{(g_i - N_i)!N_i!}$
	0	ε	2ε	3ε	4ε	
I	4	0	0	0	1	$\left(\frac{5!}{1!4!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{4!1!}\right) = 25$
II	3	1	0	1	0	$\left(\frac{5!}{2!3!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{5!}{5!0!}\right) = 250$
III	3	0	2	0	0	$\left(\frac{5!}{2!3!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{3!2!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right) = 100$
IV	2	2	1	0	0	$\left(\frac{5!}{3!2!}\right)\left(\frac{5!}{3!2!}\right)\left(\frac{5!}{4!1!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right) = 500$
V	1	4	0	0	0	$\left(\frac{5!}{4!1!}\right)\left(\frac{5!}{1!4!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right)\left(\frac{5!}{5!0!}\right) = 25$

4. Logarithm of Thermodynamic Probability

Taking logarithm

$$\ln\ W = \sum_i [\ln \quad g_i! - \ln\,(g_i - N_i)! - \ln\ N_i!]$$

Using Stirling approximation

$$\ln n! \approx n\ln n - n$$

we obtain

$$\ln\ W = \sum_i [\,g_i \ln\ g_i - (g_i - N_i)\ln\,(g_i - N_i) - N_i \ln\ N_i]$$

5. Maximization of Thermodynamic Probability

At equilibrium the most probable distribution corresponds to the maximum value of W.

Thus

$$d(\ln W) = 0$$

subject to

$$\sum_i d\,N_i = 0$$

$$\sum_i \epsilon_i\, dN_i = 0$$

6. Lagrange Multiplier Method

Introduce multipliers $\alpha$and β.

$$d\left(\ln W + \alpha \sum_i N_i - \beta \sum_i N_i \epsilon_i\right) = 0$$

Differentiating

$$\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta\epsilon_i = 0$$

Differentiation

From the expression for $\ln W$

$$\frac{\partial(\ln W)}{\partial N_i} = -\ln(g_i - N_i) + \ln N_i$$

Substituting

$$-\ln(g_i - N_i) + \ln N_i + \alpha - \beta\epsilon_i = 0$$

Thus

$$\ln\left(\frac{N_i}{g_i - N_i}\right) = \beta\epsilon_i - \alpha$$

Rearranging

Exponentiating

$$\frac{N_i}{g_i - N_i} = e^{\beta\epsilon_i - \alpha}$$

Solving for N_i

$$N_i = \frac{g_i}{e^{\beta\epsilon_i - \alpha} + 1}$$

7. Introduction of Chemical Potential

Define

$$\alpha = -\frac{\mu}{kT}$$

and

$$\beta = \frac{1}{kT}$$

Substituting

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$$

This is the **Fermi–Dirac distribution law**.

8. Properties of Fermi–Dirac Distribution

The Fermi–Dirac distribution function is

$$f(\epsilon) = \frac{1}{e^{\frac{\epsilon-\mu}{kT}} + 1}$$

where $f(\epsilon)$represents the probability that a state of energy $\epsilon$is occupied.

Behaviour at Absolute Zero

At

$$T = 0$$

the distribution becomes

$$f(\epsilon) = \begin{cases} 1 & \epsilon < \mu \\ 0 & \epsilon > \mu \end{cases}$$

Thus all energy states below a certain energy are filled.

This energy is called the **Fermi energy** E_F.

9. Fermi Energy

The **Fermi energy** is defined as the highest occupied energy level at absolute zero.

$$\mu(T = 0) = E_F$$

In metals the Fermi energy is typically several electron volts.

Because

$$E_F \gg kT$$

only a small fraction of electrons near the Fermi level participate in thermal processes.

Reduction to the Boltzmann Distribution Law

For a system in which $g_i/N_i \gg 1$, Eq. $N_i = \frac{g_i}{e^{-\alpha}e^{\beta\varepsilon_i}+1}$ may be approximated as

$$\frac{g_i}{N_i} = e^{-\alpha}e^{\beta\varepsilon_i}$$

or

$$N_i = g_i e^{\alpha} e^{-\beta\varepsilon_i}$$

Using

$$N = \sum_i N_i = \sum_i g_i\, e^{\alpha} e^{-\beta\varepsilon_i}$$

$$e^{\alpha} = \frac{N}{\sum_i g_i\, e^{-\beta\varepsilon_i}}$$

Hence,

$$N_i = N\frac{g_i e^{-\beta\varepsilon_i}}{\sum_i g_i\, e^{-\beta\varepsilon_i}} = \frac{N}{q} g_i e^{-\beta\varepsilon_i}$$

which is identical to Boltzmann distribution law.

Under the approximation $g_i \gg N_i$, the thermodynamic probability also reduces to a form identical to that of Boltzmann statistics, since we can write

$$\frac{g_i!}{(g_i - N_i)!} = g_i(g_i - 1)\cdots(g_i - N_i + 1) \approx g_i^{N_i}$$

In an atom or molecule, the number of translational energy states alone is sufficient to guarantee the number of energy states available to any atom/molecule much greater than the number of atoms/molecules in the system.

10. Application to Electrons in Metals

Electrons in a metal behave approximately like a gas of free fermions confined within the metal.

This model is known as the **free electron model**.

Assumptions of this model include

1. Electrons move freely inside the metal.
2. Electron–electron interactions are neglected.
3. The Pauli exclusion principle applies.

Density of States

For free electrons in three dimensions, the number of states between energy $\epsilon$and $\epsilon + d\epsilon$is

$$g(\epsilon)d\epsilon = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \epsilon^{\frac{1}{2}} d\epsilon$$

Electron Distribution

Using the Fermi–Dirac distribution

$$N(\epsilon)d\epsilon = g(\epsilon)f(\epsilon)d\epsilon$$

This expression gives the number of electrons with energy between $\epsilon$and $\epsilon + d\epsilon$.

Comparison of Statistical Distributions

Statistics	Particle Type	Occupation Rule
Maxwell–Boltzmann	classical particles	unlimited

Statistics	Particle Type	Occupation Rule
Bose–Einstein	bosons	unlimited
Fermi–Dirac	fermions	maximum one per state

Summary

Feature	Fermi–Dirac Statistics
Particle type	fermions
Spin	half-integer
Occupation rule	at most one particle per state
Distribution	$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$
Important concept	Fermi energy

Part VIII – Application of Fermi–Dirac Statistics to Electrons in Metals **(Free Electron Model)**

1. Introduction

In metals, a large number of electrons are not tightly bound to individual atoms. Instead, they move relatively freely throughout the crystal lattice. These mobile electrons are called **conduction electrons**.

Because electrons are fermions (spin $\frac{1}{2}$), they obey **Fermi–Dirac statistics**. The statistical behaviour of electrons in metals explains many important physical properties such as

- electrical conductivity
- thermal conductivity
- electronic heat capacity.

A simple model used to describe electrons in metals is the **free electron model**.

2. Assumptions of the Free Electron Model

The free electron model is based on the following assumptions:

1. Conduction electrons behave like **free particles** moving inside the metal.
2. The metal is treated as a **three-dimensional box** of volume V.

3. Interactions between electrons are neglected.
4. The only quantum mechanical restriction is the **Pauli exclusion principle**.
5. Each quantum state can contain **two electrons** (due to spin).

This model provides a surprisingly good approximation for many metallic properties.

3. Quantum Mechanical Description

To determine the allowed energy levels of electrons, we treat the metal as a particle confined in a three-dimensional box.

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi$$

The normalized solutions are

$$\psi_{n_x,n_y,n_z}(x,y,z) = \left(\frac{2}{L}\right)^{\frac{3}{2}} \sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right)$$

where

$$n_x, n_y, n_z = 1,2,3, \ldots$$

Energy Levels

The energy of an electron in the box is

$$E = \frac{h^2}{8mL^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

This shows that the energy depends on the squares of the quantum numbers.

Each set of quantum numbers corresponds to a **single spatial quantum state**.

Spin Degeneracy

Electrons possess spin $\frac{1}{2}$.

Thus each spatial state can accommodate two electrons:

- one with spin $+\frac{1}{2}$
- one with spin $-\frac{1}{2}$.

Therefore the degeneracy of each state is **2**.

k-Space Representation

It is convenient to describe electron states in terms of the **wave vector**

$$k$$

where

$$p = \hbar k$$

The energy becomes

$$E = \frac{\hbar^2 k^2}{2m}$$

Thus energy depends only on the magnitude of k.

Filling of Energy States

At absolute zero, electrons occupy the lowest available energy states.

According to the Pauli exclusion principle

- only two electrons can occupy each state.

Thus electrons fill states from lowest energy upward until all electrons are accommodated.

4. Fermi Energy

The **highest occupied energy level at absolute zero** is called the **Fermi energy**.

$$E_F = \frac{\hbar^2 k_F^2}{2m}$$

where

$$k_F$$

is the **Fermi wave vector**.

The occupied states in k-space form a sphere called the **Fermi sphere**.

Determination of the Fermi Energy

The number of allowed states in k-space is determined by the volume of the sphere.

Volume of sphere in k-space:

$$\frac{4}{3}\pi k_F^3$$

Volume occupied by each state:

$$\left(\frac{2\pi}{L}\right)^3$$

Thus the number of states is

$$N = 2 \times \frac{\frac{4}{3}\pi k_F^3}{\left(\frac{2\pi}{L}\right)^3}$$

where the factor 2 accounts for spin.

After simplification

$$N = \frac{V}{3\pi^2} k_F^3$$

Thus

$$k_F = (3\pi^2 n)^{\frac{1}{3}}$$

where

$$n = \frac{N}{V}$$

is the electron density.

Substituting into the energy equation

$$E_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{\frac{2}{3}}$$

Temperature Effects

At absolute zero:

- all states below E_F are filled
- all states above E_F are empty.

At finite temperature, the distribution becomes slightly smeared.

However, because

$$E_F \gg kT$$

only electrons near the Fermi energy participate in thermal processes.

Consequences of Fermi–Dirac Statistics in Metals

Fermi–Dirac statistics explains several important properties of metals:

Electrical conductivity

Electrons near the Fermi level can move easily under an applied electric field.

Electronic heat capacity

Only a small fraction of electrons contribute to heat capacity, explaining why the electronic heat capacity of metals is small.

Thermal conductivity

Mobile electrons also transport heat efficiently.

Summary

Concept	Description
Model	free electron model
Statistics	Fermi–Dirac statistics
Key energy	Fermi energy E_F
Important concept	Fermi sphere
Applications	electrical and thermal properties of metals

Part IX – Relationship Between Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac Statistics

Introduction

In statistical thermodynamics, the distribution of particles among energy levels depends on the nature of the particles.

Three statistical laws describe these distributions:

1. **Maxwell–Boltzmann statistics**
2. **Bose–Einstein statistics**
3. **Fermi–Dirac statistics**

Each of these statistics arises from different assumptions about the properties of particles, particularly their **distinguishability and quantum nature**.

1. Maxwell–Boltzmann Statistics

Maxwell–Boltzmann statistics applies to **classical particles**.

Characteristics

- Particles are **distinguishable**
- No restriction on the number of particles per state
- Applicable at **high temperature and low density**

The distribution law is

$$N_i = g_i e^{\alpha - \beta \epsilon_i}$$

or

$$N_i = \frac{N}{q} g_i e^{-\frac{\epsilon_i}{kT}}$$

2. Bose–Einstein Statistics

Bose–Einstein statistics applies to **bosons**, which are particles with integer spin.

Characteristics

- Particles are **indistinguishable**
- Any number of particles may occupy the same state

The distribution law is

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

An important consequence of Bose–Einstein statistics is **Bose–Einstein condensation**, where a large number of particles occupy the lowest energy state at very low temperatures.

3. Fermi–Dirac Statistics

Fermi–Dirac statistics applies to **fermions**, which have half-integer spin.

Characteristics

- Particles are **indistinguishable**
- The **Pauli exclusion principle** applies
- Each state can contain **at most one particle**

The distribution law is

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$$

Fermi–Dirac statistics explains the behaviour of electrons in metals and the concept of **Fermi energy**.

4. Unified Form of the Distribution Laws

The three distribution laws can be written in a **single unified expression**

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + \sigma}$$

where

$$\sigma = \begin{cases} 0 & \text{Maxwell–Boltzmann} \\ -1 & \text{Bose–Einstein} \\ +1 & \text{Fermi–Dirac} \end{cases}$$

This compact expression shows that the three statistics differ only in the value of σ.

5. Classical Limit of Quantum Statistics

In statistical thermodynamics three different distribution laws describe how particles are distributed among available energy levels:

- **Maxwell–Boltzmann statistics** — applies to classical particles
- **Bose–Einstein statistics** — applies to bosons (integer spin particles)
- **Fermi–Dirac statistics** — applies to fermions (half-integer spin particles)

Although these distributions appear different, an important result of statistical mechanics is that **quantum statistics reduce to classical Maxwell–Boltzmann statistics under appropriate conditions**.

This situation is known as the **classical limit of quantum statistics**.

Understanding this limit explains why classical statistical mechanics works very well for many systems such as **ordinary gases**, even though matter is fundamentally quantum mechanical.

Unified Form of the Distribution Laws

The three distribution laws can be written in a single unified expression:

$$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + \sigma}$$

where

- n_i= number of particles in the i^{th}energy level
- g_i= degeneracy of that level
- ϵ_i= energy of the level
- k= Boltzmann constant
- T= temperature
- μ= chemical potential

The parameter σdetermines the type of statistics:

Statistics	**Particle type**	**Value of σ**
Maxwell–Boltzmann	classical particles	0
Bose–Einstein	bosons	-1
Fermi–Dirac	fermions	$+1$

Thus

Bose–Einstein distribution

$$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

Fermi–Dirac distribution

$$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$$

Condition for the Classical Limit

Consider the exponential term in the denominator:

$$e^{\frac{\epsilon_i - \mu}{kT}}$$

If this quantity is **very large compared to unity**, then

$$e^{\frac{\epsilon_i - \mu}{kT}} \gg 1$$

Under this condition, the term ± 1in the denominator becomes negligible.

Thus

$$e^{\frac{\epsilon_i - \mu}{kT}} \pm 1 \approx e^{\frac{\epsilon_i - \mu}{kT}}$$

Substituting into the quantum distributions gives

$$n_i \approx \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}}}$$

or

$$n_i \approx g_i e^{-\frac{\epsilon_i - \mu}{kT}}$$

This expression is exactly the **Maxwell–Boltzmann distribution**.

Thus

$$\boxed{\text{Bose–Einstein and Fermi–Dirac statistics reduce to Maxwell–Boltzmann statistics}}$$

when

$$e^{\frac{\epsilon_i - \mu}{kT}} \gg 1$$

Physical Interpretation of the Condition

The condition

$$e^{\frac{\epsilon_i - \mu}{kT}} \gg 1$$

implies that the **average occupation of each quantum state is very small**.

This can also be written as

$$\frac{n_i}{g_i} \ll 1$$

Thus each quantum state contains **much less than one particle on average**.

In such cases:

- particles rarely occupy the same quantum state
- quantum restrictions become unimportant

As a result, particles behave approximately as **distinguishable classical particles**.

Physical Conditions for the Classical Limit

The classical limit usually occurs under the following conditions.

1 High Temperature

At high temperatures the thermal energy $kT$is large compared with the spacing between quantum energy levels.

Therefore many energy states are accessible and quantum effects become negligible.

---

2 Low Particle Density

If the number of particles per unit volume is small, the probability that two particles occupy the same quantum state is extremely small.

Thus the **Pauli exclusion principle** or **boson accumulation effects** become insignificant.

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3 Weak Quantum Overlap

Quantum behaviour becomes important when particle wavefunctions overlap significantly.

At low densities and high temperatures the **de Broglie wavelength of particles is small**, so wavefunctions do not overlap strongly.

Thus the particles behave classically.

Classical Limit in Terms of Thermal Wavelength

A more precise condition for the classical limit can be expressed using the **thermal de Broglie wavelength**:

$$\lambda = \frac{h}{\sqrt{2\pi mkT}}$$

where

- h= Planck constant

- m= particle mass.

The classical limit holds when

$$n\lambda^3 \ll 1$$

where $n$is the number density of particles.

This condition means that the **volume occupied by each particle is much larger than the volume associated with its quantum wavelength**.

Physical Examples

Ideal gases

For most gases at ordinary temperatures and pressures:

$$n\lambda^3 \ll 1$$

Therefore classical Maxwell–Boltzmann statistics accurately describe the distribution of molecules.

Electrons in metals

Electrons have very high density and small mass, giving a large de Broglie wavelength.

Thus

$$n\lambda^3 \gtrsim 1$$

Quantum effects become important and **Fermi–Dirac statistics must be used**.

Photons

Photons are bosons and can occupy the same state in large numbers.

Thus their behaviour follows **Bose–Einstein statistics**.

Importance of the Classical Limit

The classical limit explains why classical statistical mechanics successfully describes many macroscopic systems.

Although matter obeys quantum mechanics at the microscopic level, classical behaviour emerges when:

- temperature is sufficiently high
- particle density is low.

Under these conditions, the Maxwell–Boltzmann distribution becomes an excellent approximation to the more general quantum statistical distributions.

Summary

Distribution	Expression
Maxwell–Boltzmann	$n_i = g_i e^{-\frac{\epsilon_i - \mu}{kT}}$
Bose–Einstein	$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$
Fermi–Dirac	$n_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$

At high temperatures or low densities

$$e^{\frac{\epsilon_i - \mu}{kT}} \gg 1$$

the ± 1 in the denominator becomes negligible.

Therefore both Bose–Einstein and Fermi–Dirac distributions reduce to

$$N_i \approx g_i e^{-\frac{\epsilon_i - \mu}{kT}}$$

which is the **Maxwell–Boltzmann distribution**.

Thus classical statistics represent the **high-temperature, low-density limit of quantum statistics**.

$$e^{\frac{\epsilon_i - \mu}{kT}} \gg 1$$

Comparison of the Three Statistics

Property	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Particle type	classical particles	bosons	fermions
Spin	any	integer	half-integer
Distinguishability	distinguishable	indistinguishable	indistinguishable
Occupation of state	unlimited	unlimited	maximum 1
Distribution	exponential	BE distribution	FD distribution
Examples	ideal gases	photons, He-4 atoms	electrons, protons

Graphical Behaviour

If the three distributions are plotted as functions of energy:

- **Maxwell–Boltzmann distribution** decreases exponentially with energy

- **Bose–Einstein distribution** shows higher population at low energies
- **Fermi–Dirac distribution** has a sharp cutoff at the **Fermi energy**

These differences become especially important at **low temperatures and high densities**.

Significance in Statistical Thermodynamics

The three statistical distributions play important roles in different physical systems:

Maxwell–Boltzmann statistics

Used for classical gases and many chemical systems.

Bose–Einstein statistics

Important in phenomena such as

- blackbody radiation
- Bose–Einstein condensation
- superfluidity.

Fermi–Dirac statistics

Essential for understanding

- electrons in metals
- semiconductors
- white dwarf stars.

Final Summary of the Chapter

The central ideas of statistical thermodynamics can be summarized as follows:

1. Macroscopic properties arise from **microscopic states**.
2. The **partition function** connects microscopic energy levels to thermodynamic quantities.
3. Particle distributions follow one of three statistical laws:
    - Maxwell–Boltzmann
    - Bose–Einstein
    - Fermi–Dirac.

Thus statistical thermodynamics provides a powerful framework for understanding the relationship between **molecular behaviour and macroscopic thermodynamic properties**.

Part X – Heat Capacity of Solids

1. Introduction

The heat capacity of a solid describes how much heat energy is required to raise the temperature of the solid by one degree.

Understanding the heat capacity of solids was historically an important problem in statistical mechanics because classical physics predicted a constant value for heat capacity, while experiments showed that heat capacity decreases at low temperatures.

Statistical thermodynamics explains this behaviour using quantum theory.

Three important models are used:

1. **Dulong–Petit law (classical theory)**
2. **Einstein theory of heat capacity**
3. **Debye theory of heat capacity**

Dulong–Petit Law of Heat Capacity of Solids

One of the earliest attempts to understand the heat capacity of solids was made by **Pierre Louis Dulong** and **Alexis Thérèse Petit** in 1819.

From experimental measurements on many solid elements, they observed that the **molar heat capacities of most solids are approximately the same at ordinary temperatures**.

This observation is known as the **Dulong–Petit law**.

Statement of Dulong–Petit Law

The **molar heat capacity at constant volume** of many solid elements is approximately

$$C_V \approx 3R$$

where

- R= gas constant $= 8.314\, J\, mol^{-1} K^{-1}$

Thus

$$C_V \approx 24.9\, J\, mol^{-1} K^{-1}$$

This value is nearly the same for many metallic elements at moderate and high temperatures.

Classical Explanation Using Equipartition Theorem

In a solid, atoms are arranged in a lattice and vibrate about their equilibrium positions.

Each atom can vibrate in **three independent directions**:

- x-direction

- y-direction
- z-direction

Thus every atom behaves approximately like **three independent harmonic oscillators**.

Energy of a Harmonic Oscillator

For a harmonic oscillator the energy consists of two quadratic terms:

1. kinetic energy
2. potential energy

$$E = \frac{p^2}{2m} + \frac{1}{2}kx^2$$

According to the **equipartition theorem**, each quadratic term contributes

$$\frac{1}{2}kT$$

to the average energy.

Therefore for one vibrational mode

$$E = \frac{1}{2}kT + \frac{1}{2}kT = kT$$

Total Energy of an Atom in a Solid

Since each atom has **three vibrational modes**, the average energy per atom becomes

$$E = 3kT$$

Internal Energy of One Mole of Solid

For one mole of atoms

$$U = 3NkT$$

where

$$Nk = R$$

Thus

$$U = 3RT$$

Heat Capacity of the Solid

Heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Substituting

$$U = 3RT$$

$$C_V = 3R$$

Thus the **molar heat capacity of a solid is approximately** $3R$.

This result is the **Dulong–Petit law**.

Agreement with Experiment

The law works well for many metals at ordinary temperatures.

Examples include:

Element	Experimental C_V (J mol^{-1} K^{-1})
Copper	~24.5
Silver	~25
Aluminium	~24

These values are close to

$$3R = 24.9\, J\, mol^{-1} K^{-1}$$

Limitations of the Dulong–Petit Law

Although successful for many metals, the law fails in several situations.

1. Low temperatures

Heat capacity decreases sharply as temperature approaches absolute zero.

2. Light elements

Elements such as

- carbon
- boron
- silicon

show much smaller heat capacities than $3R$.

3. Temperature dependence

Experimental heat capacity varies with temperature, whereas the classical theory predicts a constant value.

Reason for Failure

The Dulong–Petit law is based on **classical mechanics**, which assumes that vibrational energy can vary continuously.

However, atomic vibrations are actually **quantized**, as described by quantum mechanics.

This limitation led to the development of the **Einstein theory of heat capacity**, which introduced quantum ideas into the description of lattice vibrations.

Einstein Theory of Heat Capacity of Solids

The classical **Dulong–Petit law** predicts that the molar heat capacity of solids should be constant and equal to $3R$. This prediction agrees reasonably well with experiments at high temperatures, but it fails at low temperatures where the heat capacity of solids decreases significantly.

To explain this behaviour, **Albert Einstein** proposed a quantum theory of heat capacity in 1907. Einstein introduced the idea that the vibrational energy of atoms in a solid is **quantized**, applying the concepts of quantum mechanics to lattice vibrations.

Einstein's model successfully explained why the heat capacity of solids decreases at low temperatures and represents the first quantum theory of the thermal properties of solids.

Basic Idea of Einstein's Model

In a crystalline solid, atoms are arranged in a regular lattice and vibrate about their equilibrium positions.

Einstein assumed that:

- Each atom behaves as an **independent quantum harmonic oscillator**.
- All atoms vibrate with the **same frequency** ν.
- The vibrational energy is **quantized** according to quantum mechanics.

Since an atom can vibrate in three perpendicular directions (x, y, and z), each atom has **three independent vibrational degrees of freedom**.

Thus a crystal containing N atoms can be considered as a system of $3N$ **independent harmonic oscillators**.

Energy Levels of a Quantum Harmonic Oscillator

According to quantum mechanics, the energy levels of a harmonic oscillator are

$$\epsilon_n = \left(n + \frac{1}{2}\right) h\nu$$

where

- $n = 0,1,2, \ldots$
- h= Planck's constant
- ν= frequency of vibration.

Thus the vibrational energy is **quantized** and increases in discrete steps of $h\nu$.

Even at absolute zero, the oscillator possesses energy

$$\frac{1}{2}h\nu$$

This is called the **zero-point energy**.

Vibrational Partition Function

The vibrational partition function for a quantum harmonic oscillator is

$$q_{vib} = \sum_{n=0}^{\infty} e^{-\frac{\epsilon_n}{kT}}$$

Substituting the energy levels

$$q_{vib} = \sum_{n=0}^{\infty} e^{-\frac{\left(n+\frac{1}{2}\right)h\nu}{kT}}$$

Factor out the ground-state term

$$q_{vib} = e^{-\frac{h\nu}{2kT}} \sum_{n=0}^{\infty} \left(e^{-\frac{h\nu}{kT}}\right)^n$$

The summation is a geometric series

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

where

$$r = e^{-\frac{h\nu}{kT}}$$

Thus the partition function becomes

$$q_{vib} = \frac{e^{-\frac{h\nu}{2kT}}}{1-e^{-\frac{h\nu}{kT}}}$$

Internal Energy of the Solid

For one atom there are three vibrational modes, therefore the molecular partition function is

$$q = (q_{vib})^3$$

Using statistical thermodynamics, the internal energy is given by

$$U = NkT^2\left(\frac{\partial \ln q}{\partial T}\right)$$

Substituting the partition function and differentiating gives

$$U = 3N\left(\frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} + \frac{1}{2}h\nu\right)$$

The term

$$\frac{1}{2}h\nu$$

represents the **zero-point energy** of the oscillator.

Since this term is independent of temperature, it does **not contribute to heat capacity**.

Thus the temperature-dependent internal energy is

$$U = 3N\frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$$

Heat Capacity of the Solid

The heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Differentiating the expression for internal energy gives

$$C_V = 3Nk\left(\frac{h\nu}{kT}\right)^2 \frac{e^{\frac{h\nu}{kT}}}{\left(e^{\frac{h\nu}{kT}} - 1\right)^2}$$

For one mole of atoms, the expression becomes

$$C_V = 3R\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\frac{\theta_E}{T}}}{\left(e^{\frac{\theta_E}{T}} - 1\right)^2}$$

where

$$\theta_E = \frac{h\nu}{k}$$

is called the **Einstein temperature**.

The Einstein temperature represents a characteristic temperature corresponding to the vibrational frequency of the crystal lattice.

Predictions of Einstein Theory

High Temperature Limit

When the temperature is much higher than the Einstein temperature

$$T \gg \theta_E$$

the quantity

$$\frac{h\nu}{kT}$$

becomes very small. Using the approximation $e^x \approx 1 + x$, the heat capacity expression reduces to

$$C_V \approx 3Nk$$

or for one mole

$$C_V \approx 3R$$

Thus Einstein's theory reduces to the **Dulong–Petit law at high temperatures**.

Low Temperature Limit

When the temperature is much lower than the Einstein temperature

$$T \ll \theta_E$$

the exponential term becomes very large and the heat capacity decreases rapidly.

Einstein theory predicts

$$C_V \propto e^{-\frac{\theta_E}{T}}$$

Thus the heat capacity approaches zero as temperature approaches absolute zero.

Limitations of Einstein Theory

Although Einstein's theory improved the classical description, it still has certain limitations.

1. Einstein assumed that all atoms vibrate with the **same frequency**, which is not realistic.
2. At very low temperatures the experimental heat capacity varies as

$$C_V \propto T^3$$

while Einstein theory predicts an exponential decrease.

3. The model does not consider **collective lattice vibrations**.

These limitations led to the development of the **Debye theory**, which provides a more accurate description of the heat capacity of solids.

Importance of Einstein Theory

Einstein's model was a major milestone in statistical physics because it

- introduced **quantum ideas into the study of solids**
- explained the decrease of heat capacity at low temperatures
- provided the foundation for the more accurate **Debye theory**.

Einstein's work demonstrated how quantum mechanics can be used to explain macroscopic thermodynamic properties.

Debye Theory of Heat Capacity of Solids

Although the Einstein theory successfully introduced quantum ideas into the study of heat capacity of solids, it still had limitations. The main difficulty was the assumption that all atoms in the crystal vibrate with the **same frequency**. In reality, atoms in a solid vibrate with a **range of frequencies**.

To overcome this limitation, **Peter Debye** proposed a more accurate theory in 1912. Debye treated the vibrations of atoms in a crystal as **collective lattice vibrations**, rather than independent oscillations.

These collective vibrations are now called **phonons**, which are quantized modes of vibration of the crystal lattice.

The Debye model provides a much better description of the heat capacity of solids, particularly at **low temperatures**.

Basic Idea of Debye Model

In a crystal containing N atoms, each atom has three vibrational degrees of freedom. Thus the total number of vibrational modes is

$$3N$$

Debye assumed that these vibrations propagate through the crystal as **elastic waves**, similar to sound waves in a solid.

Thus instead of treating atoms as independent oscillators, Debye treated the entire crystal as a system supporting **standing waves of vibration**.

Vibrational Frequency Spectrum

Unlike Einstein's model, which assumes a single vibration frequency, Debye assumed that the lattice vibrations have a **continuous range of frequencies**.

The frequencies vary from

$$0 \leq \nu \leq \nu_D$$

where

$$\nu_D$$

is called the **Debye frequency**.

The Debye frequency represents the **maximum vibrational frequency allowed in the crystal**.

Debye Temperature

It is convenient to express the Debye frequency in terms of temperature using

$$\theta_D = \frac{h\nu_D}{k}$$

where

- h= Planck's constant
- k= Boltzmann constant.

The quantity $\theta_D$is called the **Debye temperature**.

The Debye temperature is a characteristic property of a solid and is related to the stiffness of the crystal lattice.

Density of Vibrational States

To calculate the heat capacity, Debye determined the number of vibrational modes between frequencies $\nu$and $\nu + d\nu$.

The number of modes is given by

$$g(\nu)d\nu = \frac{9N}{\nu_D^3}\nu^2 d\nu$$

This expression is known as the **density of vibrational states**.

It ensures that the total number of vibrational modes equals

$$3N$$

which is required for a crystal containing Natoms.

Energy of Lattice Vibrations

Each vibrational mode behaves like a **quantum harmonic oscillator**.

The average energy of a mode of frequency $\nu$is

$$\epsilon = \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$$

The total internal energy of the solid is obtained by integrating over all vibrational frequencies:

$$U = \int_0^{\nu_D} \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} g(\nu)d\nu$$

Substituting the density of states

$$U = \int_0^{\nu_D} \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} \frac{9N}{\nu_D^3}\nu^2 d\nu$$

Debye Heat Capacity Formula

After changing variables and performing the integration, Debye obtained the heat capacity

$$C_V = 9Nk\left(\frac{T}{\theta_D}\right)^3 \int_0^{\frac{\theta_D}{T}} \frac{x^4 e^x}{(e^x-1)^2}\,dx$$

For one mole of atoms this becomes

$$C_V = 9R\left(\frac{T}{\theta_D}\right)^3 \int_0^{\frac{\theta_D}{T}} \frac{x^4 e^x}{(e^x-1)^2}\,dx$$

This equation is known as the **Debye heat capacity formula**.

Predictions of Debye Theory

High Temperature Limit

When

$$T \gg \theta_D$$

the integral approaches a constant value and

$$C_V \approx 3R$$

Thus the Debye model reduces to the **Dulong–Petit law at high temperatures**.

Low Temperature Limit

When

$$T \ll \theta_D$$

Debye showed that

$$C_V \propto T^3$$

Thus

$$C_V = \frac{12\pi^4}{5} Nk\left(\frac{T}{\theta_D}\right)^3$$

This result agrees extremely well with experimental measurements of heat capacity at low temperatures.

Advantages of Debye Model

The Debye theory successfully explains several important features:

1. The decrease of heat capacity at low temperatures.
2. The correct T^3dependence of heat capacity at very low temperatures.
3. The agreement with the Dulong–Petit law at high temperatures.

Thus Debye theory provides a much more accurate description of the heat capacity of solids than the Einstein model.

Significance of Debye Theory

The Debye theory represents a major success of statistical thermodynamics and quantum mechanics. It demonstrates how the thermal properties of solids arise from the quantized vibrational motion of atoms in a crystal lattice.

The concept of **phonons** introduced by Debye later became fundamental in solid-state physics and plays an important role in understanding many physical properties of solids, including thermal conductivity and lattice dynamics.

Comparison of Dulong–Petit, Einstein, and Debye Theories of Heat Capacity of Solids

The three models—classical **Dulong–Petit law**, **Einstein theory**, and **Debye theory**—represent successive improvements in understanding the heat capacity of solids. Each theory introduced new physical ideas to explain experimental observations more accurately.

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1. Dulong–Petit Law (Classical Theory)

The **Dulong–Petit law** is based on classical mechanics and the equipartition theorem.

Key Idea

Atoms in a solid vibrate about their equilibrium positions and possess **three degrees of freedom**. According to the equipartition theorem, each quadratic degree of freedom contributes $\frac{1}{2}kT$ to the average energy.

Since a harmonic oscillator has both kinetic and potential energy terms, each vibrational mode contributes kT.

Thus the total internal energy per atom becomes

$$E = 3kT$$

For one mole

$$U = 3RT$$

Therefore

$$C_V = 3R$$

Success

- Works well for many metals at **high temperatures**.

Failure

- Cannot explain the decrease of heat capacity at **low temperatures**.
- Ignores the **quantized nature of vibrations**.

2. Einstein Theory (Quantum Oscillator Model)

Einstein introduced **quantum theory** into the study of lattice vibrations.

Key Idea

Each atom in the crystal behaves as a **quantum harmonic oscillator** with a single characteristic frequency.

Energy levels are

$$\epsilon_n = \left(n + \frac{1}{2}\right) h\nu$$

Using the partition function for the oscillator, Einstein derived the heat capacity expression

$$C_V = 3R\left(\frac{\theta_E}{T}\right)^2 \frac{e^{\frac{\theta_E}{T}}}{\left(e^{\frac{\theta_E}{T}} - 1\right)^2}$$

where

$$\theta_E = \frac{h\nu}{k}$$

is the **Einstein temperature**.

Success

- Explains why heat capacity decreases at **low temperatures**.
- Reduces to $3R$ at **high temperatures**.

Limitation

- Assumes **all atoms vibrate with the same frequency**, which is unrealistic.
- Predicts incorrect behaviour at very low temperatures.

Einstein theory predicts

$$C_V \propto e^{-\frac{\theta_E}{T}}$$

but experiments show

$$C_V \propto T^3$$

3. Debye Theory (Lattice Vibration Model)

The **Debye model** provides a more accurate description of lattice vibrations.

Key Idea

Instead of independent oscillators, Debye treated vibrations as **collective waves propagating through the crystal lattice**.

These quantized lattice vibrations are called **phonons**.

The crystal supports a **continuous spectrum of frequencies** up to a maximum value called the **Debye frequency**.

The heat capacity is given by

$$C_V = 9R\left(\frac{T}{\theta_D}\right)^3 \int_0^{\frac{\theta_D}{T}} \frac{x^4 e^x}{(e^x - 1)^2} dx$$

where

$$\theta_D = \frac{h\nu_D}{k}$$

is the **Debye temperature**.

Success

- Correctly predicts the **low-temperature behaviour**:

$$C_V \propto T^3$$

- Approaches $3R$at high temperatures.

Limitation

- Mathematical treatment is more complex.
- Approximates the real phonon spectrum.

Summary

Feature	**Dulong–Petit Law**	**Einstein Model**	**Debye Model**
Nature of theory	Classical	Quantum	Quantum
Treatment of vibrations	Classical oscillations	Independent quantum oscillators	Collective lattice vibrations
Frequency assumption	Not considered	Single frequency	Continuous spectrum
Characteristic temperature	None	Einstein temperature θ_E	Debye temperature θ_D
High-temperature behaviour	$C_V = 3R$	$C_V \to 3R$	$C_V \to 3R$
Low-temperature behaviour	Incorrect	$C_V \propto e^{-\frac{\theta_E}{T}}$	$C_V \propto T^3$

Feature	Dulong–Petit Law	Einstein Model	Debye Model
Agreement with experiment	Poor at low T	Moderate	Excellent

The three models represent successive improvements in understanding the heat capacity of solids.

- The **Dulong–Petit law** provides a simple classical prediction valid at high temperatures.
- The **Einstein model** introduced quantum mechanics and explained the decrease of heat capacity at low temperatures.
- The **Debye theory** provided the most accurate description by considering the full spectrum of lattice vibrations.

Thus the Debye theory remains the most successful statistical mechanical model for explaining the heat capacity of solids.

Equipartition Theorem and Its Derivation Using Partition Functions

Degrees of Freedom

A **degree of freedom** is an independent variable required to specify the motion or configuration of a system.

In molecules, degrees of freedom correspond to independent ways in which energy can be stored.

Typical molecular motions:

Type of motion	Degrees of freedom
Translation	motion along x, y, zaxes
Rotation	rotation about molecular axes
Vibration	oscillation of atoms about equilibrium positions

For example, a molecule moving freely in space has **three translational degrees of freedom** corresponding to motion along the three spatial directions.

Quadratic Degrees of Freedom

A **quadratic degree of freedom** is one in which the energy depends on the **square of a coordinate or momentum**.

General forms:

$$E = ax^2$$

or

$$E = bp^2$$

where

- x= coordinate
- p= momentum
- a, b= constants.

A **quadratic degree of freedom** is one whose contribution to the Hamiltonian has the form

$$E = ax^2 \text{or} E = bp^2$$

where $x$is a generalized coordinate and $p$its conjugate momentum.

Equipartition applies **only when the Hamiltonian contains such quadratic terms**.

The term **quadratic** arises because the variable appears **squared**.

Equipartition theorem applies specifically to such **quadratic terms in the Hamiltonian**.

Examples of Quadratic Energy Terms

Translational Motion

The kinetic energy of a particle is

$$E = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

Each term contains the square of a momentum component.

Thus there are **three quadratic degrees of freedom**.

Rotational Motion

Rotational energy is

$$E = \frac{L^2}{2I}$$

where $L$is angular momentum and $I$is the moment of inertia.

Again the energy depends on the **square of angular momentum**.

Vibrational Motion

For a harmonic oscillator

$$E = \frac{p^2}{2m} + \frac{1}{2}kx^2$$

Thus one vibrational mode contains **two quadratic terms**:

Term	**Meaning**
$p^2/2m$	kinetic energy
$kx^2/2$	potential energy

The Equipartition Theorem

The **equipartition theorem** states:

For a system in thermal equilibrium, each independent quadratic term in the Hamiltonian contributes an average energy

$$\frac{1}{2}kT$$

per molecule.

For one mole of molecules this becomes

$$\frac{1}{2}RT$$

Thus thermal energy is **equally partitioned among quadratic degrees of freedom**.

Proof Using Partition Functions

Consider a system whose energy contains a quadratic coordinate

$$E = ax^2$$

The contribution of this coordinate to the partition function is

$$q = \int_{-\infty}^{\infty} e^{-E/kT}\, dx$$

Substituting the energy expression

$$q = \int_{-\infty}^{\infty} e^{-ax^2/kT}\, dx$$

This integral is a **Gaussian integral**

$$\int_{-\infty}^{\infty} e^{-bx^2}\, dx = \sqrt{\frac{\pi}{b}}$$

Therefore

$$q = \sqrt{\frac{\pi kT}{a}}$$

Temperature Dependence

Taking logarithm

$$\ln\, q = \frac{1}{2}\ln\, T + \text{constant}$$

Average Energy

In statistical thermodynamics

$$\langle E \rangle = kT^2 \left(\frac{\partial \ln\, q}{\partial T}\right)$$

Differentiating

$$\frac{\partial \ln\, q}{\partial T} = \frac{1}{2T}$$

Thus

$$\langle E \rangle = kT^2 \left(\frac{1}{2T}\right)$$

$$\langle E \rangle = \frac{1}{2}kT$$

This proves the **equipartition theorem**.

Example: Ideal Gas

For an ideal gas molecule

$$E = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

There are **three quadratic momentum terms**.

Thus

$$E = \frac{3}{2}kT$$

per molecule.

For one mole

$$U = \frac{3}{2}RT$$

Derivation from the Classical Partition Function

For N molecules of an ideal gas

$$Q = \frac{1}{N!\,h^{3N}} \int e^{-H/kT} d^{3N}p\, d^{3N}q$$

Evaluating the integrals gives

$$Q = \frac{V^N}{N!\,h^{3N}} (2\pi mkT)^{3N/2}$$

Taking logarithm

$$\ln\, Q = \frac{3N}{2} \ln\, T + \text{constant}$$

The internal energy is

$$U = kT^2 \left(\frac{\partial \ln\, Q}{\partial T}\right)$$

which gives

$$U = \frac{3}{2}NkT$$

confirming that each quadratic term contributes $\frac{1}{2}kT$.

Energy Contributions of Molecular Motions

Motion	**Quadratic terms**	**Energy per molecule**
Translation	3	$3/2kT$

Motion	Quadratic terms	Energy per molecule
Rotation (linear molecule)	2	kT
Rotation (non-linear molecule)	3	$3/2kT$
One vibrational mode	2	kT

Heat Capacity from Equipartition

Heat capacity at constant volume is

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

If a molecule has f quadratic degrees of freedom,

$$U = \frac{f}{2}RT$$

Therefore

$$\boxed{C_V = \frac{f}{2}R}$$

Thus heat capacity depends directly on the **number of quadratic degrees of freedom**.

Examples

Monatomic Gas

A monatomic gas molecule can only store energy as translational kinetic energy.

Energy expression:

$$E = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

Thus there are 3 quadratic terms.

$$f = 3$$

Internal energy

$$U = \frac{3}{2}RT$$

Heat capacity

$$C_V = \frac{3}{2}R$$

This agrees very well with experimental values for gases such as He, Ne, Ar.

Diatomic Molecules (Without Vibrations)

A diatomic molecule has

Translational degrees 3

Rotational degrees 2 (rotation about two axes perpendicular to bond)

Thus

$$f = 5$$

Internal energy

$$U = \frac{5}{2}RT$$

Heat capacity

$$C_V = \frac{5}{2}R$$

Examples: N_2, O_2, CO at ordinary temperatures.

Diatomic Molecules (Including Vibrations)

Each vibrational mode contributes two quadratic terms:

- kinetic energy $p^2/2m$
- potential energy $kx^2/2$

Thus

$$f = 7$$

Internal energy

$$U = \frac{7}{2}RT$$

Heat capacity

$$C_V = \frac{7}{2}R$$

Vibrational Contribution Is Often Not Observed

At ordinary temperatures

- vibrational energy levels are widely spaced
- molecules cannot easily access excited vibrational states

Thus vibrations are “frozen out”.

Therefore the observed heat capacity of diatomic gases is usually closer to

$$\frac{5}{2}R$$

rather than

$$\frac{7}{2}R$$

Molecule	f	U	C_V
Monatomic gas	3	$3/2RT$	$3/2R$
Diatomic (no vibration)	5	$5/2RT$	$5/2R$
Diatomic (with vibration)	7	$7/2RT$	$7/2R$

Temperature Dependence of Degrees of Freedom

In real molecules, not all degrees of freedom are active at all temperatures because energy levels are **quantized**.

As temperature increases, additional motions become accessible.

Temperature Active motion C_V

Very low translation only $3/2R$

Moderate translation + rotation $5/2R$

High translation + rotation + vibration $7/2R$

Vibrational modes usually require **high temperatures** because their energy level spacing is large.

Summary

The **equipartition theorem** states that every quadratic degree of freedom contributes

$$\frac{1}{2}kT$$

to the average energy per molecule.

This result follows directly from the **temperature dependence of the partition function** and provides a powerful method for predicting **internal energies and heat capacities of gases** from the microscopic structure of molecules.

Equilibrium Constants in Terms of Partition Functions

Consider the general gas phase chemical reaction represented by

$$\nu_A A + \nu_B B \rightleftharpoons \nu_C C + \nu_D D$$

where **A, B, C and D** are the reactants and products of the reaction, and $\nu_A$is the stoichiometric coefficient of chemical A, $\nu_B$is the stoichiometric coefficient of chemical B, and so on. Each of the gases involved in the reaction will eventually reach an equilibrium concentration when the forward and reverse reaction rates become equal. The distribution of reactants to products at the equilibrium point is represented by the equilibrium constant K_c:

$$Kc = \frac{[C]^{\nu C}[D]^{\nu D}}{[A]^{\nu A}[B]^{\nu B}}$$

If the system is not at equilibrium, a shift in the number of reactants and products will occur to lower the overall energy of the system. The difference in the energy of the system at this non-equilibrated point and the energy of the system at equilibrium for any particular species is termed **chemical potential**. When both temperature and volume are constant for both points aforementioned, the chemical potential of species $i$is expressed by the equation

$$\mu_i = \left(\frac{\partial A}{\partial N_i}\right)_{T,V,N_{j\neq i}}$$

where $A$is the **Helmholtz energy**, and $N_i$is the number of molecules of species i.

The Helmholtz energy can be determined as a function of the total partition function Q:

$$A = -k_B T \ln\ Q$$

where $k_B$is the Boltzmann constant and $T$is the temperature of the system.

The total partition function is given by

$$Q = \frac{q_i(V,T)^{N_i}}{N_i!}$$

where $q_i$is the molecular partition function of chemical species i.

Substituting the molecular partition function into the equation for Helmholtz energy yields:

$$A = -k_B T \ln\left(\frac{q_i(V,T)^{N_i}}{N_i!}\right)$$

and then further substituting this equation into the definition of chemical potential (Equation 26.8.2) yields:

$$\mu_i = \left(\frac{\partial}{\partial N_i}\left[-k_B T \ln\left(\frac{q_i(V,T)^{N_i}}{N_i!}\right)\right]\right)_{T,V,N_{j\neq i}}$$

Rearranging this equation the following derivative can be set-up:

$$\mu_i = -k_B T \left[\left(\frac{\partial N_i \ln\ q_i(V,T)}{\partial N_i} \right) - \left(\frac{\partial \ln\ (N_i!)}{\partial N_i} \right) \right]_{T,V,N_{j \neq i}}$$

From this point **Stirling's approximation**

$$\ln\ N! = N \ln\ N - N$$

can be substituted into the derivative to yield:

$$\mu_i = -k_B T \left[\left(\frac{\partial N_i \ln\ q_i(V,T)}{\partial N_i} \right) - \left(\frac{\partial (N_i \ln\ N_i - N_i)}{\partial N_i} \right) \right]_{T,V,N_{j \neq i}}$$

From here the derivative can be rearranged and solved:

$$\mu_i = -k_B T \left[\ln\ (q_i(V,T)) \frac{\partial N_i}{\partial N_i} + N_i \frac{\partial \ln\ (q_i(V,T))}{\partial N_i} - \ln\ (N_i) \frac{\partial N_i}{\partial N_i} - N_i \frac{\partial \ln\ (N_i)}{\partial N_i} + \frac{\partial N_i}{\partial N_i} \right]_{T,V,N_{j \neq i}}$$

$$= -k_B T \left(\ln\ (q_i(V,T)) + 0 - \ln\ (N_i) - N_i \frac{1}{N_i} + 1 \right)$$

$$= -k_B T (\ln(q_i(V,T)) - \ln(N_i))$$

$$= -k_B T \ln \left(\frac{q_i(V,T)}{N_i} \right)$$

(26.8.3)

Helmholtz Energy with Respect to Equilibrium Shifts

A variable $\lambda$is then defined such that

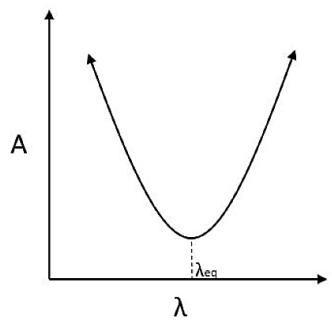
$$dN_j = \nu_j d\lambda$$

where $j = A, B, C, D$and $\nu_j$is taken to be positive for products and negative for reactants.

A change in λtherefore corresponds to a change in the concentrations of the reactants and products.

Thus, at equilibrium

$$\left(\frac{\partial A}{\partial \lambda} \right)_{T,V} = 0$$



Helmholtz Energy with respect to Equilibrium Shifts

From classical thermodynamics, the total differential of $A$is

$$dA = -S\,dT - p\,dV + \sum_j \mu_j\,dN_j$$

For a reaction at a fixed volume and temperature (such as in the canonical ensemble),

$$dT = 0, dV = 0$$

Therefore,

$$dA = \sum_j \mu_j\,dN_j$$

Substituting $dN_j = \nu_j d\lambda$

$$dA = \sum_j \mu_j\,\nu_j d\lambda$$

$$dA = d\lambda \sum_j \mu_j\,\nu_j$$

At equilibrium

$$\sum_j \mu_j\,\nu_j = 0$$

Substituting the expanded form of chemical potential (Equation 26.8.3):

$$-k_B T \sum_j \ln\left(\frac{q_j}{N_j}\right)\nu_j = 0$$

$$\sum_j \ln\left(\frac{q_j}{N_j}\right)\nu_j = 0$$

$$\sum_j \nu_j \ln(q_j) - \sum_j \nu_j \ln(N_j) = 0$$

For the reaction in Equation (26.8.1):

$$\nu_C \ln\,q_C - \nu_C \ln\,N_C + \nu_D \ln\,q_D - \nu_D \ln\,N_D - \nu_A \ln\,q_A + \nu_A \ln\,N_A - \nu_B \ln\,q_B + \nu_B \ln\,N_B = 0$$

This equation simplifies to

$$\frac{q_C^{\nu_C} q_D^{\nu_D}}{q_A^{\nu_A} q_B^{\nu_B}} = \frac{N_C^{\nu_C} N_D^{\nu_D}}{N_A^{\nu_A} N_B^{\nu_B}}$$

Dividing all terms by volume and using the number density relationship

By dividing all terms by volume, and noting the relationship $\frac{N_A}{V} = \# \frac{molecules}{volume} = \rho_A = [A]$ where ρ is referred to a numberdensity, the following equation is obtained

$$\rho_i = \frac{N_i}{V}$$

we obtain

$$K_c = \frac{\rho_C^{\nu_C} \rho_D^{\nu_D}}{\rho_A^{\nu_A} \rho_B^{\nu_B}} = \frac{\left(\frac{q_C}{V}\right)^{\nu_C} \left(\frac{q_D}{V}\right)^{\nu_D}}{\left(\frac{q_A}{V}\right)^{\nu_A} \left(\frac{q_B}{V}\right)^{\nu_B}}$$

(26.8.4)

Hence, knowledge of the **molecular partition function** for all species in the reaction allows calculation of the equilibrium constant.

Example: Reacting Diatomic Molecules

Calculate the equilibrium constant $(K_c)$for the reaction of $H_2(g)$and $Cl_2(g)$at **650 K**.

$$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$$

Solution

We will use **Equation 26.8.4** that uses Molecular Partition Functions to evaluate K_c.

$$K_c(T) = \frac{\left(\frac{q_{HCl}}{V}\right)^2}{\left(\frac{q_{H_2}}{V}\right)\left(\frac{q_{Cl_2}}{V}\right)}$$

This equation is expanded into the rotational, vibrational, translational and electronic molecular partition functions of each species.

$$K_c(T) = \frac{\left(\frac{q_{trans\ HCl}\ q_{rot\ HCl}\ q_{vib\ HCl}\ q_{elec\ HCl}}{V}\right)^2}{\left(\frac{q_{trans\ H_2}\ q_{rot\ H_2}\ q_{vib\ H_2}\ q_{elec\ H_2}}{V}\right)\left(\frac{q_{trans\ Cl_2}\ q_{rot\ Cl_2}\ q_{vib\ Cl_2}\ q_{elec\ Cl_2}}{V}\right)}$$

A simple problem solving strategy for finding equilibrium constants via statistical mechanics is to separate the equation into the molecular partition functions of each of the reactant and product species, solve for each one, and recombine them to arrive at a final answer (e.g., for HCl).

$$\frac{q_{HCl}}{V} = \underbrace{\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}}_{\text{translation}} \times \underbrace{\frac{2k_B T\ \mu r_e^2}{\sigma h^2}}_{\text{rotation}} \times \underbrace{\frac{1}{1 - \exp\left(\frac{-h\nu}{k_B T}\right)}}_{\text{vibration}} \times \underbrace{g_1 \exp\left(\frac{D_0}{k_B T}\right)}_{\text{electronic}}$$

To simplify the calculations of molecular partition functions, the characteristic temperature of rotation (Θ_r)and vibration (Θ_v)are used.

$$\frac{q_{HCl}}{V} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} \times \frac{T}{\sigma \Theta_r} \times \frac{1}{1 - \exp\left(\frac{-\Theta_v}{T}\right)} \times g_1 \exp\left(\frac{D_0}{RT}\right)$$

(26.8.6)

These values are constants that incorporate the physical constants found in the rotational and vibrational partition functions of the molecules. Tabulated values $\Theta_r$and $\Theta_v$for select molecules can be found here.

Table

Species	Θ_v**(K)**	Θ_r**(K)**	D_0**(kJ mol⁻¹)**
Cl2 (g)	6125	0.351	239.0
H2 (g)	808	87.6	431.9
HCl (g)	4303	15.2	427.7

We need to evaluate **Equation 26.8.6** for each species then then evaluate **Equation 26.8.5** directly.

For HCl:

$$\frac{q_{HCl}}{V} = \left(\frac{2\pi(2.1957 \times 10^{-24}\ kg)(1.38065 \times 10^{-23}\ JK^{-1})(650\ K)}{(6.62607 \times 10^{-34}\ Js)^2}\right)^{3/2} \times \frac{650\ K}{(1)(15.2\ K)}$$
$$\times \frac{1}{1 - \exp\left(\frac{-4303\ K}{650\ K}\right)}$$
$$\times (1)\exp\left(\frac{427{,}700\ J\ mol^{-1}}{(8.3145\ J\ K^{-1} mol^{-1})(650\ K)}\right)$$

$$\frac{q_{HCl}}{V} = (1.4975 \times 10^{35}\ m^{-3})(42.76)(1.0013)(2.3419 \times 10^{34})$$

For H_2:

$$\frac{q_{H_2}}{V} = \left(\frac{2\pi(1.2140 \times 10^{-25}\,kg)(1.38065 \times 10^{-23}\,JK^{-1})(650\,K)}{(6.62607 \times 10^{-34}\,Js)^2}\right)^{3/2} \times \frac{650\,K}{(2)(87.6\,K)}$$
$$\times \frac{1}{1 - \exp\left(\frac{-808\,K}{650\,K}\right)} \times (1)\exp\left(\frac{431{,}900\,J\,mol^{-1}}{(8.3145\,J\,K^{-1}mol^{-1})(650\,K)}\right)$$

$$\frac{q_{H_2}}{V} = (1.9468 \times 10^{33}\,m^{-3})(3.71)(1.41)(5.0942 \times 10^{34})$$

For Cl_2:

$$\frac{q_{Cl_2}}{V} = \left(\frac{2\pi(4.2700 \times 10^{-24}\,kg)(1.38065 \times 10^{-23}\,JK^{-1})(650\,K)}{(6.62607 \times 10^{-34}\,Js)^2}\right)^{3/2} \times \frac{650\,K}{(2)(0.351\,K)}$$
$$\times \frac{1}{1 - \exp\left(\frac{-6125\,K}{650\,K}\right)}$$
$$\times (1)\exp\left(\frac{239{,}000\,J\,mol^{-1}}{(8.3145\,J\,K\,mol^{-1})(650\,K)}\right)$$

$$\frac{q_{Cl_2}}{V} = (5.4839 \times 10^{35}\,m^{-3})(925.9)(1.00)(1.606 \times 10^{19})$$

Combining the terms from each species, the following expression is obtained:

$$K_c = \frac{(1.9468 \times 10^{33}\,m^{-3})^2}{(1.9468 \times 10^{33}\,m^{-3})(5.4839 \times 10^{35}\,m^{-3})} \times \frac{(42.76)^2}{(3.71)(925.9)} \times \frac{(1.0013)^2}{(1.41)(1.00)}$$
$$\times \frac{(2.3419 \times 10^{34})^2}{(1.606 \times 10^{19})(5.0942 \times 10^{34})}$$
$$= (0.003550)(0.1333)(0.711)(6.7037 \times 10^{14})$$
$$= (2.26 \times 10^{11})$$

At **650 K**, the reaction between **H_2(g)** and **Cl_2(g)** proceeds spontaneously towards the products. From a statistical mechanics point of view, the product **HCl(g)** molecule has more states accessible to it than the reactant species. The spontaneity of this reaction is largely due to the electronic partition function: two very strong **H–Cl bonds** are formed at the expense of a very strong **H–H bond** and a relatively weak **Cl–Cl bond**.

Calculate the equilibrium constant $K_c$for the reaction of **H_2(g)** and **Cl_2(g)** at **650 K**.

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

Solution

We will use Equation **26.8.4** that uses molecular partition functions to evaluate K_c:

$$K_c(T) = \frac{(q_{HCl}/V)^2}{(q_{H_2}/V)(q_{Cl_2}/V)}$$

(26.8.5)

This equation is expanded into the **rotational, vibrational, translational and electronic molecular partition functions** of each species.

A simple problem solving strategy for finding equilibrium constants via statistical mechanics is to separate the equation into the molecular partition functions of each of the reactant and product species, solve for each one, and recombine them to arrive at a final answer.

To simplify the calculations of molecular partition functions, the **characteristic temperatures** of rotation $\theta_r$and vibration $\theta_v$are used.

These values are constants that incorporate the physical constants found in the rotational and vibrational partition functions of the molecules.

Example table:

Species	$\theta_r(K)$	$\theta_v(K)$	D_0 **(kJ mol^{-1})**
Cl_2(g)	6125	0.351	239.0
H_2(g)	808	87.6	431.9
HCl(g)	4303	15.2	427.7

The partition function per unit volume becomes

$$\frac{q}{V} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} \times \frac{T}{\sigma\theta_r} \times \frac{1}{1 - e^{-\theta_v/T}} \times g_i e^{D_0/(RT)}$$

(26.8.6)

After evaluating the partition functions for **HCl, H_2, and Cl_2**, substituting them into equation (26.8.5) gives

$$K_c \approx 2.26 \times 10^{11}$$

At **650 K**, the reaction between $H_2(g)$and $Cl_2(g)$proceeds spontaneously toward the products. From a statistical mechanics point of view, the product molecule **HCl** has more accessible states than the reactant species.

The spontaneity of the reaction is largely due to the **electronic partition function**: two very strong **H–Cl** bonds are formed at the expense of a very strong **H–H** bond and a relatively weak **Cl–Cl** bond.

Appendix A

Stirling's Approximation

A.1 Introduction

In statistical thermodynamics, factorial expressions such as $N!$frequently appear in the calculation of thermodynamic probability and entropy. For macroscopic systems the number of particles $N$is extremely large (typically of the order of 10^{23}), and direct evaluation of factorials becomes impractical.

To simplify such expressions, an approximation known as **Stirling's approximation** is used.

For large n,

$$\ln n! \approx n\ln n - n$$

This approximation is widely used in deriving statistical distribution laws such as the Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac distributions.

A.2 Logarithmic Form of the Factorial

The factorial of a positive integer $n$is

$$n! = 1 \times 2 \times 3 \times \cdots \times n$$

Taking the natural logarithm,

$$\ln n! = \ln 1 + \ln 2 + \ln 3 + \cdots + \ln n$$

Thus

$$\ln n! = \sum_{k=1}^{n} \ln k$$

This converts the factorial product into a summation.

A.3 Approximation of the Sum by an Integral

For large values of n, the summation

$$\sum_{k=1}^{n} \ln k$$

can be approximated by the area under the curve $y = \ln x$.

Thus the sum may be replaced approximately by an integral:

$$\sum_{k=1}^{n} \ln k \approx \int_{1}^{n} \ln x \, dx$$

A.4 Evaluation of the Integral

The integral of $\ln x$is

$$\ln \int x\, dx = \mathrm{x}\ln x - x$$

Evaluating between the limits $1$and n:

$$\int_1^n \ln x\, dx = [\mathrm{n}\ln n - n] - [1 \ln 1 - 1]$$

Since

$$\ln 1 = 0$$

we obtain

$$\int_1^n \ln x\, dx = \mathrm{n}\ln n - n + 1$$

For very large n, the constant term $+1$is negligible compared with $\mathrm{n}\ln n$.

Thus

$$\ln n! \approx \mathrm{n}\ln n - n$$

This is the **logarithmic form of Stirling's approximation**.

A.5 Improved Stirling Approximation

A more accurate form of Stirling's approximation is

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

Taking the logarithm,

$$\ln n! = \mathrm{n}\ln n - n + \frac{1}{2}\ln(2\pi n)$$

For very large n, the term

$$\frac{1}{2}\ln(2\pi n)$$

is small compared with $\mathrm{n}\ln n$, and is usually neglected.

Thus the simplified expression used in statistical thermodynamics is

$$\boxed{\ln n! \approx \mathrm{n}\ln n - n}$$

A.6 Importance in Statistical Thermodynamics

Stirling's approximation is essential in statistical mechanics because factorial terms frequently appear in expressions for **thermodynamic probability**.

For example, in Maxwell–Boltzmann statistics

$$W = \frac{N!}{\prod_i N_i\,!}$$

Using Stirling's approximation allows this expression to be simplified for large N, enabling the derivation of important results such as

- Maxwell–Boltzmann distribution
- Bose–Einstein distribution
- Fermi–Dirac distribution.

1. Statistical Thermodynamics – Formula Summary Sheet

1. Boltzmann Entropy Relation

$$S = k_B \ln W$$

where

- S= entropy
- k_B= Boltzmann constant
- W= thermodynamic probability (number of microstates)

---

2. Stirling Approximation

For large n:

$$\ln n! \approx n\ln n - n$$

More accurate form:

$$\ln n! \approx n\ln n - n + \frac{1}{2}\ln(2\pi n)$$

---

3. Boltzmann Distribution Law

Number of particles in energy level i:

$$N_i = N\frac{g_i e^{-\beta\epsilon_i}}{\sum g_i e^{-\beta\epsilon_i}}$$

where

$$\beta = \frac{1}{k_B T}$$

---

4. Molecular Partition Function

$$q = \sum_i g_i\, e^{-\frac{\epsilon_i}{k_B T}}$$

Physical meaning:

Partition function represents the effective number of thermally accessible states.

---

5. Relation Between Molecular and Molar Partition Function

For Nindistinguishable particles:

$$Q = \frac{q^N}{N!}$$

where

Q= canonical partition function of system

---

6. Thermodynamic Quantities from Partition Function

Helmholtz Free Energy

$$F = -k_B \, \mathrm{T} \ln Q$$

Internal Energy

$$U = -\frac{\partial}{\partial \beta} \ln Q$$

or

$$U = k_B T^2 \frac{\partial \ln Q}{\partial T}$$

Entropy

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

Pressure

$$P = k_B T \left(\frac{\partial \ln Q}{\partial V}\right)_T$$

Heat Capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

---

7. Equipartition Theorem

Each quadratic degree of freedom contributes

$$\frac{1}{2} k_B T$$

to the average energy.

Examples:

Degree of freedom	Contribution
translation	$\frac{3}{2\,kT}$
rotation (linear molecule)	kT
vibration	kT

---

8. Maxwell–Boltzmann Distribution

$$N_i = g_i e^{\alpha - \beta \epsilon_i}$$

or

$$N_i = N \frac{g_i e^{-\beta \epsilon_i}}{q}$$

---

9. Bose–Einstein Distribution

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{k_B T}} - 1}$$

---

10. Fermi–Dirac Distribution

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{k_B T}} + 1}$$

---

11. Unified Distribution Formula

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{k_B T}} + \sigma}$$

σ	Statistics
0	Maxwell–Boltzmann
-1	Bose–Einstein
+1	Fermi–Dirac

---

12. Bose–Einstein Condensation Temperature

$$T_c = \frac{2\pi\hbar^2}{mk_B}\left(\frac{N}{V\zeta\left(\frac{3}{2}\right)}\right)^{\frac{2}{3}}$$

---

13. Fermi Energy

$$E_F = \frac{\hbar^2 k_F^2}{2m}$$

where k_F is the **Fermi wave vector**.

---

2. Comparison of Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac Statistics

Feature	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Particle type	classical particles	bosons	fermions
Spin	any	integer spin	half-integer spin
Distinguishability	distinguishable	indistinguishable	indistinguishable
Occupation of state	unlimited	unlimited	maximum 1
Example particles	ideal gas molecules	photons, He-4 atoms	electrons, protons
Distribution law	$e^{-\frac{\epsilon}{kT}}$	$\frac{1}{e^{\frac{\epsilon-\mu}{kT}}-1}$	$\frac{1}{e^{\frac{\epsilon-\mu}{kT}}+1}$
High-temperature limit	valid	reduces to MB	reduces to MB
Important phenomenon	classical gases	Bose condensation	Fermi energy

Derivation Roadmap for MB, BE, and FD Statistics

Step 1 – Define the Physical System

Consider a system of N particles distributed among energy levels.

Each level i has

$$\epsilon_i = \text{energy of level}$$

$$g_i = \text{degeneracy}$$

$$N_i = \text{number of particles in level}$$

Constraints:

$$\sum_i N_i = N$$

$$\sum_i N_i \, \epsilon_i = U$$

These represent conservation of particles and total energy.

Step 2 – Write Thermodynamic Probability $\boldsymbol{W}$

The expression for W depends on the **nature of the particles**.

Maxwell–Boltzmann Statistics

Particles are **distinguishable**.

$$W = \frac{N!}{\prod N_i!} \prod g_i^{N_i}$$

Bose–Einstein Statistics

Particles are indistinguishable bosons.

$$W = \prod \frac{(g_i + N_i - 1)!}{(g_i - 1)! \, N_i!}$$

Fermi–Dirac Statistics

Particles are fermions obeying Pauli exclusion.

$$W = \prod \frac{g_i!}{(g_i - N_i)! \, N_i!}$$

Step 3 – Apply Boltzmann Entropy Relation

$$S = \mathrm{k}\ln W$$

Equilibrium corresponds to the **maximum entropy** or **maximum** W.

Thus we maximize

$$\ln W$$

Step 4 – Use Stirling Approximation

For large numbers:

$$\ln n! \approx \mathrm{n}\ln n - n$$

This simplifies the factorial expressions in W.

Step 5 – Maximize Thermodynamic Probability

We apply the condition:

$$d(\ln W) = 0$$

subject to the constraints

$$\sum dN_i = 0$$

$$\sum \epsilon_i dN_i = 0$$

Step 6 – Introduce Lagrange Multipliers

To include the constraints we write

$$d(\ln W + \alpha \sum N_i - \beta \sum N_i \epsilon_i) = 0$$

where

$$\alpha, \beta$$

are Lagrange multipliers.

Step 7 – Obtain the General Distribution

After differentiation:

$$\frac{\partial(\ln W)}{\partial N_i} + \alpha - \beta \epsilon_i = 0$$

Solving this equation gives the particle distribution.

Step 8 – Final Results

Maxwell–Boltzmann Distribution

$$N_i = g_i e^{\alpha - \beta \epsilon_i}$$

or

$$N_i = N \frac{g_i e^{-\beta \epsilon_i}}{\sum g_i e^{-\beta \epsilon_i}}$$

Bose–Einstein Distribution

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} - 1}$$

Fermi–Dirac Distribution

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + 1}$$

Step 9 – Unified Distribution Formula

All three statistics can be written in one equation:

$$N_i = \frac{g_i}{e^{\frac{\epsilon_i - \mu}{kT}} + \sigma}$$

σ	Statistics
0	Maxwell–Boltzmann
−1	Bose–Einstein
+1	Fermi–Dirac

Conceptual Flow (Students Should Remember)

Particle distribution problem

→ Write thermodynamic probability W

→ Apply $S = \mathrm{k} \ln W$

→ Use Stirling approximation

→ Maximize W with constraints

→ Obtain statistical distribution.

Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac Statistics-Derivation and Comparison

---

1. System Description

Consider a system of **N particles** distributed among energy levels ε_i.

Definitions

- n_i= number of particles in energy level i
- g_i= degeneracy of level i

Constraints

$$\sum_i n_i = N$$

$$\sum_i n_i \, \varepsilon_i = E$$

The equilibrium distribution is obtained by **maximizing the number of microstates** W subject to these constraints.

---

2. Particle Characteristics

Property	**Maxwell–Boltzmann**	**Bose–Einstein**	**Fermi–Dirac**
Particle type	Classical	Bosons	Fermions
Spin	Any	Integer (0,1,2,...)	Half-integer (1/2,3/2,...)
Distinguishability	Distinguishable	Indistinguishable	Indistinguishable
Wavefunction symmetry	—	Symmetric	Antisymmetric
State occupancy	Unlimited	Unlimited	Maximum 1

Examples

- MB: dilute ideal gas molecules
- BE: photons, phonons, ^4He atoms

- FD: electrons, protons, neutrons

---

3. Number of Microstates W

Statistics	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Number of Microstates W	$W \propto \prod_i \frac{g_i^{n_i}}{n_i!}$	$W = \prod_i \frac{(n_i + g_i - 1)!}{n_i!\,(g_i - 1)!}$	$W = \prod_i \frac{g_i!}{n_i!\,(g_i - n_i)!}$

For Maxwell–Boltzmann, $W = \frac{N!}{\prod_i n_i!} \prod_i g_i^{n_i}$. The factor $N!$ is constant during maximization and is therefore omitted:

$$W \propto \prod_i \frac{g_i^{n_i}}{n_i!}$$

4. Logarithm of Microstates

Maximization is easier using $\ln W$.

MB

$$\ln W = \sum_i [n_i \ln g_i - \ln n_i!]$$

BE

$$\ln W = \sum_i [\ln (n_i + g_i - 1)! - \ln n_i! - \ln (g_i - 1)!]$$

FD

$$\ln W = \sum_i [\ln g_i! - \ln n_i! - \ln (g_i - n_i)!]$$

---

5. Stirling Approximation

For large numbers

$$\ln n! \approx n \ln n - n$$

Applying this:

Maxwell–Boltzmann

$$\ln W = \sum_i [n_i \ln g_i - n_i \ln n_i + n_i]$$

Bose–Einstein

$$\ln W = \sum_i [(n_i + g_i - 1)\ln (n_i + g_i - 1) - n_i \ln n_i - (g_i - 1)\ln (g_i - 1)]$$

For large n_i and g_i:

$$n_i + g_i - 1 \approx n_i + g_i$$

$$g_i - 1 \approx g_i$$

Thus

$$\ln\ W = \sum_i [(n_i + g_i)\ln\,(n_i + g_i) - n_i \ln\ n_i - g_i \ln\ g_i]$$

Fermi–Dirac

$$\ln\ W = \sum_i [\,g_i \ln\ g_i - n_i \ln\ n_i - (g_i - n_i)\ln\,(g_i - n_i)]$$

---

6. Maximization Using Lagrange Multipliers

Maximize

$$\Phi = \ln\ W + \alpha(N - \textstyle\sum_i n_i) + \beta(E - \sum_i n_i\,\varepsilon_i)$$

Condition for equilibrium

$$\frac{\partial \Phi}{\partial n_i} = 0$$

---

7. Resulting Equations

Maxwell–Boltzmann

Derivative

$$\frac{\partial}{\partial n_i}(\ln\ W) = \ln\ g_i - \ln\ n_i$$

Condition

$$\ln\ g_i - \ln\ n_i - \alpha - \beta\varepsilon_i = 0$$

Bose–Einstein

$$\ln\,(n_i + g_i) - \ln\ n_i = \alpha + \beta\varepsilon_i$$

Fermi–Dirac

$$\ln\,(g_i - n_i) - \ln\ n_i = \alpha + \beta\varepsilon_i$$

---

8. Distribution Functions

Statistics	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Final distribution	$n_i = g_i e^{-(\alpha+\beta\varepsilon_i)}$	$n_i = \frac{g_i}{e^{\alpha+\beta\varepsilon_i} - 1}$	$n_i = \frac{g_i}{e^{\alpha+\beta\varepsilon_i} + 1}$

9. Thermodynamic Meaning of Multipliers

$$\beta = \frac{1}{kT}$$

$$\alpha = -\frac{\mu}{kT}$$

where

- k= Boltzmann constant
- T= absolute temperature
- μ= chemical potential

---

10. Final Distribution Laws

Statistics	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Final distribution	$n_i = g_i e^{-(\varepsilon_i-\mu)/kT}$	$n_i = \frac{g_i}{e^{(\varepsilon_i-\mu)/kT} - 1}$	$n_i = \frac{g_i}{e^{(\varepsilon_i-\mu)/kT} + 1}$

---

11. Mean Occupation Number

Average number of particles per state

$$f(\varepsilon_i) = \frac{n_i}{g_i}$$

Thus

$$n_i = g_i f(\varepsilon_i)$$

Statistics	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac
Mean occupation	$f(\varepsilon) = e^{-(\varepsilon-\mu)/kT}$	$f(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/kT} - 1}$	$f(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/kT} + 1}$

---

12. Allowed Occupation Numbers

Statistics	Maxwell–Boltzmann	Bose–Einstein	Fermi–Dirac

Possible values of n_i	0,1,2,3,...	0,1,2,3,...	0 or 1

The restriction in Fermi–Dirac statistics arises from the **Pauli exclusion principle**.

---

13. Classical Limit

If

$$e^{(\varepsilon-\mu)/kT} \gg 1$$

then

Bose–Einstein → Maxwell–Boltzmann

Fermi–Dirac → Maxwell–Boltzmann

This corresponds to

- **high temperature**
- **low particle density**

---

14. Characteristic Behaviour

Property	**Bose–Einstein**	**Fermi–Dirac**
Low-energy occupation	Very large accumulation	Limited to one particle
High-energy behaviour	Exponential decay	Exponential decay
Special phenomenon	Bose–Einstein condensation	Fermi energy

For fermions at $T = 0$

$$f(\varepsilon) = \begin{cases} 1 & \varepsilon < E_F \\ 0 & \varepsilon > E_F \end{cases}$$

where E_F is the **Fermi energy**.

---

15. Unified Representation

The quantum distributions can be written as

$$f(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/kT} \pm 1}$$

where

- $+\rightarrow$ **Fermi–Dirac statistics**

- $\longrightarrow$ **Bose–Einstein statistics**

Maxwell–Boltzmann statistics arises when

$$e^{(\varepsilon-\mu)/kT} \gg 1$$

so that

$$f(\varepsilon) \approx e^{-(\varepsilon-\mu)/kT}$$

---

16. Fundamental Conceptual Difference

Statistics	Physical origin
Maxwell–Boltzmann	Classical distinguishable particles
Bose–Einstein	Indistinguishable bosons with symmetric wavefunctions
Fermi–Dirac	Indistinguishable fermions with antisymmetric wavefunctions leading to Pauli exclusion

Determination of $\boldsymbol{\beta = \frac{1}{kT}}$

(Concise Lecture-Note Version for Statistical Thermodynamics)

1. Boltzmann Relation

Boltzmann related entropy to the number of accessible microstates:

$$S = k\ln\ \Omega$$

At equilibrium the system is overwhelmingly found in the **most probable macrostate**, therefore

$$S = k\ln\ W$$

where

- k= Boltzmann constant
- W= number of microstates of the most probable distribution.

---

2. Number of Microstates of a Distribution

Consider Nparticles distributed among energy levels ε_i.

Let

- N_i= number of particles in level i
- g_i= degeneracy of level i

The number of microstates for the distribution is

$$W = \frac{N!}{\prod_i N_i\,!} \prod_i g_i^{N_i}$$

Taking logarithm

$$\ln\ W = \ln\ N! + \sum_i N_i \ln\ g_i - \sum_i \ln\ N_i!$$

Using **Stirling's approximation**

$$\ln\ N! \approx N \ln\ N - N$$

we obtain

$$\ln\ W = N \ln\ N + \sum_i N_i \ln\ g_i - \sum_i N_i \ln\ N_i$$

---

3. Entropy Expression

Using $S = k \ln\ W$

$$S = kN \ln\ N + k \sum_i N_i \ln\ g_i - k \sum_i N_i \ln\ N_i$$

---

4. Most Probable Distribution

For a canonical ensemble the number of particles in level i is

$$N_i = \frac{N g_i e^{-\beta \varepsilon_i}}{q}$$

where

$$q = \sum_i g_i\, e^{-\beta \varepsilon_i}$$

is the **molecular partition function**.

Taking logarithm

$$\ln\ N_i = \ln\ N - \ln\ q + \ln\ g_i - \beta \varepsilon_i$$

---

5. Substituting into Entropy

Substituting the above expression for $\ln N_i$ into the entropy equation gives

$$S = kN \ln\ q + k\beta E$$

where

$$E = \sum_i N_i \, \varepsilon_i$$

is the total energy.

---

6. Differentiation

Differentiating

$$S = kN\ln \, q + k\beta E$$

$$dS = kN\frac{dq}{q} + kE \, d\beta + k\beta \, dE$$

Using the partition function

$$q = \sum_i g_i \, e^{-\beta\varepsilon_i}$$

it can be shown that

$$\frac{dq}{q} = -\frac{E}{N}d\beta$$

The partition function is

$$q = \sum_i g_i \, e^{-\beta\varepsilon_i}$$

Taking the differential:

$$dq = \sum_i g_i \left(\quad -\varepsilon_i e^{-\beta\varepsilon_i} d\beta - \beta e^{-\beta\varepsilon_i} d\varepsilon_i \right)$$

If the energy levels are fixed ($d\varepsilon_i = 0$)*:*

$$dq = -\sum_i g_i \, \varepsilon_i e^{-\beta\varepsilon_i} d\beta$$

Using

$$N_i = \frac{N g_i e^{-\beta\varepsilon_i}}{q}$$

we obtain

$$\frac{dq}{q} = -\frac{\sum_i \varepsilon_i \, N_i}{N} d\beta$$

Since

$$E = \sum_i N_i \, \varepsilon_i$$

$$\frac{dq}{q} = -\frac{E}{N} d\beta$$

Substituting,

$$dS = k\beta dE$$

Thermodynamic Relation

From classical thermodynamics

$$dE = TdS - PdV$$

At constant volume

$$dE = TdS$$

or

$$dS = \frac{dE}{T}$$

Expression of $\boldsymbol{\beta}$

Comparing

$$dS = k\beta dE$$

with

$$dS = \frac{dE}{T}$$

gives

$$k\beta = \frac{1}{T}$$

Therefore

$$\boxed{\beta = \frac{1}{kT}}$$

Alternate derivation of $\boldsymbol{\beta}$

For a monatomic ideal gas:

$$U = \sum_i N_i \, \varepsilon_i$$

Using Boltzmann distribution:

$$U = \frac{N}{q} \sum_i g_i \, \varepsilon_i e^{-\beta \varepsilon_i}$$

For translational motion:

$$\varepsilon_i = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2)$$

Replacing summation with integrals:

$$U = \frac{1}{2} mN \frac{\int_{-\infty}^{+\infty} (v_x^2 + v_y^2 + v_z^2) e^{-\frac{\beta m}{2}(v_x^2+v_y^2+v_z^2)} dv_x dv_y dv_z}{\int_{-\infty}^{+\infty} e^{-\frac{\beta m}{2}(v_x^2+v_y^2+v_z^2)} \, dv_x dv_y dv_z}$$

Since components are independent:

$$U = \frac{3}{2} mN \frac{\int_{-\infty}^{+\infty} v^2 \, e^{-\beta m v^2/2} dv}{\int_{-\infty}^{+\infty} e^{-\beta m v^2/2} \, dv}$$

Using Gaussian integrals:

$$U = \frac{3N}{2\beta}$$

The expression of average kinetic energy becomes

$$\overline{U} = \frac{U}{N} = \frac{3}{2\beta}$$

From kinetic theory:

$$\bar{U} = \frac{3}{2} kT$$

Hence

$$\frac{3}{2\beta} = \frac{3}{2} kT$$

Thus,

$$\beta = \frac{1}{kT}$$

Worked out examples on partition functions

1 Translational partition function

For a temperature of **273 K** and a volume of **2.24 × 10^{-2} m^3** determine the translational partition function for **Argon gas**. The translational partition function is

$$z_{tr} = \left(\frac{2\pi mkT}{h^2}\right)^{3/2} V$$

$$m = 40\ g\ mol^{-1} = 6.64 \times 10^{-26}\ kg \text{ per atom}$$

$$kT = 3.77 \times 10^{-21}\ J$$

$$z_{tr} = 4.79 \times 10^{30} \text{ states per atom}$$

2 Vibrational partition function

The vibrational frequency of oxygen is **1580 cm^{-1}**. Determine the vibrational partition function for oxygen gas. The vibrational partition function is

$$z_{vib} = \frac{\exp\left(-\frac{h\nu}{2kT}\right)}{1 - \exp\left(-\frac{h\nu}{kT}\right)}$$

The vibrational frequency in Hz is

$$4.737 \times 10^{13}$$

so that

$$\theta_{vib} = \frac{h\nu}{k} = 2274\ K$$

At room temperature **(300 K)**

$$\theta_{vib}/T = 7.58$$

which indicates that the vibrational energy is almost **8 times higher than the thermal energy**. This would imply that the vibrational partition function will predominantly have contribution from the ground state.

Inserting these quantities into the vibrational partition function expression gives a value of

$$1.0005$$

3 Rotational partition function

The length of the bond in oxygen molecule is **1.2074 Å**. Determine the rotational partition function for oxygen at **300 K**.

The rotational partition function for a diatomic is

$$z_{rot} = \frac{8\pi^2 IkT}{\sigma h^2} = \frac{T}{\sigma\theta_{rot}}$$

where **I** is the moment of inertia

$$I = \mu R^2$$

σ is a symmetry factor which accounts for equivalent orientations of the molecule and is **1 for a heteronuclear diatomic and 2 for a homonuclear molecule**.

For a polyatomic molecule

$$z_{rot} = \left(\frac{8\pi^2 IkT}{h^2}\right)^{3/2} \frac{\pi I_a I_b I_c}{\sigma}$$

Note that for a polyatomic the rotational partition function is a product of three terms corresponding to rotation about three perpendicular axes.

Returning to the problem at hand, the moment of inertia of oxygen molecule is

$$I = \mu R^2 = \frac{mR^2}{2}$$

$$= \frac{(0.016\ kg\ mol^{-1})(1.2074 \times 10^{-10} m)^2}{2 \times 6.023 \times 10^{23}\ molecules\ mol^{-1}}$$

$$= 1.937 \times 10^{-46}\ kg\ m^2$$

from which we determine the rotational temperature to be

$$\theta_{rot} = \frac{h^2}{8\pi^2 Ik} = 2.08\ K$$

At room temperature

$$T/\theta_{rot} \approx 150$$

which indicates that the thermal energy is almost **150 times higher than the rotational energy** and we expect the rotational partition function to have extensive contribution from the excited states.

As expected we find that the rotational partition function to be

$$z_{rot} = \frac{T}{\sigma\theta_{rot}} = 72$$

4 Entropy of a monoatomic ideal gas

The entropy may be obtained from its relation to the Helmholtz energy

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,N}$$

which in turn is related to the partition function as

$$A = -kT \ln\ Z$$

If the energy is known, as is the case here, it is easier to obtain the entropy from the classical thermodynamical definition of the Helmholtz energy

$$A = U - TS$$

For a monoatomic ideal gas there is only a translational contribution to the entropy and the average energy is given by the equipartition principle

$$S = \frac{U - A}{T} = \frac{(3/2NkT - A_{tr})}{T}$$

with

$$A_{tr} = -NkT\ln\left[\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\frac{V}{N}e\right]$$

In writing out the last equality we have used the Stirling's approximation for

$$N! = N^N e^{-N}\sqrt{2\pi N}$$

The entropy of a monoatomic ideal gas is hence

$$S = Nk\ln\left[\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\frac{V}{N}e^{5/2}\right]$$

Absolute entropy of argon

The expression for the entropy may be used to obtain the absolute entropy of monoatomic gases, say argon at **300 K and 1 atm**.

The translational partition function of argon at **300 K** is

$$z_{tr} = \left(\frac{2\pi mkT}{h^2}\right)^{3/2} V = 2.47 \times 10^{32} V$$

so that

$$\frac{S}{N} = k\ln\left(2.47 \times 10^{32}\frac{V}{N}e^{5/2}\right)$$

Using the equation of state to replace

$$V/N = kT/p = 4.10 \times 10^{-26}$$

we get that the absolute entropy of argon is

$$S/N = 158.4\, J\, K^{-1}\, mol^{-1}$$

5 Chemical equilibrium of the water-gas reaction

The water-gas reaction is

$$CO_2 + H_2 \rightleftharpoons H_2O + CO$$

for which the experimentally determined equilibrium constants are

$$2.0,\ 2.8,\ 3.5$$

at

$$1295,\ 1565,\ 1823\ K$$

respectively.

Determine the equilibrium constant at **1565 K** from the partition function.

The basic equation for the statistical thermodynamic calculation of the equilibrium constant is

$$K_c = \frac{q_{H_2O} q_{CO}}{q_{CO_2} q_{H_2}} \exp\left(-\frac{\Delta E_0}{RT}\right)$$

where $\mathbf{\Delta E_0}$ is the difference in energy between the ground rotational/vibrational states of the reactants and products.

The q's are the unit-volume partition function and may be written as

$$q = q_{trans} \times q_{rot} \times q_{vib}$$

Thus

$$K_c = K_c(trans) \times\ K_c(rot) \times K_c(vib) \times \exp\left(-\frac{\Delta E_0}{RT}\right)$$

Where K_c(trans) is

$$K_{c(trans)} = \frac{q_{H_2O(trans)} \times q_{CO(trans)}}{q_{CO_2(trans)} \times q_{H_2(trans)}}$$

The equilibrium constant is dimensionless, and thus all fundamental constants in $K_{c(trans)}$ cancel. But they do not in Kc(rot) because of the difference in the number of rotational degrees of freedom on the right, five, and the left, four.

Because the numbers of molecules on the two sides of the equation are the same

$$K_c(trans) = \left(\frac{M_{H_2O} M_{CO}}{M_{CO_2} M_{H_2}}\right)^{3/2}$$

The rotational factor in $K_c$is

$$K_c(\text{rot}) = \frac{\sqrt{\pi (I_a I_b I_c)_{H_2O}} \left(\frac{8\pi^2 kT}{h^2}\right)^{3/2} \times I_{CO} \left(\frac{8\pi^2 kT}{h^2}\right)}{I_{CO_2} \left(\frac{8\pi^2 kT}{h^2}\right) \times I_{H_2} \left(\frac{8\pi^2 kT}{h^2}\right)} \times \frac{\sigma_{CO_2} \sigma_{H_2}}{\sigma_{H_2O} \sigma_{CO}}$$

$$= \left(\frac{\sqrt{\pi (I_a I_b I_c)_{H_2O} I_{CO}}}{I_{CO_2} I_{H_2}}\right) \times \left(\frac{8\pi^2 kT}{h^2}\right)^{1/2} \times 2$$

If all moments of inertia are expressed in **amu Å²** a change is necessary in the factor containing the fundamental constants

$$K_c(\text{rot}) = \left(\frac{2\sqrt{\pi(I_a I_b I_c)_{H_2O} I_{CO}}}{I_{CO_2} I_{H_2}}\right)\left(\frac{8\pi^2 kT \times 10^{-20}}{h^2 L_A}\right)^{1/2}$$

The second factor

$$\left(\frac{8\pi^2 kT \times 10^{-20}}{h^2 L_A}\right) = 114.8$$

The vibrational factor in $K_c$is

$$K_c(\text{vib}) = \frac{q_{vib}(H_2O,\ 3\ modes)\ q_{vib}(CO,\ 1\ mode)}{q_{vib}(CO_2,\ 4\ modes)\ q_{vib}(H_2,\ 1\ mode)}$$

The molecular and other parameters are given in the table below.

Molecular and other parameters

Property	**CO_2**	**H_2**	**H_2O**	**CO**
Heat of formation at 0 K (kJ mol^{-1})	−393.165	0.000	−238.935	−113.813
Bond length (10^{-10} m)	1.160	0.741	0.960	1.128
Bond angle (degrees)	180		104.5	
Moment of inertia (amu Å^2)	43.0	0.277	1.27	8.73
Vibration frequencies (cm^{-1})	2350	4160	3652	2168
	1320		3756	
	668		1595	
	668			

[1] $I_a I_b I_c$for **H_2O** is **1.128**

The vibrational partition functions, the energies of the states being measured from their respective ground state, are listed below.

Vibrational partition functions, the energies of the states being measured from their respective ground state, are listed below

Molecule	**ω (cm^{-1})**	q_{vib}
CO_2	2350	1.13
	1320	1.43
	668	2.18

Molecule	ω (cm^{-1})	q_{vib}
	668	2.18
H_2	4160	1.02
H_2O	3652	1.04
	3756	1.04
	1595	1.30
CO	2168	1.16

The heat of reaction at **0 K** is

$$\Delta E_0 = -113.813 - 238.935 + 393.165 + 0.000 = +40.417\ kJ\ mol^{-1}$$

Finally we obtain

$$K_c = \left(\frac{18.02 \times 28.01}{44.01 \times 2.016}\right)^{3/2} \times \left(\frac{2\sqrt{\pi}\ 1.27 \times 8.73}{43.0 \times 0.277}\right) \times \sqrt{114.8} \times 0.208 \times \exp\left(-3.115\right)$$

$$K_c = 2.98$$

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[i] **Electronic Partition Function and Degeneracy**

The **electronic partition function** accounts for the population of different electronic energy levels of a molecule or atom.

The electronic partition function is

$$q_{ele} = \sum_i g_i\, e^{-\beta \varepsilon_{ele,i}}$$

where

- g_i= degeneracy of the i^{th}electronic state
- $\varepsilon_{ele,i}$= energy of that electronic state
- $\beta = \frac{1}{kT}$
- k= Boltzmann constant
- T= temperature.

It is convenient to measure energies relative to the **ground electronic state**.

Thus

$$\varepsilon_0 = 0$$

Substituting into the partition function

$$q_{ele} = g_0 e^{-\beta\varepsilon_0} + g_1 e^{-\beta\varepsilon_1} + g_2 e^{-\beta\varepsilon_2} + \cdots$$

Since

$$e^{-\beta\varepsilon_0} = 1$$

the expression becomes

$$q_{ele} = g_0 + g_1 e^{-\varepsilon_1/kT} + g_2 e^{-\varepsilon_2/kT} + \cdots$$

Electronic excitation energies are typically very large.

Typical values:

Quantity	Approximate Value
Electronic excitation energy	1–10 eV
Thermal energy at 300 K	$kT \approx 0.026$ eV

Thus

$$\varepsilon_1 \gg kT$$

The contribution of excited electronic states is determined by the Boltzmann factor

$$e^{-\varepsilon_i/kT}$$

If

$$\varepsilon_i \gg kT$$

then

$$e^{-\varepsilon_i/kT} \approx 0$$

Thus the contributions of excited electronic states are negligible at ordinary temperatures.
Neglecting the excited-state terms

$$q_{ele} \approx g_0$$

where g_0 is the degeneracy of the ground electronic state.

Value of g_0

For most stable molecules the ground electronic state is **non-degenerate**.
Thus

$$g_0 = 1$$

Therefore

$$q_{ele} \approx 1$$

This means that almost all molecules remain in the ground electronic state at ordinary temperatures.

Where $g_0 \neq 1$

For some atoms or radicals the ground state is degenerate.
Examples:

Species	Ground-state degeneracy
O atom	5
N atom	4
NO molecule	2

In such cases

$$q_{ele} \approx g_0$$

but g_0 is greater than 1.

Thermodynamic Contribution

The internal energy contribution from electronic motion is

$$E_{ele} = NkT^2 \left(\frac{\partial \ln\ q_{ele}}{\partial T}\right)$$

Since

$$q_{ele} \approx g_0$$

and g_0 does not depend on temperature,

$$\frac{\partial \ln\ q_{ele}}{\partial T} = 0$$

Thus

$$E_{ele} \approx 0$$

Similarly

$$C_{V,ele} \approx 0$$

At ordinary temperatures:

$$q_{ele} \approx g_0$$

and for most molecules

$$q_{ele} \approx 1$$

Therefore electronic motion contributes negligibly to thermodynamic properties under normal conditions.